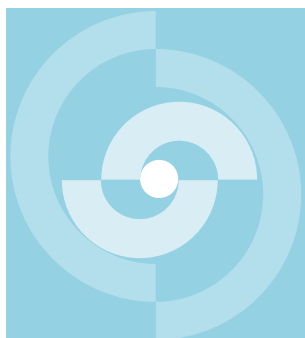




TECHNICAL REPORT 03-07

**Diffusion of HTO, $^{36}\text{Cl}^-$, $^{125}\text{I}^-$ and
 $^{22}\text{Na}^+$ in Opalinus Clay:**

**Effect of confining pressure,
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This report was prepared on behalf of Nagra. The viewpoints presented and conclusions reached are those of the author(s) and do not necessarily represent those of Nagra.

PREFACE

The Laboratory for Waste Management of the Nuclear Energy and Safety Research Department at the Paul Scherrer Institut is performing work to develop and test models as well as to acquire specific data relevant to performance assessments of planned Swiss nuclear waste repositories. These investigations are undertaken in close co-operation with, and with the financial support of, the National Cooperative for the Disposal of Radioactive Waste (Nagra). The present report is issued simultaneously as a PSI-Bericht and a Nagra Technical Report.

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ABSTRACT

The Opalinus Clay (OPA) formation in the Zürcher Weinland is a potential host rock for a repository for spent fuel, vitrified high-level waste and long-lived intermediate-level waste in Switzerland. Owing to its small hydraulic conductivity (10^{-14} - 10^{-13} m·s⁻¹), it is expected that transport of solutes will be dominated by diffusion. This study addresses the diffusion of tritiated water (HTO), ³⁶Cl⁻, ¹²⁵I⁻ and ²²Na⁺ through Opalinus Clay samples. The samples were collected in the Mont Terri Underground Rock Laboratory, where the OPA formation is located at a depth between -200 and -300 m below the surface, and in the deep borehole in Benken (Zürcher Weinland), where the OPA layer is located at a depth between -539 and -652 m.

Effective diffusion coefficients (D_e), rock capacity factors (α) and diffusion-accessible porosities (ϵ) were measured using the through-diffusion technique. Transport (diffusion) was measured both perpendicular and parallel to the bedding. Special cells that allowed the application of an axial confining pressure were designed. The pressures applied ranged from 1 to 5 MPa for Mont Terri samples and between 4 and 15 MPa for Benken samples, the upper values representing the in-situ confining pressure at both locations. The test solutions used in the experiments were synthetic Opalinus Clay pore water, which has Na and Cl as main components (Mont Terri: $I = 0.39$ M; Benken: $I = 0.20$ M).

Pressure only had a small effect on the value of the effective diffusion coefficients. In the case of Mont Terri samples, increasing the pressure from 1 to 5 MPa resulted in a decrease of the effective diffusion coefficient of 20% for HTO, 27% for ³⁶Cl⁻, 29% for ¹²⁵I⁻ and 17 % for ²²Na⁺. In the case of Benken samples, increasing the pressure from 4 to 15 MPa resulted in a decrease of D_e of 17% for HTO, 22% for ³⁶Cl⁻, 32% for ¹²⁵I⁻ and 17 % for ²²Na⁺. Moreover, the effective diffusion coefficients for ³⁶Cl⁻ are smaller than for HTO, which is consistent with an effect arising from anion exclusion. This ion exclusion effect is smaller in samples from Mont Terri than in samples from Benken, which can be explained by the higher ionic strength of the Mont Terri water used in the experiments. The diffusion of ²²Na⁺ is similar to that of HTO in the case of Mont Terri OPA. For Benken OPA, the D_e value of ²²Na⁺ is a factor of 2 higher than that of HTO. This last observation cannot be explained so far but is comparable to experimental data from ANDRA (1999) on Callovo-Oxfordian claystones from the Meuse/Haute Sarne site.

¹²⁵I⁻ is retarded with respect to ³⁶Cl⁻. This is caused by a weak sorption of ¹²⁵I⁻ on the Opalinus Clay. The distribution coefficients, calculated from the rock capacity factor under the assumption that the diffusion-accessible porosity of ¹²⁵I⁻ is the same as for ³⁶Cl⁻, range between 0.01 and 0.02 cm³·g⁻¹. The effective diffusion coefficients of ¹²⁵I⁻ are comparable with those of ³⁶Cl⁻.

Out-diffusion data of HTO, $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$ are in good agreement with the through-diffusion data. In the case of ^{125}I the agreement is less. The flux calculated with D_e and α derived from through-diffusion measurements is smaller than the observed flux. This indicates that other (unknown) processes are taking place.

The diffusion coefficients measured in this study on Mont Terri samples are in good agreement with recent measurements of three other laboratories, within the framework of a laboratory comparison exercise. The values of the diffusion-accessible porosities, however, show a larger degree of scatter, indicating that through-diffusion is not the method of choice for obtaining reliable porosity values.

Diffusion parallel to the bedding is higher than diffusion perpendicular to the bedding. The effective diffusion coefficient for diffusion parallel to bedding is a factor of 4 – 6 larger than for diffusion perpendicular to the bedding. This is due to the layered structure of the Opalinus Clay, resulting in a smaller tortuosity factor for diffusion along the bedding planes. The observed effect was similar for HTO, $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$. This anisotropy is more pronounced for the Opalinus Clay from Benken than for Mont Terri, indicating that the clay platelets are more preferentially oriented in the case of Benken OPA.

The temperature dependence of diffusion of HTO in OPA is of an Arrhenius type. The activation energy ($22 \text{ kJ}\cdot\text{mol}^{-1}$), however, is larger than for diffusion in bulk water ($18 \text{ kJ}\cdot\text{mol}^{-1}$). This indicates that confined water in the narrow pores of the Opalinus Clay has partly a different structure.

Diffusion measurements with HTO on OPA samples from different depths showed that the effective diffusion coefficients for diffusion perpendicular to the bedding decrease with increasing depth. The difference between the top and the bottom of the OPA layer, however, is not more than a factor of 1.5. For diffusion parallel to the bedding, no difference between top and bottom could be observed. It can be concluded that the Opalinus Clay layer is very homogeneous with respect to its diffusion properties.

The effective diffusion coefficient measured for the HTO in OPA is in good agreement with values measured in other sedimentary rocks and can be related to the porosity using Archie's Law with exponent $m=2.5$.

RESUME

Les Argiles à Opalinus (OPA) constituent la formation d'accueil envisagée pour un dépôt destiné aux assemblages combustibles usés (AC), aux déchets de haute activité vitrifiés (DHA) issus du retraitement des assemblages combustibles, ainsi qu'aux déchets de moyenne activité à vie longue (DMAL) en Suisse. En raison de la faible conductivité hydraulique de ces argiles (10^{-14} - 10^{-13} m·s⁻¹), les transports de solutés au travers de la formation s'effectuent en majorité par diffusion. La présente étude aborde la diffusion de l'eau tritiée (HTO), ainsi que de ³⁶Cl⁻, ¹²⁵I⁻ et ²²Na⁺ au travers d'échantillons d'Argiles à Opalinus. Les échantillons proviennent du laboratoire souterrain du Mont Terri, où la couche d'OPA est située à une profondeur de -200 à -300 m par rapport à la surface du sol, et du forage de Benken (Weinland zurichois), où la couche d'OPA est située à une profondeur de -539 à -652 m.

Les coefficients de diffusion effectifs (D_e), les capacités de la roche (α) et les porosités accessibles à la diffusion (ε) ont été mesurés par le biais de la technique de la "through-diffusion". On a mesuré le transport (c'est-à-dire la diffusion) perpendiculairement et parallèlement au litage. Des cellules spéciales permettant d'appliquer une contrainte de confinement axiale ont été mises au point. Les contraintes appliquées étaient comprises entre 1 et 5 MPa pour les échantillons du Mont Terri, et entre 4 et 15 MPa pour les échantillons de Benken, les valeurs supérieures correspondant aux contraintes de confinement in-situ pour chaque site respectif. Pour les expériences, on a utilisé des équivalents synthétiques de l'eau interstitielle des Argiles à Opalinus, dont les principaux composants sont Na⁺ et Cl⁻ (Mont Terri: I = 0.39 M; Benken: I = 0.20 M).

On a observé que les coefficients de diffusion effectifs n'étaient que marginalement affectés par les contraintes appliquées. Dans le cas des échantillons du Mont Terri, une augmentation de la pression de 1 à 5 MPa a entraîné une diminution du coefficient de diffusion de 20% pour HTO, 27% pour ³⁶Cl⁻, 29% pour ¹²⁵I⁻ et 17 % pour ²²Na⁺. Pour les échantillons de Benken, le passage de 4 à 15 MPa a entraîné une baisse du D_e de 17% pour HTO, 22% pour ³⁶Cl⁻, 32% pour ¹²⁵I⁻ et 17 % pour ²²Na⁺. Par ailleurs, les coefficients de diffusion pour ³⁶Cl⁻ se sont avérés inférieurs à ceux observés pour HTO, ce qui peut être interprété comme une conséquence d'un phénomène d'exclusion anionique. Toutefois, ce phénomène d'exclusion ionique est moins marqué pour les échantillons du Mont Terri que pour ceux de Benken, ce qui peut s'expliquer par le fait que l'eau du Mont Terri utilisée pour les expériences avait une force ionique supérieure. La diffusion de ²²Na⁺ est du même ordre que celle de HTO dans le cas des argiles du Mont Terri. Pour celles de Benken, la valeur observée pour ²²Na⁺ est deux fois supérieure à celle de HTO. Cette différence n'a pas encore pu être expliquée. Des observations semblables ont été faites par ANDRA (1999) pour le Callovo-Oxfordien au site de Meuse/Haute-Marne.

Pour $^{125}\text{I}^-$, on observe un effet de retard par rapport à $^{36}\text{Cl}^-$, causé par la faible sorption de $^{125}\text{I}^-$ sur les Argiles à Opalinus. Si l'on calcule les coefficients de diffusion à partir de la capacité de la roche en postulant que la porosité accessible à la diffusion de $^{125}\text{I}^-$ est la même que pour $^{36}\text{Cl}^-$, on obtient des valeurs comprises entre 0.01 et 0.02 $\text{cm}^3 \cdot \text{g}^{-1}$. Les coefficients de diffusion effectifs pour $^{125}\text{I}^-$ sont du même ordre que ceux observés pour $^{36}\text{Cl}^-$.

Les données de la "out-diffusion" pour HTO, $^{36}\text{Cl}^-$ et $^{22}\text{Na}^+$ concordent largement avec les données de la "through-diffusion". Dans le cas de $^{125}\text{I}^-$, la concordance est moindre. Le flux calculé en dérivant D_e et α des paramètres de la "through-diffusion" est inférieur au flux effectivement observé. Ceci montre que d'autres processus (encore indéterminés) sont à l'oeuvre. Des recherches supplémentaires s'avèrent ici nécessaires.

Les coefficients de diffusion mesurés au cours de cette étude sur les échantillons du Mont Terri concordent avec les mesures récentes effectuées par trois autres laboratoires dans le cadre d'un exercice de comparaison. Pour les porosités accessibles à la diffusion, on observe cependant une plus large répartition des valeurs. On peut en conclure que la technique de la "through-diffusion" n'est pas la méthode la plus sûre pour mesurer la porosité.

La diffusion parallèle au litage est plus rapide que la diffusion perpendiculaire au litage. Les coefficients de diffusion effectifs pour la diffusion parallèle au litage sont un facteur 4 - 6 plus larges que pour la diffusion perpendiculaire au litage. Ceci est dû à la structure en feuillets des Argiles à Opalinus, où le facteur d'obstruction est plus faible pour la diffusion dans le sens du litage. Ceci a été observé aussi bien pour HTO, $^{36}\text{Cl}^-$ que $^{22}\text{Na}^+$. Cette anisotropie est plus prononcée pour les argiles de Benken que pour celles du Mont Terri, ce qui montre qu'à Benken, l'orientation des feuillets est plus marquée.

La sensibilité à la température de la diffusion de HTO dans les Argiles à Opalinus est de type Arrhenius. Toutefois, l'énergie d'activation (22 $\text{kJ} \cdot \text{mol}^{-1}$) est supérieure aux valeurs observées pour la diffusion dans de l'eau (bulk water) (18 $\text{kJ} \cdot \text{mol}^{-1}$). Ceci indique que l'eau confinée dans les pores étroits des Argiles à Opalinus a, pour une part, une structure différente.

La mesure de la diffusion de HTO sur des échantillons d'OPA prélevés à des profondeurs différentes montre que les coefficients de diffusion effectifs pour la diffusion perpendiculaire au litage décroissent au fur et à mesure que la profondeur augmente. Cependant, la différence observée entre les limites supérieures et inférieures de la couche d'OPA ne dépasse pas un facteur d'1,5. Pour la diffusion parallèle au litage, on n'a pas observé de différence entre les différentes profondeurs. Ceci pourrait indiquer que les Argiles à Opalinus sont très homogènes du point de vue de leurs propriétés de diffusion.

Le coefficient de diffusion effective mesuré pour le HTO en OPA est en bon accord avec des valeurs mesurés en cas d'autres roches sédimentaires et peut être lié à la porosité en utilisant la loi d'Archie avec l'exponent $m=2.5$.

ZUSAMMENFASSUNG

Der Opalinuston im Zürcher Weinland ist ein potentiell Wirtgestein für ein geologisches Tiefenlager für abgebrannte Brennelemente, hochaktive verglaste Abfälle und langlebige mittelaktive Abfälle. Aufgrund seiner niedrigen hydraulischen Leitfähigkeit (10^{-14} - 10^{-13} m·s⁻¹) ist zu erwarten, daß der Transport von gelösten Stoffen in erster Linie durch Diffusion erfolgt. Die vorliegenden Untersuchungen beschreiben die Diffusion von tritiiertem Wasser (HTO), ³⁶Cl⁻, ¹²⁵I⁻ und ²²Na⁺ durch Proben von Opalinuston. Die Proben stammen aus dem Mont Terri Felslabor, wo sich die Opalinustonformation in einer Tiefe zwischen -200 und -300 m unterhalb der Oberfläche befindet, sowie aus der Tiefenbohrung bei Benken (Zürcher Weinland). Die Opalinustonformation liegt dort in einer Tiefe zwischen -539 und -652 m.

Effektive Diffusionskoeffizienten (D_e), „rock-capacity“-Faktoren (α) und diffusionszugängliche Porositäten (ϵ) wurden mittels der Durchdiffusionstechnik ermittelt. Der Stofftransport (Diffusion) wurde senkrecht und parallel zur Gesteinsschichtung in speziellen Diffusionszellen gemessen, in welchen ein axial wirkender Druck auf die Probe übertragen wurde. Der Druck wurde in den Proben vom Mont Terri zwischen 1 und 5 MPa und in den Proben der Tiefbohrung Benken zwischen 4 und 15 MPa variiert. Der obere Wert entspricht jeweils dem in der entsprechenden Gesteinsschicht herrschenden Bergdruck. Als Testlösung wurde synthetisches Opalinustonporenwasser mit den Hauptkomponenten Na und Cl verwendet (Mont Terri: I = 0.39 M; Benken: I = 0.2 M).

Der Druck hatte nur einen kleinen Effekt auf die Werte der effektiven Diffusionskoeffizienten (D_e). Bei den Mont Terri Proben führte eine Erhöhung des Drucks von 1 auf 5 MPa zu einer Abnahme von D_e um 20 % für HTO, 27 % für ³⁶Cl⁻, 29 % für ¹²⁵I⁻ und 17 % für ²²Na⁺. Bei den Proben aus der Tiefbohrung Benken wurde bei einer Erhöhung des Drucks von 5 auf 15 MPa jeweils eine Abnahme der effektiven Diffusionskoeffizienten beobachtet (17 % für HTO, 22 % für ³⁶Cl⁻, 32 % für ¹²⁵I⁻ und 17 % für ²²Na⁺). Ferner sind die D_e Werte für ³⁶Cl⁻ kleiner als für HTO, was im Einklang mit einem Anionenausschlusseffekt ist. Allerdings ist dieser Effekt bei den Mont Terri Proben weniger stark ausgeprägt als bei den Proben der Tiefbohrung Benken, dies kann mit der höheren Ionenstärke des in den Experimente verwendeten Mont Terri Wassers erklärt werden. Im Fall der Mont Terri Proben ist das Diffusionsverhalten von ²²Na⁺ jenem von HTO ähnlich. Bei den Proben der Tiefbohrung Benken liegt der D_e Wert von ²²Na⁺ um den Faktor 2 höher als jener von HTO. Für diesen Befund gibt es bislang keine Erklärung. Er steht aber im Einklang mit experimentellen Daten der ANDRA (1999), welche an Callovo-Oxfordian Tongestein vom Meuse/Haute Sarne Fundort gewonnen wurden.

Die Diffusion von $^{125}\text{I}^-$ ist gegenüber jener von $^{36}\text{Cl}^-$ verzögert, was mit der schwachen Sorption von $^{125}\text{I}^-$ an Opalinuston erklärt werden kann. Unter der Annahme identischer diffusionszugänglicher Porositäten für $^{125}\text{I}^-$ und $^{36}\text{Cl}^-$, variieren die aus den "rock-capacity" Faktoren berechneten Sorptionsverteilungskoeffizienten für $^{125}\text{I}^-$ zwischen 0.01 und $0.02 \text{ cm}^3 \cdot \text{g}^{-1}$. Die D_e -Werte für $^{125}\text{I}^-$ sind mit jenen von $^{36}\text{Cl}^-$ vergleichbar.

Die Rückdiffusionsdaten von HTO, $^{36}\text{Cl}^-$ und $^{22}\text{Na}^+$ stimmen gut mit den Durchdiffusionsdaten überein. Im Fall von $^{125}\text{I}^-$ ist die Übereinstimmung weniger befriedigend. Der aus den D_e und α Werten berechnete Durchfluss ist kleiner als im Experiment beobachtet. Dies deutet auf das Vorhandensein anderer (unbekannter) Prozesse hin.

Die in dieser Studie ermittelten Diffusionskoeffizienten stimmen gut mit jenen überein, die im Rahmen einer Laborvergleichsstudie zwischen drei anderen Laboratorien gemessen wurden. Die Werte für die diffusionszugängliche Porosität weisen jedoch etwas grössere Abweichungen auf, was einmal mehr zeigt, dass Durchdiffusionsmessungen nicht unbedingt die Methode der Wahl sind, um Porositäts-Parameter zuverlässig zu ermitteln.

Die Diffusion parallel zur Gesteinsschichtung verlief schneller als jene parallel zur Gesteinsschichtung. Der effektive Diffusionskoeffizient für die Diffusion parallel zur Gesteinsschichtung ist um Faktor 4-6 höher als bei der Diffusion senkrecht zur Gesteinsschichtung. Dies lässt sich mit der geschichteten Struktur des Opalinuston erklären, welche einen kleineren Tortuositätsfaktor für Diffusionsvorgänge entlang der Schichtungen zur Konsequenz hat. Der beobachtete Effekt war ähnlich für HTO, $^{36}\text{Cl}^-$ und $^{22}\text{Na}^+$. Die Anisotropie der Opalinustonproben aus der Tiefbohrung Benken ist stärker ausgeprägt als bei den Proben vom Mont Terri. Offensichtlich sind die Tonplättchen bei den erstgenannten Proben stärker orientiert.

Die Temperaturabhängigkeit der Diffusion von HTO in Opalinuston weist ein Arrhenius-Verhalten auf. Die Aktivierungsenergie ($22 \text{ kJ} \cdot \text{mol}^{-1}$) ist größer als jene von freiem Wasser ($18 \text{ kJ} \cdot \text{mol}^{-1}$). Dies deutet darauf hin, dass das Porenwasser des Opalinustongesteins eine andere Struktur als freies Wasser hat.

Diffusionsmessungen von HTO in Opalinustonproben aus unterschiedlicher Tiefe zeigten, dass die effektiven Diffusionskoeffizienten für die Diffusion senkrecht zur Gesteinsschichtung mit zunehmender Tiefe abnehmen. Die gemessenen D_e -Wert der oberen und unteren Opalinustonschicht unterscheiden sich jedoch höchstens um den Faktor 1.5. Der parallel zur Gesteinsschichtung gemessene effektive Diffusionskoeffizient ist unabhängig von der Tiefe. Dies ist ein Hinweis auf die Homogenität des Opalinustons, was sich in den Diffusionseigenschaften zeigt.

Der effektive Diffusionskoeffizient für HTO in Opalinuston stimmt gut mit Werten von anderen Sedimentgesteine überein und lässt sich über das *Archie Gesetz* (Exponent $m = 2.5$) mit der Porosität verknüpfen.

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1 INTRODUCTION AND MOTIVATION

The Opalinus Clay (OPA) is currently under investigation to demonstrate the basic feasibility of disposing of spent fuel, vitrified high-level waste and long-lived intermediate-level waste (NAGRA, 2002). The OPA was deposited about 180 million years ago as a marine sediment consisting of fine mud particles. It has a thickness of somewhat more than 100 m and contains between 40 and 80 % clay minerals, about 10 % are illite/smectite mixed layer minerals capable of swelling (NAGRA, 2002; MAZUREK, 1999). This formation is located in the northern part of Switzerland.

OPA has a very low hydraulic conductivity ($10^{-14} \text{ m}\cdot\text{s}^{-1} < K_h < 10^{-12} \text{ m}\cdot\text{s}^{-1}$; NAGRA, 2002). Therefore molecular diffusion is considered to be the main transport mechanism for radionuclides released from a repository. Because there are little or no diffusion data available for OPA, a diffusion programme with samples from the Mont Terri Underground Rock Laboratory (URL) and the deep borehole in Benken was started at the Paul Scherrer Institute. The Mont Terri URL (THURY & BOSSART, 1999) is situated in the Canton Jura and was constructed in order to investigate rock mechanical, geochemical and geohydrological properties of argillaceous rocks. The deep borehole in Benken is situated in the Zürcher Weinland (Canton Zürich). This area is a potential candidate to host a repository for high-level and long-lived intermediate-level radioactive waste.

The questions to be answered with the help of the diffusion experiments relate to:

- The effect of overburden pressure on diffusion: sampling of rock material is accompanied by a pressure release that can lead to the formation of micro-cracks. Such micro cracks can affect laboratory measurements.
- The effect of anisotropy on diffusion: OPA is a sedimentary rock with a layered fabric. Diffusion can take place parallel and perpendicular to the orientation of the clay platelets (See Figure 1). It was unknown whether there is a difference between diffusion perpendicular and diffusion parallel to the bedding.
- The effect of temperature on the diffusion: can diffusion coefficients measured at room temperature be extrapolated to other temperatures?
- The transferability of diffusion data from Mont Terri to other OPA sites such as Benken.
- Homogeneity/heterogeneity of OPA with respect to the diffusion properties.

This study focuses on the effect of overburden pressure on the through-diffusion of HTO,

$^{36}\text{Cl}^-$, $^{125}\text{I}^-$ and $^{22}\text{Na}^+$ in OPA from Mont Terri and Benken. The direction of diffusion was perpendicular to bedding. For this purpose, special high-pressure cells that allow the application of an uniaxial pressure on the samples were constructed. The pressure was applied perpendicular to the bedding of the clay. In addition, the anisotropy of diffusion was also investigated. Therefore, a new radial through-diffusion technique was introduced that allows the measurement of effective diffusion coefficients for diffusion parallel to the bedding of a rock. Finally, measurements on the effect of temperature on diffusion were performed in order to explore the temperature dependence of diffusion.

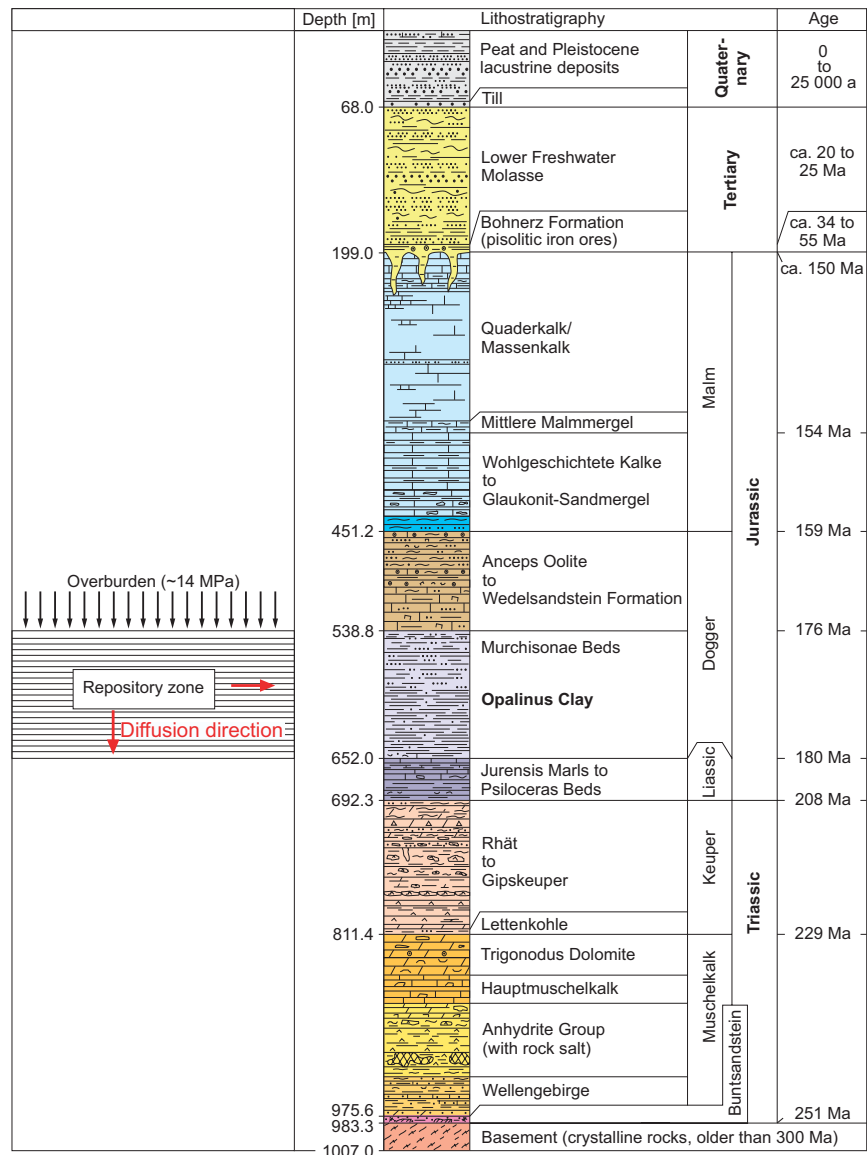


Figure 1: Schematic view of the OPA-formation in the Zürcher Weinland (Benken deep borehole).

2 DIFFUSION PERPENDICULAR TO THE BEDDING

2.1 Through-diffusion experiments

2.1.1 Equipment for one-dimensional through-diffusion experiments

In a through-diffusion experiment, one is looking at the flux and the amount of a solute that is transported (diffused) through a piece of rock during a time interval Δt . The equipment for this kind of measurements basically comprises two compartments that are separated by this piece of rock. The compartments are filled with a solution in equilibrium with the rock. This solution is a real or often a synthetic pore water. The solution of one compartment is labeled with a given amount of tracer and the solution of the other one contains no tracer. By monitoring the solution of the non-labeled compartment, diffusion of the solute can be followed as a function of time.

For the diffusion studies with the Opalinus Clay, the same principle as described above is used. However, because unconfined clay rocks will disintegrate when in contact with the pore water and, because the rock has to be put under a well-defined pressure, a special cell had to be designed. Figure 2 schematically represents the new diffusion cell (PSI-HPDIF-PER/1.0)

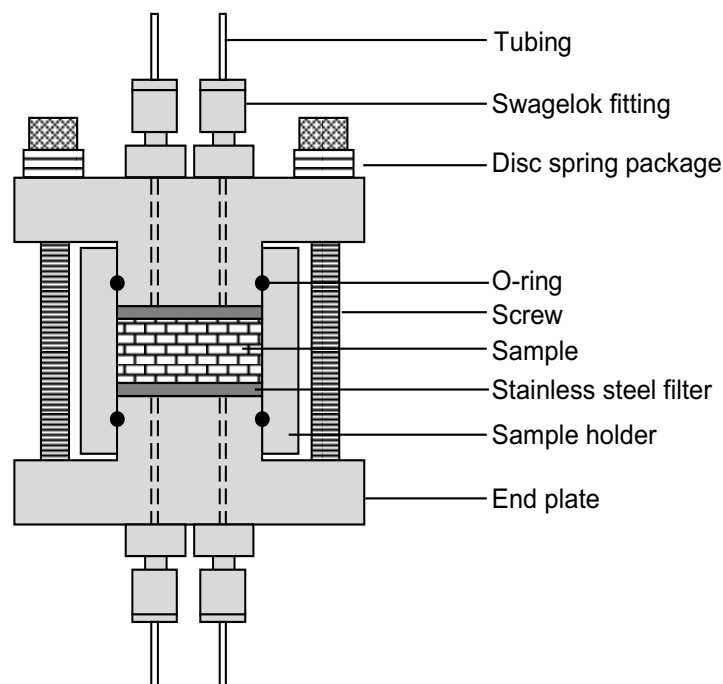


Figure 2: Cross section view of the PSI high-pressure diffusion cell (type PSI-HPDIF-PER/1.0) for measuring diffusion perpendicular to the bedding.

The cell comprises a sample holder, two end pieces and a set of bolts. The sample is sandwiched between two stainless steel filter plates (Stainless steel: 316L, pore diameter: 10 μm ; MOTT industrial division, Farmington, USA). The sample and filters are placed in the sample holder and two end-pieces are put on top of the filters. The two end-pieces are held together by a set of 4 bolts. Between the surface of the end plate and the head of the bolt, a package of 6 disc springs is placed (disc spring A10 DIN 2093). These disc spring packages allow the application of a well-defined stress on the sample when tightening the bolts. The stress is defined over the torque by the following relationship:

$$P_{axial} = k_{ax} \cdot N \quad (1)$$

where P_{axial} is the pressure on the sample [MPa], k_{ax} is a system factor [$\text{MPa} \cdot \text{Nm}^{-1}$] and N is the torque [Nm]. k_{ax} is a factor including the friction coefficient, the pitch of the screw-thread and the diameter of the screw-head. The factor k_{ax} was determined by calibrating the cell. For this purpose, the cell was equipped with a manometer and filled with water. The screws were greased with a MoS_2 screw paste (Molykote 1000, Molykote, Dow Corning GmbH, Germany) to obtain a well-defined friction coefficient of 0.125. The screws were tightened by a torque screwdriver connected to a torque indicator (AFG-500N, Mecmesin Corporation, Santa Rosa, USA).

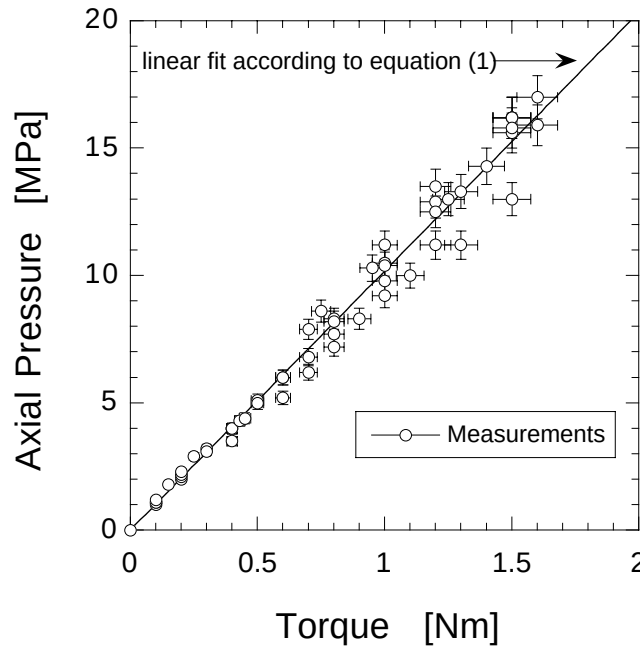


Figure 3: Relationship between the torque and the axial pressure on the sample.

Figure 3 shows the measurements and the linear relationship between the torque and the axial pressure as given by Eqn. (1). The calibration factor ($k_{ax} = 10.165$) was calculated from the slope of the line. In order to be able to monitor the pressure during an experiment, a second slightly modified type of cell was designed (PSI-HPDIF-PER/2.0). This cell was additionally equipped with a load sensor (SENSY, Belgium) connected to a force indicator (FMD-6, Penko, the Netherlands). Figure 4 gives a schematic view of the cell.

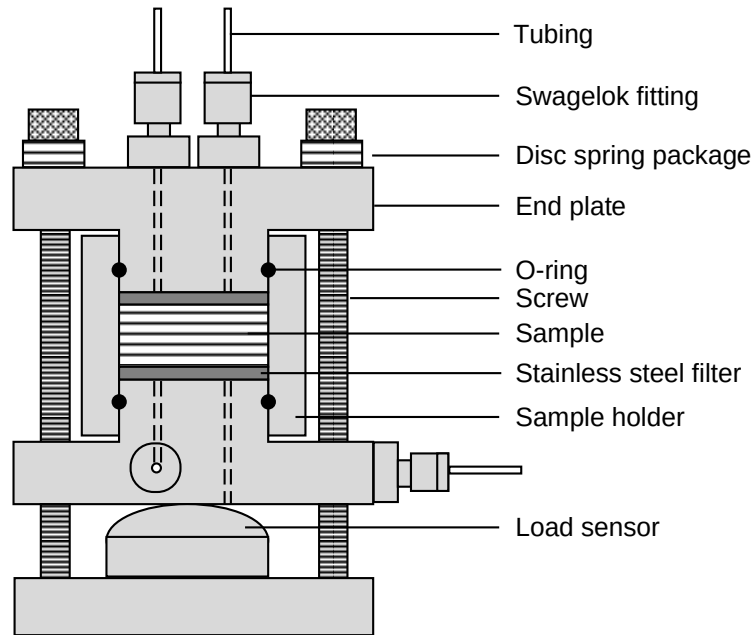


Figure 4: Cross section view of the modified PSI high-pressure diffusion cell (PSI-HPDIF-PER/2.0). The cell is equipped with a load sensor for monitoring the pressure during the experiments.

2.1.2 Samples and sample preparation

A bore core of Opalinus Clay was sliced into 2 cm thick pieces using a diamond saw (no lubricating fluid was used). Each piece was cut in 4 parts as shown in Figure 5 (A). Each part was placed in a plexiglas cylinder (diameter = 5.4 cm, height = 5 cm) and embedded (B) in an epoxy resin (Epofix, Struers GmbH). After hardening of the resin, the cylinder was placed on a lathe and a cylindrical sample (diameter = 2.57 cm, thickness = 1.1 cm) was prepared (C). The samples used were OPA samples from Mont Terri (MT) and from the deep borehole in Benken (BE):

- Benken: a borecore originating from a depth between - 636.67 and - 636.82 m was used for the experiments. The core turned out to be broken in three pieces. The center piece was used for the experiments, i.e. 636.72 – 636.77 m.
- Mont Terri: a borecore BDI-2/7: 5.60 – 5.80 m was used for the experiments. The core turned out to be broken in three pieces. The center piece was used for the experiments: 5.67-5.71 m.

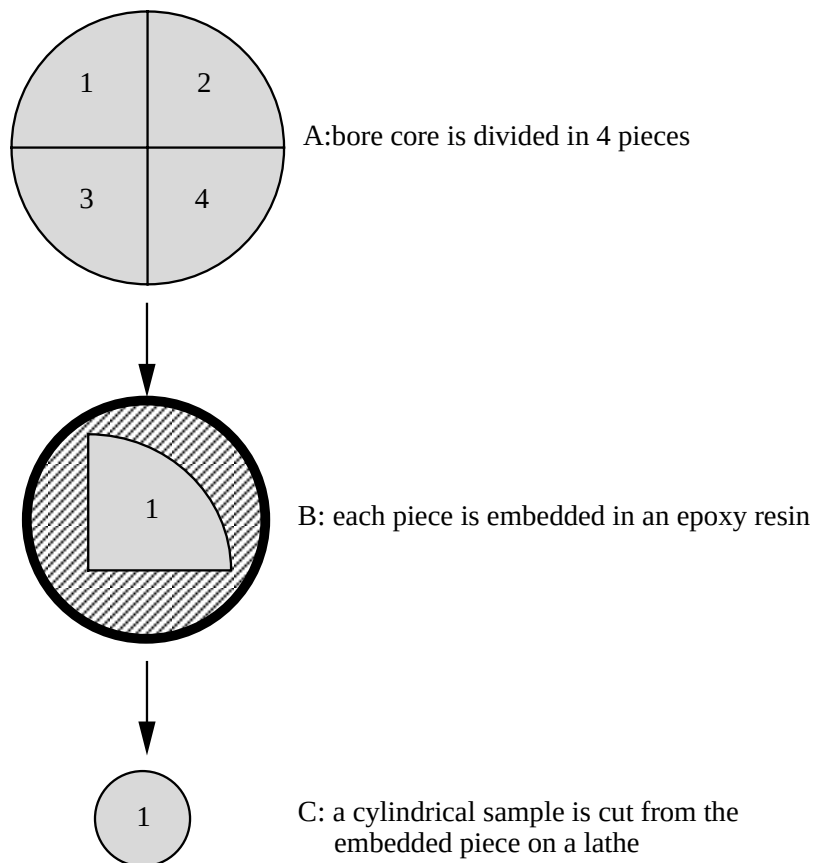


Figure 5: Preparation of small cylindrical samples from a large bore core. The samples are used for diffusion measurements perpendicular to the bedding.

2.1.3 Synthetic pore water

The solutions used for the diffusion experiments were synthetic pore waters. Their composition is given in Table 1. The pore waters are prepared by mixing NaCl with minor amounts of Ca-, Mg- and SrCl₂ and Na₂SO₄. The water of Mont Terri is approximately twice as concentrated as the water of Benken. Both waters were prepared under oxic conditions and in the presence of CO₂.

Table 1: Composition of the synthetic OPA pore waters used in the diffusion experiments.

Element	¹ Mont Terri [mol·dm ⁻³]	² Benken [mol·dm ⁻³]
Na	2.40·10 ⁻¹	1.50·10 ⁻¹
K	1.61·10 ⁻³	4.31·10 ⁻³
Mg	1.69·10 ⁻²	5.21·10 ⁻³
Ca	2.58·10 ⁻²	7.24·10 ⁻³
Sr	5.05·10 ⁻⁴	3.97·10 ⁻⁴
Cl	3.00·10 ⁻¹	1.60·10 ⁻¹
SO ₄	1.41·10 ⁻²	1.00·10 ⁻²
CO ₃ /HCO ₃	4.76·10 ⁻⁴	3.11·10 ⁻⁴
pH	7.6	7.9
Σcations	3.28·10 ⁻¹ eq·dm ⁻³	1.80·10 ⁻¹ eq·dm ⁻³
Σanions	3.28·10 ⁻¹ eq·dm ⁻³	1.80·10 ⁻¹ eq·dm ⁻³
Ionic strength	0.39 M	0.20 M

¹ Type A1 (PEARSON, 1998)

² PEARSON (2000)

2.1.4 Experimental set-up

The experimental set-up of the diffusion experiments is given schematically in Figure 6. The set-up comprises a diffusion cell (PSI-HPDIF-PER/1.0) as described in 2.1.1, an 8-channel peristaltic pump (IPC, Ismatec, IDEX corporation, USA), a 250 cm³ and a 25 cm³ container. The sample is put in the sample holder together with two stainless steel filters and is closed with two end plates. The bolts are greased with Molykote 1000 paste and tightened until the desired torque. Table 2 gives an overview of the torques and pressures applied to the different samples.

The large container was filled with 200 cm³ of synthetic OPA pore water and the small one with 20 cm³. The solutions were circulated in order to saturate the samples. The saturation time was 5 weeks. From independent experiments with a pressure

surveillance¹, it was found that this time was long enough to resaturate the samples. After these 5 weeks, the solutions were replaced by fresh ones and the solutions in the large containers were labeled with the desired tracer. Table 3 gives an overview of the tracers used, as well as their activity and their concentration. The activities in Table 3 represent average values of the tracer during the whole experiment.

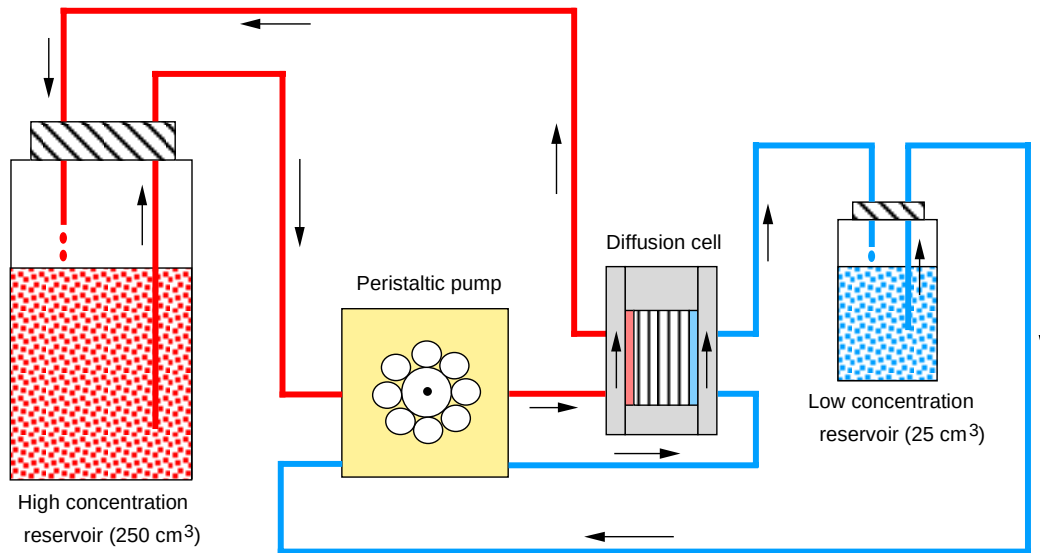


Figure 6: Experimental set-up of a through-diffusion experiment for diffusion perpendicular to the bedding.

The solutions in the small containers (low concentration) were regularly replaced after a given time interval Δt to keep the concentration of the tracer in this compartment as low as possible, i.e. $< 1\%$ of the concentration in the high-concentration compartment. After finishing a diffusion experiment, the solutions in both containers were replaced by synthetic pore water without tracer and out-diffusion was started (see also chapter 2.2). At given time intervals, the activity in the solutions was measured and the solutions were replaced by fresh ones. This procedure was repeated until all activity was diffused out of the samples. As soon as all activity had been diffused out, a new radiotracer was added to the large container and diffusion of the new tracer was started. The sequence of radiotracers studied in this work was:

¹ During the first few days the samples exhibited some swelling and, because the volume of the cell was kept constant, the pressure on the sample increased. After one week, however, the pressure stabilized and no further significant changes in pressure were observed

Table 2: Torque and resulting confining pressures for various OPA samples

Sample	Torque [Nm]	Pressure [MPa]
BE-2a	0.4	4
BE-3a	0.8	8
BE-4a	1.5	15
MT-1a	0.1	1
MT-3a	0.5	5

BE: Benken samples; MT: Mont Terri samples

- 1) tritiated water (HTO; series 1)
- 2) $^{36}\text{Cl}^-$
- 3) $^{125}\text{I}^-$
- 4) tritiated water (HTO; series 2)
- 5) $^{22}\text{Na}^+$

Table 3: Overview of the tracers used and their activity/concentration. The activities given represent average values.

Sample	Activity x 10^{-6} [Bq m $^{-3}$]				
	HTO-1	$^{36}\text{Cl}^-$	$^{125}\text{I}^-$	HTO-2	$^{22}\text{Na}^+$
BE-2a	610±40	1066±46	2642±150	1207±82	995±67
BE-3a	621±40	1064±45	2748±150	1205±82	993±65
BE-4a	547±35	1064±45	2680±150	1197±81	994±65
MT-1a	624±40	1073±46	2731±150	1213±83	993±65
MT-3a	627±40	1072±46	2701±150	1217±83	1003±67

HTO-1 = HTO (series 1): $5.7 \cdot 10^{-10}$ mol·dm $^{-3}$; $^{36}\text{Cl}^-$: $2.4 \cdot 10^{-5}$ mol·dm $^{-3}$; $^{125}\text{I}^-$: $3.3 \cdot 10^{-11}$ mol·dm $^{-3}$;

HTO-2 = HTO (series 2): $1.0 \cdot 10^{-9}$ mol·dm $^{-3}$; $^{22}\text{Na}^+$: $2.0 \cdot 10^{-10}$ mol·dm $^{-3}$

2.1.5 Activity measurements

The activities of the radiotracers were measured by liquid scintillation counting. 10 cm 3 of sample (V_{sample}) were placed in a 20 cm 3 polyethylene (PE) counting vial and 10 cm 3

scintillation cocktail were added (Ultima Gold XR, Canberra-Packard). The samples were vigorously shaken and placed in a liquid scintillation counter (Tri-carb 2250 CA, Canberra-Packard). The counting efficiency for each tracer was measured using synthetic pore water with known amounts of activities. Background measurements were performed in a similar way using artificial pore water without radiotracers. Table 4 gives the counting conditions (counting window: LL and UL), typical background values and counting efficiency for the three tracers used. In the case of $^{125}\text{I}^-$ and $^{22}\text{Na}^+$, the measured activities were corrected for decay.

Table 4: Counting conditions, typical background values, half-life and counting efficiency (liquid scintillation counting) for the tracers used in the diffusion experiments.

Tracer	$T_{1/2}$	Counting conditions			BG [cpm]	Efficiency [cps·Bq ⁻¹]
		LL	UL	Time (T)		
		[keV]	[keV]	[min]		
HTO	12.35 y	0.0	18.6	30	15	0.35
$^{36}\text{Cl}^-$	$3 \cdot 10^5$ y	0.0	710.0	30	39	1.00
$^{125}\text{I}^-$	60.1 d	0.0	70.0	30	22	0.57
$^{22}\text{Na}^+$	2.602 y	0.0	2000	30	36	0.95

$T_{1/2}$: half-life; LL: lower level ; UL: upper level, BG: background;

2.1.6 Data processing

After a time t_i , the low activity container, having a volume V_{low} , is replaced by the same volume of solution, but containing no radiotracer. A given volume (V_{sample}) of this sampled solution is used for the activity measurements as described in 2.1.5. (counting time is T) and gives a count rate, $N_T^{t_i}$ [cpm]. The background is measured under similar conditions resulting in a count rate $N_o^{t_i}$ [cpm]. From these two measurements, a net count rate, $N_s^{t_i}$ [cpm], is calculated as:

$$N_s^{t_i} = N_T^{t_i} - N_o^{t_i} \quad (2)$$

The total amount of tracer [cpm] diffused through the sample during a time interval $\Delta t_i = t_i - t_{i-1}$ is calculated by:

$$N_{dif}^{\Delta t_i} = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{low} + \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} - \frac{N_s^{t_{i-1}}}{V_{sample}} \cdot V_{dead} \quad (3)$$

where V_{dead} is the volume that is not replaced when changing the solution in the low concentration reservoir and represents the dead volume, i.e. the volume of the tubings, the grooves in the end plates and the filter pore space. The negative term in Eqn. 3 represents the correction for the activity left in this dead volume when replacing the solution in the container. At time t_o , the count rate, $N_s^{t_o}$, is zero. So for $i=1$, i.e. the first sampling, the total amount of tracer [cpm] diffused is:

$$N_{dif}^{\Delta t_i} = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{low} + \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} \quad (4)$$

The activity [Bq] diffused through the rock during a time Δt_i is now given by:

$$A_{dif}^{\Delta t_i} = \frac{N_{dif}^{\Delta t_i}}{60 \cdot f} \quad (5)$$

where f [cps·Bq⁻¹] is the counting efficiency. The average flux during Δt_i , $J_L^{t_i}$ [Bq·m⁻²·s⁻¹], is calculated by:

$$J_L^{t_i} = \frac{A_{dif}^{\Delta t_i}}{S \cdot \Delta t_i} \quad (6)$$

where S represents the cross section area [m²]. The accumulated activity diffused through the rock sample after diffusion time t_n is:

$$A_{dif}^{t_n} = \sum_{i=1}^n A_{dif}^{\Delta t_i} \quad (7)$$

2.1.7 Uncertainty estimation

The uncertainty on the values of the fluxes and on the accumulated diffused activity in the through-diffusion measurements results from the uncertainty on the parameters involved to calculate these values. The main sources are the uncertainty on the count rates ($N_T^{t_i}$, $N_o^{t_i}$) and the counting efficiency (f), the uncertainty on the dead volume (V_{dead}), and the uncertainty on pipetting (V_{sample}) and weighing procedures (V_{low}). Figure 7

represents a cause and effect diagram of the different uncertainty sources. The uncertainty on J^{t_i} and $A_{dif}^{t_i}$ is calculated by combining the uncertainties of the parameters involved (EURACHEM, 2000).

The uncertainty, u , on the net count rate is given by (BRANDT, 1976):

$$u(N_s^{t_i}) = \sqrt{\frac{N_T^{t_i}}{T} + \frac{N_o^{t_i}}{T}} \quad (8)$$

with T = counting time. The relative uncertainty ($r.u$) on the parameters V_{low} , V_{sample} , V_{dead} , f , S and Δt_i is given in Table 5. These values are in-house values based on experience and represent rather pessimistic estimates.

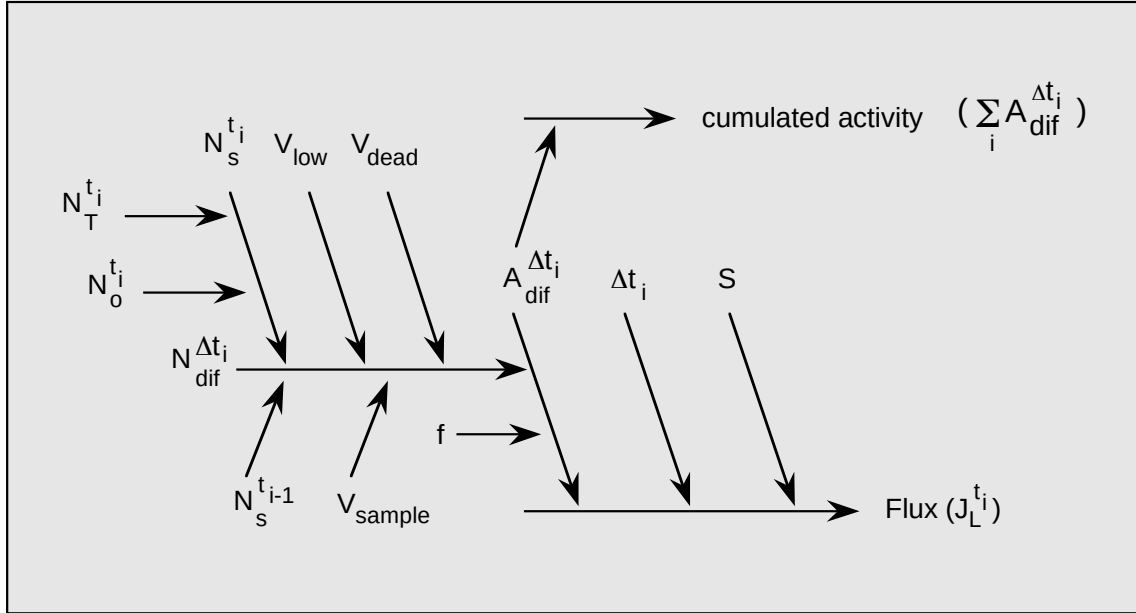


Figure 7: Cause and effect diagram for the uncertainty estimations.

The numerator of Eqn. 5 represents $N_{dif}^{\Delta t_i}$ of Eqn. 3 and has three terms. For each term, the uncertainty is calculated and combined to an uncertainty for the nominator ($N_{dif}^{\Delta t_i}$). The uncertainty of the different terms are:

$$u\left(\frac{N_s^{t_i}}{V_{sample}} \cdot V_{low}\right) = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{low} \cdot \sqrt{(r.u(N_s^{t_i}))^2 + (r.u(V_{low}))^2 + (r.u(V_{sample}))^2} \quad (9)$$

$$u\left(\frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead}\right) = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} \cdot \sqrt{(r.u(N_s^{t_i}))^2 + (r.u(V_{dead}))^2 + (r.u(V_{sample}))^2} \quad (10)$$

$$u\left(\frac{N_s^{t_{i-1}}}{V_{sample}} \cdot V_{dead}\right) = \frac{N_s^{t_{i-1}}}{V_{sample}} \cdot V_{dead} \cdot \sqrt{(r.u(N_s^{t_{i-1}}))^2 + (r.u(V_{dead}))^2 + (r.u(V_{sample}))^2} \quad (11)$$

and can be combined as

$$u(N_{dif}^{\Delta t_i}) = \sqrt{\left(u\left(\frac{N_s^{t_i}}{V_{sample}} \cdot V_{low}\right)\right)^2 + \left(u\left(\frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead}\right)\right)^2 + \left(u\left(\frac{N_s^{t_{i-1}}}{V_{sample}} \cdot V_{dead}\right)\right)^2} \quad (12)$$

Table 5: Relative uncertainty (r.u) on the parameters used in Eqn. 3, 5 and 6.

Parameter	r.u
V_{low}	0.01
V_{sample}	0.05
V_{dead}	0.1
f	0.03 to 0.06
S	0.01
Δt_i	0.02

The uncertainty on the flux $J_L^{t_i}$ is given according to Eqn. 5 and 6 by:

$$u(J_L^{t_i}) = J_L^{t_i} \cdot \sqrt{(r.u(N_{dif}^{\Delta t_i}))^2 + (r.u(f))^2 + (r.u(S))^2 + (r.u(\Delta t_i))^2} \quad (13)$$

The uncertainty on the diffused activity is (see Eqn. 5):

$$u(A_{dif}^{\Delta t_i}) = A_{dif}^{\Delta t_i} \cdot \sqrt{(u(N_{dif}^{\Delta t_i}))^2 + (u(f))^2} \quad (14)$$

and the uncertainty on the total diffused activity:

$$u(A_{dif}^{t_n}) = u\left(\sum_{i=1}^n A_{dif}^{\Delta t_i}\right) = \sqrt{(u(A_{dif}^{\Delta t_1}))^2 + \dots + (u(A_{dif}^{\Delta t_i}))^2 + \dots + (u(A_{dif}^{\Delta t_n}))^2} \quad (15)$$

The use of filters might have a potential impact on the calculated D_e and α , resulting in a systematic error. The impact will depend on the relative thicknesses of filter and sample, and on the diffusivity and porosity of sinter and clay. An estimation of the relative error introduced by the use of filters is given in 2.3.2.

2.2 Out-diffusion experiments

2.2.1 Experimental set-up

The out-diffusion experiments were performed using the same equipment as described for the through-diffusion (see chapter 2.1.1). As soon as steady state was achieved in a through-diffusion experiment, the solutions in both containers were replaced by synthetic pore water without tracer, and out-diffusion was started. At given time intervals, the activity in the solutions was measured (see 2.1.5) and the solutions were replaced by fresh ones. This procedure was repeated until all the activity was diffused out of the samples.

2.2.2 Data processing

After a time t_i , the solutions in both containers, having a volume V_{out}^x ($x=0$ and $x=L$), are replaced by the same volume of solution, but containing no radiotracer. 10 cm³ of this sampled solution are used for the activity measurements as described in 2.1.5 (counting time is T) and gives a count rate, $N_T^{t_i}$ [cpm]. The background is measured under similar conditions resulting in a count rate $N_0^{t_i}$ [cpm]. From this two measurements, a net count rate, $N_s^{t_i}$ [cpm], is calculated as:

$$N_s^{t_i} = N_T^{t_i} - N_0^{t_i} \quad (16)$$

The total amount of tracer diffused out of the sample through the boundary at location x during time interval $\Delta t_i = t_i - t_{i-1}$ is given by:

$$N_{x,dif}^{\Delta t_i} = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{out}^x + \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} - \frac{N_s^{t_{i-1}}}{V_{sample}} \cdot V_{dead} \quad (17)$$

V_{dead} is the rest volume that is not replaced when changing the solution in the reservoirs and represents the volume of the tubings, the grooves in the end plates and the filter pore space. The negative term in Eqn. 17 represents the correction for the activity left in this dead volume when replacing the solution in the container. At time t_0 , the starting time of the out-diffusion, the solution in the dead volume at boundary $x=0$ is the same solution as

in the high-concentration container of the through-diffusion. The activity of this solution is C_o [cpm·cm⁻³]. So for $i=1$, i.e. the first sampling, the activity [cpm] diffused through the boundary $x=0$ is given by:

$$N_{0,dif}^{\Delta t_i} = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{out}^0 + \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} - C_o \cdot V_{dead} \quad (18)$$

The solution in the dead volume at boundary $x=L$ has a composition equal to the solution at the end of the through-diffusion. The activity of this solution is C_n (cpm·cm⁻³). So for $i=1$, i.e. the first sampling, the amount [cpm] diffused out through the boundary at location $x=L$ is:

$$N_{L,dif}^{\Delta t_i} = \frac{N_s^{t_i}}{V_{sample}} \cdot V_{out}^L + \frac{N_s^{t_i}}{V_{sample}} \cdot V_{dead} - C_n \cdot V_{dead} \quad (19)$$

The activity diffused out of the rock through boundary x during a time Δt_i is given by:

$$A_{x,dif}^{\Delta t_i} = \frac{N_{x,dif}^{\Delta t_i}}{60 \cdot f} \quad (20)$$

where f (cps·Bq⁻¹) is the counting efficiency. The flux at time t_i at boundary x , $J_x^{t_i}$, is calculated by:

$$J_x^{t_i} = \frac{A_{x,dif}^{\Delta t_i}}{S \cdot \Delta t_i} \quad (21)$$

where S is the cross section area (m²).

2.2.3 Uncertainty estimation

The uncertainty on the values of the flux $J_x^{t_i}$ [Bq·m²·s⁻¹] was calculated in a similar way as described in chapter 2.1.7:

$$u(J_x^{t_i}) = J_x^{t_i} \cdot \sqrt{(r \cdot u(N_{x,dif}^{\Delta t_i}))^2 + (r \cdot u(f))^2 + (r \cdot u(S))^2 + (r \cdot u(\Delta t_i))^2} \quad (22)$$

2.3 Results and discussion of the through-diffusion experiments

2.3.1 Theoretical background of one-dimensional (planar) through-diffusion

For a one-dimensional diffusion process through a plane with thickness L the flux, J , is given by Fick's first law:

$$J = -D_e \cdot \frac{\partial C}{\partial x} \quad (23)$$

If the concentration within the system is changing, Fick's second law applies:

$$\frac{\partial C}{\partial t} = D_a \cdot \frac{\partial^2 C}{\partial x^2} \quad (24)$$

where

$$D_a = \frac{D_e}{\alpha} \quad (25)$$

$[\text{m}^2 \cdot \text{s}^{-1}]$ is the apparent diffusion coefficient. The rock capacity factor, α , is defined as:

$$\alpha = \varepsilon + \rho \cdot K_d \quad (26)$$

where K_d = the equilibrium distribution coefficient ($\text{m}^3 \cdot \text{kg}^{-1}$)

ρ = the dry bulk density of the rock ($\text{kg} \cdot \text{m}^{-3}$)

For non-sorbing tracers ($K_d = 0$), the rock capacity factor equals the transport porosity. For $K_d > 0$, the diffusion accessible porosity can be deduced from the rock capacity factor by:

$$\varepsilon = \alpha - \rho \cdot K_d \quad (27)$$

This, however, requires knowledge of the K_d value. The main problem hereby is the fact that K_d values are in general measured on powdered samples suspended in a relatively large volume of solution. Whether it is justified to use these values for intact samples is questionable. So far, the uncertainty on porosity values derived from diffusion studies with sorbing tracers is relatively large.

D_e represents the effective diffusion coefficient and is related to the diffusion coefficient in free water D_w [$\text{m}^2 \cdot \text{s}^{-1}$] by:

$$D_e = \frac{\varepsilon \cdot \delta}{\tau^2} \cdot D_w \quad (28)$$

ε is the porosity of the solid phase in which the diffusion process takes place, δ [-] is the constrictivity and accounts for the fact that the pore diameter varies along the pathway, and τ [-] is the tortuosity of the sample and takes account for path lengthening. The three parameters are mostly lumped together in a geometric factor G [-]:

$$G = \frac{\varepsilon \cdot \delta}{\tau^2} \quad (29)$$

Figure 8 is a schematic representation of one-dimensional diffusion through a homogeneous and isotropic porous medium (rock) with thickness L . The initial condition for the system is:

$$C(x,t) = 0 ; x \in [0,L], t = 0$$

meaning that the Opalinus Clay is originally free of tracer.

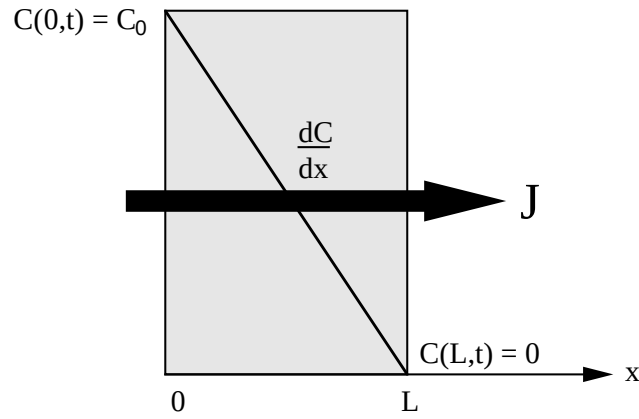


Figure 8: One-dimensional diffusion through a homogeneous and isotropic porous medium with thickness L . The solid line represents the concentration profile of the tracer in the pore water at steady state.

The boundary conditions for the system in our experimental set-up are:

$$C(0,t) = C_o = \text{constant} ; t > 0$$

$$C(L,t) = 0 ; t > 0$$

where $C(0,t) = C_o$ [Bq·m⁻³] is the concentration of tracer at the high-concentration side and $C(L,t)$ [Bq·m⁻³] is the concentration of tracer at the low-concentration side (measurement cell). At steady-state the concentration gradient in the sample along the x-direction is assumed to be linear (Figure 8). The analytical solution to the through-diffusion problem is obtained by solving Eqn. 24 with the initial and boundary conditions given above and making use of the fact that the flux of the tracer at the low-concentration boundary at x=L according to Eqn. 23 is:

$$J(L,t) = -D_e \cdot \left. \frac{\partial C}{\partial x} \right|_{x=L} \quad (30)$$

The analytical solution is (CRANK, 1975; JAKOB et al., 1999):

$$A_{dif}^t = S \cdot L \cdot C_o \cdot \left(\frac{D_e \cdot t}{L^2} - \frac{\alpha}{6} - \frac{2 \cdot \alpha}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cdot \exp \left\{ -\frac{D_e \cdot n^2 \cdot \pi^2 \cdot t}{L^2 \cdot \alpha} \right\} \right) \quad (31)$$

Eqn. 31 gives the cumulative mass or activity of a tracer in the low-concentration reservoir as a function of time. One of the boundary conditions used to solve the differential equation specifies that the concentration of the tracer at the low-concentration boundary of the cell must be equal to zero. However, as long as $C(L,t) \ll C_o$, the analytical solution approximates well Eqn. 31. As time increases, the exponential term reduces towards zero and Eqn. 31 approximates:

$$A_{dif}^t = S \cdot L \cdot C_o \cdot \left(\frac{D_e \cdot t}{L^2} - \frac{\alpha}{6} \right) \quad (32)$$

Figure 9 shows data of a hypothetical through-diffusion experiment. Both the flux and the total accumulated mass are presented. Two stages can be distinguished. In a first stage the flux increases. As long as the flux increases, the diffusion process is in a transient phase. As soon as the flux reaches a constant value, the process is in steady state. In this stage, the total accumulated mass increases linearly with time. From the data acquired in the steady state phase, a diffusion coefficient (D_e) and a transport relevant porosity value (ϵ) can be derived. When the diffusion process is in the steady state phase, the total diffused mass thus becomes a linear function of time:

$$A_{dif}^t = a \cdot t + b \quad (33)$$

where

$$a = \frac{S \cdot C_o \cdot D_e}{L} \quad (34)$$

and

$$-b = \frac{S \cdot L \cdot C_o \cdot \alpha}{6} \quad (35)$$

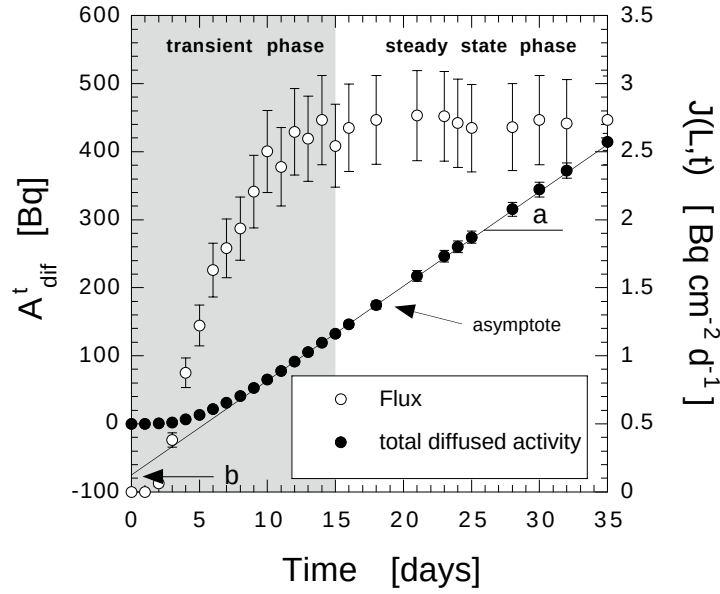


Figure 9: Evolution of flux and total diffused mass as a function of time in a through-diffusion experiment.

The straight line in Figure 9 shows this linear equation. From the slope of the line and the intercept with the y-axis, both D_e and α can be calculated:

$$D_e = \frac{a \cdot L}{S \cdot C_o} \quad (36)$$

and

$$\alpha = -\frac{6 \cdot b}{S \cdot L \cdot C_o} \quad (37)$$

The uncertainty on D_e and ε were calculated by combining the relative errors on the individual parameters a , b , L , S and C_o :

$$u(D_e) = D_e \cdot \sqrt{(r \cdot u(a))^2 + (r \cdot u(L))^2 + (r \cdot u(S))^2 + (r \cdot u(C_o))^2} \quad (38)$$

$$u(\alpha) = \alpha \cdot \sqrt{(r \cdot u(b))^2 + (r \cdot u(L))^2 + (r \cdot u(S))^2 + (r \cdot u(C_o))^2} \quad (39)$$

2.3.2 Results for HTO (series 1)

Figures A1 in the Appendix 1 show the results of the diffusion measurements for HTO in OPA from Mont Terri and Benken at different confining pressures. Both data on the flux [$\text{Bq} \cdot \text{cm}^{-2} \cdot \text{d}^{-1}$] and the accumulated activity diffused through the rock samples [Bq] are shown. Diffusion through Mont Terri samples is in steady state after 5 days. Diffusion through Benken samples, however, reaches steady state after 12 days. This is an indication that the diffusion in OPA from Benken is slower than in OPA from Mont Terri. The values of the flux decrease with increasing confining pressure, indicating that the confining pressure has an effect on the diffusion process.

From the steady state part of the accumulative plot, an effective diffusion coefficient D_e , and a rock capacity factor, α , were calculated using Eqn. 36 and 37. For Mont Terri, data of the accumulated diffused activity between 5 - 39 days and in the case of Benken, data between 12 - 39 days were used for linear regression analysis. The evolution of the activity concentration ($\text{Bq} \cdot \text{cm}^{-3}$) of HTO in the high-concentration container is also shown in Figures A1 in Appendix 1. As can be seen, the activity is more or less constant during the course of the experiment. The activity of HTO in the low-concentration compartment never exceeded 1% of the value in the high-concentration compartment. The assumed zero-concentration boundary condition for modeling purposes was therefore appropriate for the whole time period of the experiments.

The results are summarised in Table 6. Because HTO is considered as a non-sorbing tracer, i.e. $K_d=0$, the rock capacity factor, α , equals the diffusion accessible porosity, ε . The quality of the parameters (D_e and α) was tested by using them as input parameters for the full analytical solution (Eqn. 32) and calculating the whole diffusion curve. As can be seen from Figures A1 in Appendix 1, it was possible to reproduce the data (fluxes) in both the initial transient and the final steady state stages of the experiments. The values of the effective diffusion coefficient of HTO for Mont Terri samples are in good agreement with the values measured by SCK, AEA and CIEMAT (VAN LOON et al., 2001) on similar samples. These values range between $(1.0 - 2.3) \cdot 10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$. The values for the diffusion accessible porosity, ε , show a larger variety and range between 0.12 and 0.26. The values determined in this work lie in the lower part of this range. The large variety in the values indicates that the applied method for determining porosity is not very accurate.

Small disturbances in the transient phase can cause large effects on the intercept with the Y-axis from which α is calculated.

Table 6: Effective diffusion coefficients and rock capacity factors for the diffusion of HTO (series 1) through Opalinus Clay.

	C_0 [Bq·m ⁻³]	D_e [m ² ·s ⁻¹]	α [-]
BE-2a	(610±40)×10 ⁶	(6.71±0.41)×10 ⁻¹²	0.15±0.01
BE-3a	(621±40)×10 ⁶	(5.92±0.36)×10 ⁻¹²	0.14±0.01
BE-4a	(547±35)×10 ⁶	(5.60±0.34)×10 ⁻¹²	0.13±0.01
MT-1a	(624±40)×10 ⁶	(1.49±0.09)×10 ⁻¹¹	0.10±0.01
MT-3a	(627±40)×10 ⁶	(1.18±0.07)×10 ⁻¹¹	0.10±0.01

The results measured at two different confining pressures further show that there is only a small effect of pressure on diffusion. Increasing the pressure from 1 to 5 MPa results in a decrease of D_e of 20 %. The porosity values are hardly affected at all.

The diffusion coefficients measured for the Benken samples are lower by a factor of 2 - 3. The porosity values are slightly higher than those observed for Mont Terri. This is surprising because the porosity determined from the loss of weight after drying the samples results in lower values (NAGRA, 2001). In view of the large uncertainty inherent to the method applied here, however, the porosity values can be assumed to be more or less similar to those of Mont Terri. The smaller values of the diffusion coefficients might be explained by a smaller mean pore diameter in Benken OPA (2 - 10 nm; GABORIAU & SERON, 2000) compared to Mont Terri OPA (10 - 50 nm; MAIS, 1998). This could be due to the larger overburden in the past in Benken (ca. 1500 m) than in Mont Terri (estimated overburden ca. 1000 m; NAGRA, 2002). As could be observed for Mont Terri OPA, increasing the confining pressure results in a small decrease of D_e . Increasing the pressure from 4 to 15 MPa results in a decrease of D_e of 16 %. Again, porosity values were hardly affected at all. This indicates that the structure of the clay samples was not affected by the pressure release induced by the sampling campaign. Moreover, the fact that an external pressure has hardly an effect on the diffusion process is an indication that the Opalinus Clay is overconsolidated². This is in contrast with

² Rocks and soils with a level of compaction corresponding to their present burial depth are said to be normally consolidated. Where erosion has occurred, rocks may be compacted much more than expected for their current depth of burial. These rocks and soils are said to be overconsolidated.

earlier work on the plastic Boom Clay (HENRION et al., 1991) where confining pressure has a significant effect on diffusion and diffusion accessible porosity.

The formation factor, F , is defined as:

$$F = \frac{\tau^2}{\varepsilon \cdot \delta} \quad (40)$$

and can be calculated:

$$F = \frac{D_w}{D_e} \quad (41)$$

where ε is the diffusion accessible porosity, δ is the constrictivity and τ the tortuosity of the sample. The constrictivity represents the reduction in diffusivity due to pore narrowing, whereas tortuosity represents the reduction in diffusivity owing to path lengthening (See Figure 10).

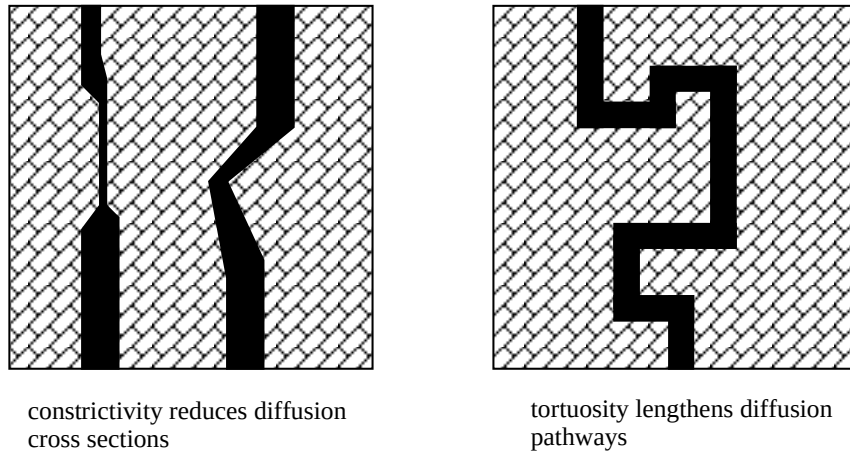


Figure 10: Schematic representation of constrictivity and tortuosity (FRICK, 1993).

Both parameters are dimensionless. With D_w equal to $2.2 \cdot 10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$ at 25 °C (MILLS, 1973), the formation factor F ranges between 317 - 380 for Benken OPA and 143 - 181 for Mont Terri OPA. The larger value of the formation factor for Benken, if porosity (ε) is about the same in Mont Terri and Benken, may be due to a larger tortuosity and/or a smaller constrictivity³.

³ It has to be noted that it is assumed that the pore water in OPA has similar properties as bulk liquid water and that the value of D_w is therefore applicable for pore water. As will be discussed later (chapter 4), there is some evidence that the properties of confined water in OPA are different from bulk liquid water.

The effective diffusion coefficient of the tracers in the filters is $1.2 \times 10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$ (BOURKE et al., 1990). The transport times in the filter and the rock were calculated using $\langle d^2 \rangle = 4 \cdot D_a \cdot t$, where d is the average diffusion length [m], D_a is the apparent diffusion coefficient [$\text{m}^2 \cdot \text{s}^{-1}$] and t is time [s]. The diffusion times of HTO calculated for the filter and OPA (Mont Terri) are 3.2 hr and 76.9 hr, respectively. Because HTO has the largest D_e value, this case represents the fastest one. The diffusion time through the filter is only 4% of that through the rock. The relative error on the porosity and D_e caused by the use of a filter is at maximum 4%. For $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$, the diffusion times in the rock are longer so that the relative error on the rock capacity factor and effective diffusion coefficient caused by the use of the filter is $< 4 \%$.

2.3.3 Results for $^{36}\text{Cl}^-$

The results of the diffusion of $^{36}\text{Cl}^-$ are given in Figures A2 in Appendix 2. Similar observations can be made as for HTO, i.e. the confining pressure has only a small effect on the tracer diffusion and - in general - diffusion in Benken OPA is slower than in Mont Terri OPA. The flux of $^{36}\text{Cl}^-$ through OPA was constant after 10 days in the case of Mont Terri OPA. For Benken OPA, a constant flux could be observed after 25 days.

Table 7: Effective diffusion coefficient and rock capacity factor for $^{36}\text{Cl}^-$.

	C_0 [Bq m^{-3}]	D_e [$\text{m}^2 \text{s}^{-1}$]	α [-]
BE-2a	$(1066 \pm 46) \times 10^6$	$(9.11 \pm 0.55) \times 10^{-13}$	0.044 ± 0.003
BE-3a	$(1064 \pm 45) \times 10^6$	$(7.92 \pm 0.35) \times 10^{-13}$	0.039 ± 0.002
BE-4a	$(1064 \pm 45) \times 10^6$	$(7.12 \pm 0.43) \times 10^{-13}$	0.039 ± 0.002
MT-1a	$(1073 \pm 46) \times 10^6$	$(5.70 \pm 0.34) \times 10^{-12}$	0.073 ± 0.004
MT-3a	$(1072 \pm 46) \times 10^6$	$(4.11 \pm 0.25) \times 10^{-12}$	0.072 ± 0.004

Data of the accumulated diffused activity between 25 - 84 days were used for linear regression analysis. The values of the effective diffusion coefficient, D_e , and rock capacity factor, α , are summarised in Table 7. Because $^{36}\text{Cl}^-$ is assumed not to sorb on OPA, i.e. $K_d=0$, the rock capacity factor equals the diffusion accessible porosity, ϵ . The quality of the parameters (D_e and α) was tested by using them as input parameters for the full analytical solution (Eqn. 31) and calculating the whole diffusion curve. As can be

seen from Figures A2 in Appendix 2, it was possible to reproduce the flux data in both the initial transient and the final steady state stages of the experiments. The values of D_e and α of $^{36}\text{Cl}^-$ are smaller than the values of HTO. This can be explained by the anion exclusion effect (BOLT & DE HAAN, 1979) (Figure 11). The exclusion effect, however, is smaller in Mont Terri samples than in Benken samples. This can clearly be seen from the values of the diffusion coefficients and the diffusion accessible porosities. For Benken samples, the D_e for $^{36}\text{Cl}^-$ is 7.5 times smaller than D_e for HTO. In Mont Terri samples, the difference is only a factor of 2.7. This, in turn, can be explained by the difference in pore water chemistry used in the experiments (see Table 1). The ionic strength of the Mont Terri pore water is twice that of the Benken pore water. Hence the permanent charges on the clay surface are more shielded (thinner diffuse double layers) in Mont Terri OPA than in Benken OPA, resulting in a smaller anion exclusion effect in Mont Terri OPA.

The formation factor calculated from the results in Table 7, using $D_w(^{36}\text{Cl}^-) = 2 \cdot 10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$ (LI & GREGORY, 1974), ranges between 2195 - 2809 for OPA from Benken and between 351 - 487 for OPA from Mont Terri. This larger value for Benken is probably caused by i) the smaller pore diameter and ii) the lower ionic strength of the pore water resulting in a larger anion exclusion effect. A more direct evidence of the anion exclusion can be seen from the diffusion accessible porosity ε .

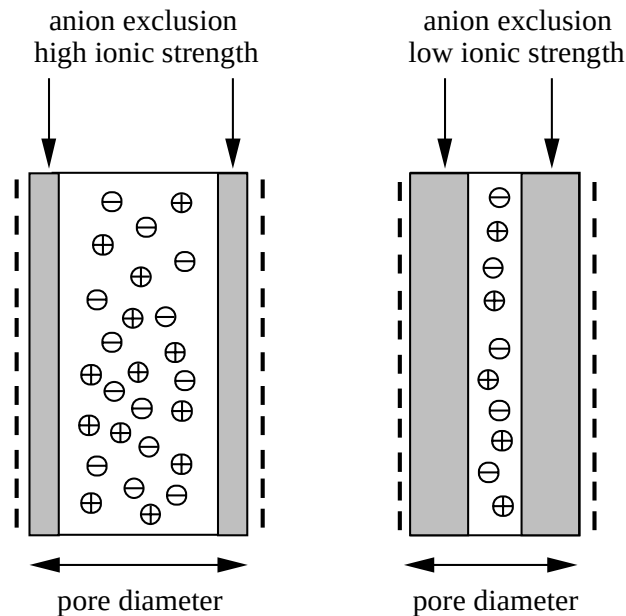


Figure 11: Effect of anion exclusion on the effective pore diameter (and pore volume).

According to OCHS et al. (1998), the constrictivity is the product of two terms:

$$\delta = \delta_g \cdot \delta_{el} \quad (42)$$

with δ = total constrictivity [-]

δ_g = the geometrical constrictivity [-]

δ_{el} = a term considering electrostatic interaction between clay surface and ions [-]

Anions will be repulsed from the surface and $\delta_{el} < 1$. So far, Eqn. 42 gives a qualitative explanation of the observed effect (Figure 11). A quantitative description of the anion exclusion effect is not possible.

Similar to HTO, the confining pressure has only a small effect on the diffusion.

2.3.4 Results for $^{125}\text{I}^-$

The results of $^{125}\text{I}^-$ are given in Figures A3 in Appendix 3. In the case of Mont Terri steady state is reached after 20 days. For Benken, it takes almost 90 days to reach steady state. The increase in flux after 50 days in the case of Mont Terri is caused by a sudden increase in the average room temperature from 20 to 25 °C. The data between 20 and 40 days were used for the calculations. In the case of Benken OPA, such an increase could not be observed because the diffusion was still in the transient stage. The data between 93 and 165 days were used to calculate the diffusion parameters. The best-fit diffusion parameters are summarized in Table 8. From the evolution of the flux with time it can thus be seen that it takes much more time for $^{125}\text{I}^-$ to reach steady state. It can further be observed that iodide ($^{125}\text{I}^-$) is retarded compared to chloride ($^{36}\text{Cl}^-$). This can be deduced from the larger values of the rock capacity factor α . Because I^- and Cl^- have similar ionic potentials⁴, the diffusion accessible porosity for $^{36}\text{Cl}^-$ and $^{125}\text{I}^-$ can be assumed to be similar. The larger rock capacity factor must be explained by the sorption of $^{125}\text{I}^-$ on Opalinus Clay. LAUBER (2001) studied the sorption of $^{125}\text{I}^-$ on crushed Opalinus Clay from Mont Terri in artificial pore water. He found a linear sorption behaviour of $^{125}\text{I}^-$ in a concentration range of $10^{-5} \text{ M} < [\text{I}^-] < 10^{-3} \text{ M}$. The K_d value for I-sorption on crushed Opalinus Clay from Mont Terri was $(4 \pm 2) \times 10^{-5} \text{ m}^3 \cdot \text{kg}^{-1}$. The mechanism(s) of I-sorption, however, is (are) still unknown, but there are some indications in the literature that I^- sorbs on illite (KAPLAN et al., 2000). Illite is a key clay mineral present in OPA. Similar observations were also made by FUHRMANN et al. (1998). Further, microbial oxidation of I(-1) to I(+1), followed by sorption of I(+1) on the clay is also possible (LIU & VON GUNTEN, 1988 and references therein). Assuming that the diffusion-accessible

⁴ ionic potential = $|z|/r$ where z is the charge of the ion and r is the crystal ionic radius; ionic potential = 0.55 for Cl^- and 0.45 for I^- , (LI & GREGORY, 1974; CRC Handbook of Physics and Chemistry, 1979).

porosities for Γ^- and Cl^- are similar, the K_d value for $^{125}\text{I}^-$ calculated from the rock capacity factor is $1 \times 10^{-5} \text{ m}^3 \cdot \text{kg}^{-1}$ for Opalinus Clay from Mont Terri and $2 \times 10^{-5} \text{ m}^3 \cdot \text{kg}^{-1}$ for Benken. The value for Benken is somewhat larger than the value for Mont Terri. This might be explained by the smaller concentration of Cl^- in the artificial pore water of Benken. KAPLAN et al. (2000) showed that the higher the Cl^- concentration in solution, the lower the sorption of Γ^- on illite.

Table 8: Effective diffusion coefficient and rock capacity factor for $^{125}\text{I}^-$.

	C_o [Bq m ⁻³]	D_e [m ² s ⁻¹]	α [-]
BE-2a	(2642±150)×10 ⁶	(6.57±0.40)×10 ⁻¹³	0.083±0.006
BE-3a	(2748±150)×10 ⁶	(5.02±0.32)×10 ⁻¹³	0.063±0.006
BE-4a	(2680±150)×10 ⁶	(4.47±0.28)×10 ⁻¹³	0.076±0.006
MT-1a	(2731±150)×10 ⁶	(4.76±0.34)×10 ⁻¹²	0.105±0.022
MT-3a	(2701±150)×10 ⁶	(3.37±0.24)×10 ⁻¹²	0.077±0.016

2.3.5 Results for HTO (series 2)

After the diffusion of $^{125}\text{I}^-$, diffusion of HTO was repeated in order to check whether the samples had significantly changed after 1.5 years. The results of these measurements are given in Figures A4 in Appendix 4. A summary of the effective diffusion coefficients and rock capacity factors are given in Table 9 below.

Table 9: Effective diffusion coefficients and rock capacity factors for the diffusion of HTO (series 2) through Opalinus Clay.

	C_o [Bq·m ⁻³]	D_e [m ² ·s ⁻¹]	α [-]
BE-2a	(1207±82)×10 ⁶	(7.41±0.47)×10 ⁻¹²	0.16±0.02
BE-3a	(1205±82)×10 ⁶	(6.37±0.40)×10 ⁻¹²	0.15±0.02
BE-4a	(1197±81)×10 ⁶	(6.15±0.39)×10 ⁻¹²	0.16±0.02
MT-1a	(1213±83)×10 ⁶	(1.53±0.10)×10 ⁻¹¹	0.11±0.02
MT-3a	(1217±83)×10 ⁶	(1.21±0.08)×10 ⁻¹¹	0.09±0.02

In the case of Benken OPA, slightly larger values for both the effective diffusion coefficients and rock capacity factors were measured. However, these differences are not significant. For Mont Terri, the measured values for the effective diffusion coefficient and the rock capacity factor are identical with the previous measurements (see Table 6). These observations clearly show that after 1.5 years the samples had not changed significantly with respect to their transport (diffusion) properties for HTO and probably also for other radionuclides.

2.3.6 Results for $^{22}\text{Na}^+$

The results for $^{22}\text{Na}^+$ are given in Figures A5 in Appendix 5. Steady state is reached after roughly 15 days and 30 days for Mont Terri and Benken, respectively. The data between 16 and 70 days were used for the analysis in the case of Mont Terri and between 32 and 70 days in the case of Benken. An overview of the best-fit values for D_e and α are given in Table 10. Once again the confining pressure has a negligible effect on the diffusion. Unlike the other radionuclides, there is almost no difference between the effective diffusion coefficient for OPA from Benken and Mont Terri. Moreover, the effective diffusion coefficient is similar to that of HTO for OPA from Mont Terri and is twice the value of HTO for OPA from Benken. Why the values of D_e for $^{22}\text{Na}^+$ are larger than those of HTO in the case of Benken OPA cannot be explained yet. Similar observations were made in diffusion studies on Callovo-Oxfordian claystones from the Meuse/Haute Sarne site in France (ANDRA, 1999).

Table 10: Effective diffusion coefficient (D_e), rock capacity factor (α) and sorption values (K_d) for $^{22}\text{Na}^+$.

	C_0 [Bq m ⁻³]	D_e [m ² s ⁻¹]	α	K_d [m ³ kg ⁻¹]
BE-2a	(995±66)×10 ⁶	(1.49±0.09)×10 ⁻¹¹	0.87±0.07	2.9×10 ⁻⁴
BE-3a	(993±65)×10 ⁶	(1.25±0.08)×10 ⁻¹¹	0.83±0.06	2.8×10 ⁻⁴
BE-4a	(994±66)×10 ⁶	(1.23±0.08)×10 ⁻¹¹	0.82±0.06	2.7×10 ⁻⁴
MT-1a	(993±65)×10 ⁶	(1.64±0.10)×10 ⁻¹¹	0.33±0.03	0.9×10 ⁻⁴
MT-3a	(1000±67)×10 ⁶	(1.36±0.08)×10 ⁻¹¹	0.34±0.02	1.0×10 ⁻⁴

The quality of the parameters (D_e and α) was tested by using them as input parameters for the full analytical solution (Eqn. 31) and calculating the whole diffusion curve. As can be

seen from Figures A5 in Appendix 5, it was possible to reproduce the data (fluxes) in both the initial transient and the final steady state stages of the experiments.

$^{22}\text{Na}^+$ is more retarded than HTO, $^{36}\text{Cl}^-$ and $^{125}\text{I}^-$, what can be explained by sorption. This effect is reflected in the larger rock capacity factors measured for $^{22}\text{Na}^+$. Assuming that the diffusion accessible porosity for $^{22}\text{Na}^+$ is similar to that for HTO ($\varepsilon = 0.14 \pm 0.01$ and 0.10 ± 0.01 for Benken and Mont Terri respectively), the sorption values of $^{22}\text{Na}^+$ can be calculated using equation (26). Using a value $\rho = 2.5 \text{ g cm}^{-3}$ for the dry bulk density, the K_d values for $^{22}\text{Na}^+$ are $(2.8 \pm 0.1) \times 10^{-4}$ and $(0.9 \pm 0.1) \times 10^{-4} \text{ m}^3 \text{ kg}^{-1}$ in the case of Benken and Mont Terri OPA, respectively. The lower K_d -value for Mont Terri can be explained by the larger concentration of Na^+ in the pore water and a slightly lower loading of Na^+ on the surface.

2.3.7 Diffusion of HTO in other argillaceous formations: a comparison

Figure 12 shows effective diffusion coefficients of HTO measured in several sedimentary argillaceous rocks of marine origin, i.e. Valanginian Marl from Wellenberg (LINEHAM & STONE, 1991), London Clay (BOURKE et al., 1990, 1993), Boom Clay (HENRION et al., 1991, PUT & DE CANNIERE, 1994), Tournemire (MELKIOR, 2000, MELKIOR et al., 2000), Opalinus Clay from Mont Terri (this work). In general, the effective diffusion coefficient decreases with decreasing diffusion accessible porosity, ε . In order to predict the diffusivity in porous media many equations, based on pore space models or empirical correlations, have been proposed (VAN BRAKEL & HEERTJES, 1974); the general tendency is to give the diffusivity as a unique function of porosity. For diffusion in porous rocks the diffusivity, defined as the ratio between the effective diffusion coefficient, D_e , measured for the rock and the diffusion coefficient in bulk water, D_w , is often defined as an empirical function of the porosity, ε (BOVING & GRATHWOHL, 2001):

$$\frac{D_e}{D_w} = \varepsilon^m \quad (43)$$

where m is an empirical parameter. Eqn. 43 is analogous to Archie's law for relating the electrical conductivity of a porous medium, R_{rock} , to the fluid resistivity, R_{fluid} , and porosity, ε (ARCHIE, 1942):

$$R_{rock} = R_{fluid} \cdot \varepsilon^{-m} \quad (44)$$

The parameter m , also called the Archie cementation factor, depends on the pore geometry and varies significantly, depending on the type of medium (SALEM, 2001). The solid curve in Figure 12 was calculated using Eqn. 43 with $D_w = 2.2 \cdot 10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$ at 25 °C (MILLS, 1973) and $m=2.5$. The shaded area represents calculations with $m=1.5$ and $m=3$.

As can be seen, the effective diffusion coefficients measured for different argillaceous rocks can be satisfactorily described using Archie's law with an exponent of ~ 2.5 . For performance assessment purposes the apparent diffusion coefficient is of interest:

$$D_a = \frac{D_e}{\alpha} \quad (45)$$

For non-sorbing tracers, $\alpha = \varepsilon$ and D_a is equal to D_p , the pore diffusion coefficient. Combining Eqn. 43 and 45 gives:

$$D_a = D_w \cdot \varepsilon^{(m-1)} \quad (46)$$

This equation, however, is only valid for neutral and non-reactive species that show no interaction (repulsion, attraction) with the mineral surfaces. The effective and apparent diffusion coefficients of neutral species can thus roughly be estimated for argillaceous rocks using Archie's law when the total porosity of the rock is known. The value for D_w is readily available from the literature (e.g. MILLS, 1973).

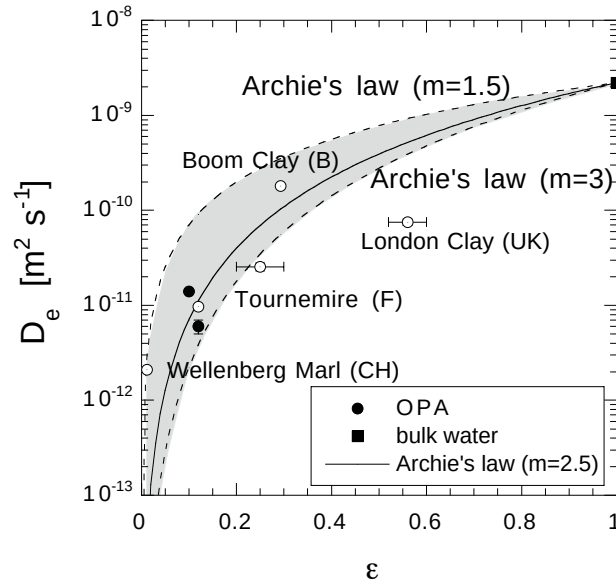


Figure 12: Relationship between the effective diffusion coefficient, D_e , of HTO for a series of argillaceous rocks and the corresponding diffusion accessible porosity, ε .

2.4 Results and discussion of the out-diffusion experiments

2.4.1 Theoretical background of out-diffusion

The out-diffusion experiments, as was the case for the through-diffusion, can be described by a one-dimensional single porosity medium model. The fluxes at the interfaces at $x = 0$ and $x = L$ are described by Fick's first law:

$$J = -D_e \cdot \frac{\partial C}{\partial x} \quad (47)$$

If the concentration gradient across the sample is linear, the concentration profile is:

$$C(x) = C_0 \cdot \left(1 - \frac{x}{L}\right) \quad (48)$$

The magnitude of diffusive flux caused by that concentration gradient would be equal to:

$$J_L = D_e \cdot \frac{C_0}{L} \quad (49)$$

To model the out-diffusion with an initial tracer distribution according to equation (48), the initial tracer concentration, C_0 , of the high-concentration side has to be replaced by a concentration $C_0 = 0$.

Again Eqn. 24 has to be solved with the corresponding boundary conditions:

$$C(0, t) = C(L, t) = 0 ; t > 0 \quad (50)$$

The initial condition is given by Eqn. 48. The solution is described in detail by JAKOB et al. (1999). In the following, only key equations are summarized. The tracer concentration in the pore water within the clay sample is given by:

$$C(x, t) = 2 \cdot C_0 \cdot \sum_{n=1}^{\infty} \frac{1}{n \cdot \pi} \cdot e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e}{\alpha} \cdot t} \cdot \sin\left(n \cdot \pi \cdot \frac{x}{L}\right) \quad (51)$$

Figure 13 shows a calculated concentration profile according to Eqn. 51 of HTO in the pore water of the BE-2a sample for different times during an out-diffusion experiment (parameters used: $D_e = 6.7 \cdot 10^{-12} \text{ m}^2 \cdot \text{s}^{-1}$; $\alpha = 0.15$).

The spatial derivative of Eqn. 51 is proportional to the flux:

$$\frac{\partial C(x,t)}{\partial x} = 2 \cdot C_0 \cdot \frac{1}{L} \cdot \sum_{n=1}^{\infty} e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e \cdot t}{\alpha}} \cdot \cos\left(n \cdot \pi \cdot \frac{x}{L}\right) \quad (52)$$

Hence, the absolute values of the diffusive fluxes at both boundaries are:

$$\text{at } x=0: \quad J(0,t) = 2 \cdot C_0 \cdot \frac{D_e}{L} \cdot \sum_{n=1}^{\infty} e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e \cdot t}{\alpha}} = 2 \cdot J_L \cdot \sum_{n=1}^{\infty} e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e \cdot t}{\alpha}} \quad (53)$$

and

$$\text{at } x=L: \quad J(L,t) = 2 \cdot C_0 \cdot \frac{D_e}{L} \cdot \sum_{n=1}^{\infty} (-1)^n \cdot e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e \cdot t}{\alpha}} = 2 \cdot J_L \cdot \sum_{n=1}^{\infty} (-1)^n \cdot e^{-\left(\frac{n \cdot \pi}{L}\right)^2 \cdot \frac{D_e \cdot t}{\alpha}} \quad (54)$$

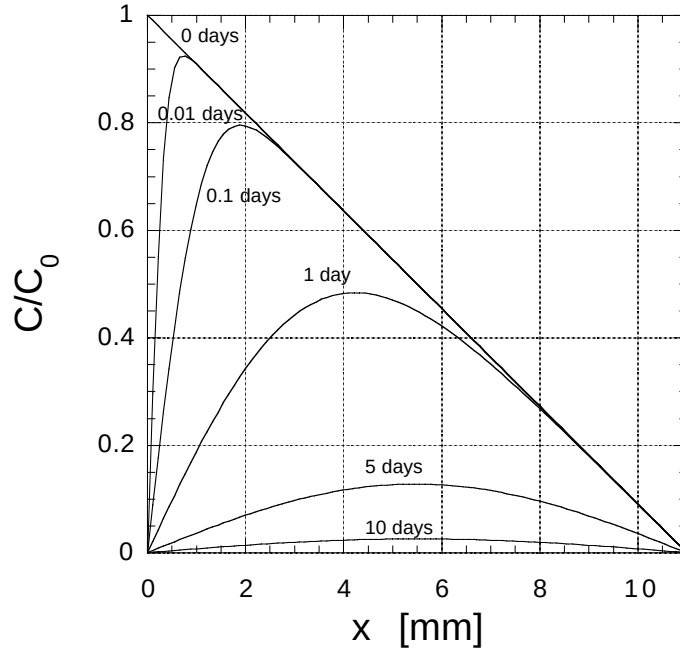


Figure 13: Concentration profile of HTO in the pore water of the OPA as a function of time during an out-diffusion experiment.

At the beginning of the out-diffusion, at $t \rightarrow 0$, the flux at $x=0$ is infinitely high, and at $x=L$ it is equal to J_L , i.e. the steady state flux in the former through-diffusion phase. However, for $t \rightarrow \infty$, the two fluxes become equal to each other, hence their ratio becomes -1 :

$$\lim_{t \rightarrow \infty} \frac{J(0,t)}{J(L,t)} = -1 \quad (55)$$

2.4.2 Results

The out-diffusion results are summarized in Figures A6 - A10 in Appendices 6 - 10. The experimental and modelled data are in good agreement with each other in the case of HTO (series1), $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$, indicating that a similar diffusion process is taking place as in the case of the through-diffusion, characterised by the same values for the effective diffusion coefficient and rock capacity (porosity). Moreover, the ratio of the fluxes at both boundaries becomes -1 as was predicted by Eqn. 55. The observation that the out-diffusion of $^{22}\text{Na}^+$ can be perfectly described by Eqn. 53 and 54 is a confirmation of the assumption of linear and reversible sorption of $^{22}\text{Na}^+$. This is not unexpected because the sorption of Na^+ on clays is an ion exchange process which is known to be a reversible sorption process. In the case of $^{125}\text{I}^-$, however, the situation is different. Experimental and expected data are not in agreement, especially in the case of the high concentration boundary. For the latter, the calculated flux is by a factor of ten lower than the measured flux, indicating that an additional process is taking place. The most probable process is a slow release of sorbed iodine, resulting in a higher diffusive flux. This means that in the through-diffusion experiments, part of the water soluble iodine (I^-) was transformed to a surface-bound iodine. The release of this bound iodine, however, seems to be a slow process. So far, it is not clear at all what kind of reactions occur and further investigations to clarify this phenomenon are underway.

The out-diffusion results for HTO (series 2) also show a small discrepancy between predicted and observed fluxes in the case of the high-concentration boundary, i.e. at $x=0$. The decrease in flux becomes slower than predicted. This would mean that a second, slow diffusion process is taking place. This phenomenon, however, could not be observed in the through-diffusion experiments. So, it seems that out-diffusion measurements are more sensitive and give more detailed information about the diffusion/sorption process. The question now is whether this observation can be interpreted as a small change in the pore structure of the OPA samples. Note that it was discussed earlier in chapter 2.3.5 that through-diffusion of HTO indicated that no change in pore structure had happened.

In order to properly explain the observed phenomena, a detailed picture of the microstructure of the OPA is necessary.

3 DIFFUSION PARALLEL TO THE BEDDING

3.1 Introduction

The diffusion of charged and neutral species in rocks depends strongly on geometric parameters of the rocks such as tortuosity and constrictivity. Tortuosity takes into account path lengthening because of the particulate nature of the porous medium (SCHACKELFORD & DANIEL, 1991) and constrictivity is concerned with pore narrowing in the system. For aqueous phase diffusion the relationship between the effective diffusion coefficient, D_e , the geometric parameters and the diffusion coefficient in water, D_w , is given by:

$$D_e = \frac{\delta \cdot \varepsilon}{\tau^2} \cdot D_w \quad (56)$$

where δ represents the constrictivity, τ is the tortuosity and ε , the diffusion accessible porosity.

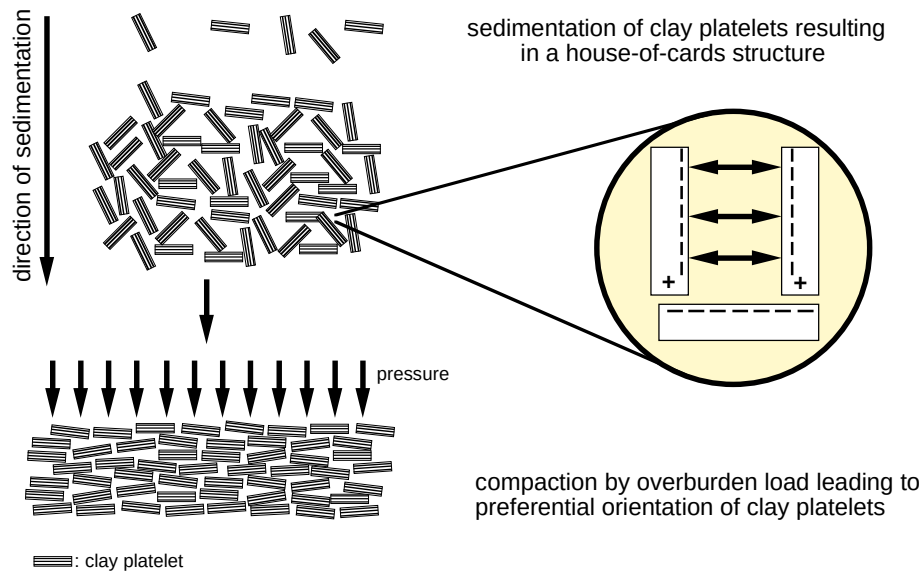


Figure 14: Structure of a mud rock during sedimentation (house-of-cards structure) and after compaction (preferential orientation of clay platelets). The orientation of the clay platelets after compaction is perpendicular to the direction of sedimentation.

Argillaceous rocks are mainly composed of clay platelets that settled in an aqueous

environment to form a mud deposit. BENNETT et al. (1981) reported that the clay platelets in marine deposits of low porosity are oriented preferentially perpendicular to the direction of sedimentation. During sedimentation, the grains form “house-of-cards” type structures because of the electrostatic repulsion by like-charged basal planes, and attraction by oppositely charged edge and basal planes (Figure 14). Also solution composition and ionic strength have an effect on the structure (BERNER 1980). After deposition, the layers of clay mud are compacted by the weight of younger superimposed sediments (overburden pressure) resulting in water being squeezed out. During this process, the mutually repelling particles are forced closer together and the porosity of the mud decreases. Compaction has been reported to increase the preferential orientation of clay platelets (VON ENGELHARDT & GAIDA 1963, MEADE 1964). Owing to this preferentially layered structure of argillaceous rocks, tortuosity (path lengthening) is expected to be anisotropic and consequently also the effective diffusion in such rocks. Figure 15 represents the case of Opalinus Clay in the Zürcher Weinland in Switzerland.

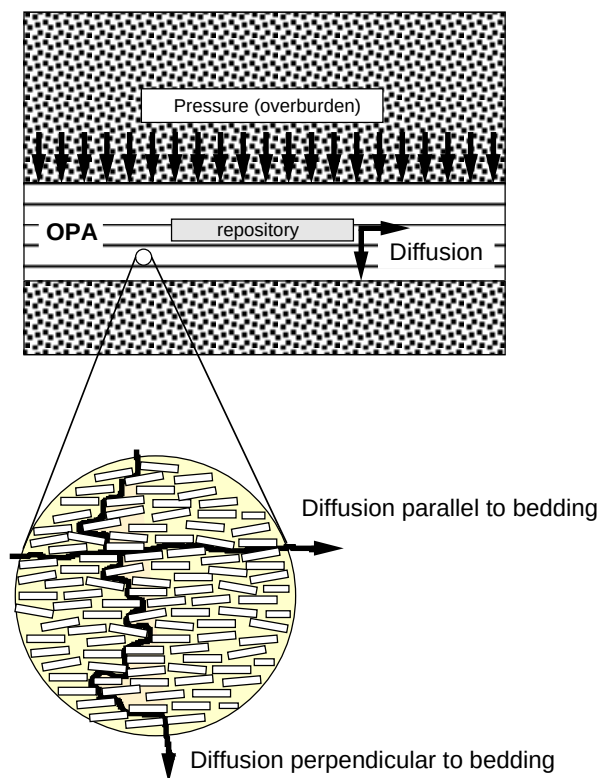


Figure 15: Schematic view of the anisotropy of a layered rock with respect to diffusion (Opalinus clay, Zürcher Weinland, Switzerland).

Anisotropic diffusion has been observed in many different media, e.g. crystals, textile fibres, polymers in which the molecules have a preferential direction of orientation (CRANK 1975, SEO et al. 1999). When a species diffuses parallel to the fabric, tortuosity is expected to be smaller than in the case where diffusion is perpendicular to the bedding (Figure 15) and therefore, the effective diffusion rate will be faster in the former case.

So far, in geochemistry little attention has been paid to these potential anisotropic properties with respect to diffusion. Systematic studies comparing diffusion perpendicular and parallel to the fabric on the same material and for different species are, to our knowledge, not available with the exception of two studies. DE WINDT & PALUT (1999) studied the diffusion of HTO and I⁻ through a piece of Opalinus Clay perpendicular and parallel to the fabric. Although they observed a larger value for diffusion parallel to the bedding, they could not exclude experimental artefacts so that their results are not beyond all question. GILLING et al. (1987) measured the horizontal and vertical component for the diffusion of I⁻ in London Clay using a radial diffusion cell. They found that the horizontal component is two times larger than the vertical component. The main part of the studies on natural argillaceous materials published in the open literature, however, are on diffusion properties perpendicular to bedding (BOURKE et al., 1993; PUT et al., 1998; MELKIOR, 2000).

In this work, diffusion was studied perpendicular and parallel to the bedding of Opalinus clay (OPA), a layered argillaceous rock. For diffusion parallel to the bedding, we used a radial through-diffusion technique recently developed at PSI. For the diffusion perpendicular to the bedding, a one-dimensional through-diffusion method as described earlier was used. The experimental set-up is arranged in such a way that the pressure applied to the samples is always perpendicular and the diffusion is parallel or perpendicular to the bedding, representing the most common situation in natural systems.

3.2 Radial through-diffusion experiments

3.2.1 Equipment for radial through-diffusion experiments

A view of the radial through-diffusion concept is given in Figure 16. A similar concept was used earlier by VAN DER KAMP et al. (1996) for determining isotopic composition, chemistry, and effective porosities for groundwater in aquitards and by NOVAKOWSKI & VAN DER KAMP (1996) for determining effective diffusion coefficients, porosity and adsorption from in-diffusion experiments in porous geological materials. The technique developed by VAN DER KAMP (1996) operates in the transient mode (in-diffusion, out-

diffusion) and was extended by us for operating in the steady state mode (through-diffusion). The advantage of the steady state through-diffusion is that a flux is measured yielding directly the effective diffusion coefficient. A concentration gradient is set up across the sample in such a way that diffusion takes place from the centre of the sample, (high concentration side) to the outer boundary of the sample (low concentration side). Because the layering of the sample is parallel to the concentration gradient, diffusion will take place along the layering. A similar cell was also used by GILLING et al. (1989).

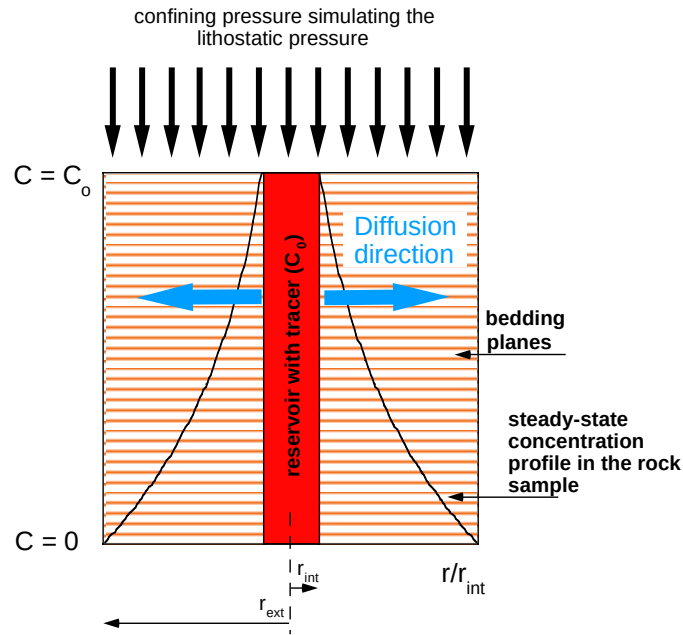


Figure 16: Schematic view of the radial through-diffusion concept.

The diffusion cell is schematically presented in Figure 17. The cell comprises a sample holder, two end plates and two cylindrical stainless steel filters (Figure 18). A cylindrical sample with a central hole ($r_{\text{ext}} = 25.4$ mm, $r_{\text{int}} = 6.58$ mm, $h = 52$ mm) is used in this set-up (Figure 18). One stainless steel filter ($r_{\text{ext}} = 6.58$ mm, $r_{\text{int}} = 4.78$ mm, $h = 61$ mm, porosity = 41.8 ± 0.8 %, sintered stainless steel 316L with pore diameter $10 \mu\text{m}$, MOTT industrial division, Farmington, USA) is put in the centre along the axes of the sample and another ($r_{\text{ext}} = 26.98$ mm, $r_{\text{int}} = 25.4$ mm, $h = 61$ mm, porosity = 36.7 ± 0.6 %) contacts the outer boundary of the sample (Figure 16). A stainless steel pin ($r = 4.78$ mm, $h = 61$ mm) with a spiral groove is placed in the centre of the inner filter. The sample is put in the sample holder ($r_{\text{int}} = 27$ mm, $L = 61$ mm) and the two end-plates are placed in position. The wall of the sample holder has a spiral groove at the inner surface. The two end-plates are held together by a set of 4 bolts. Between the surface of the end plate and the head of the bolt, packages of 12 disc springs (GR1, DIN 2093 – A16) are placed.

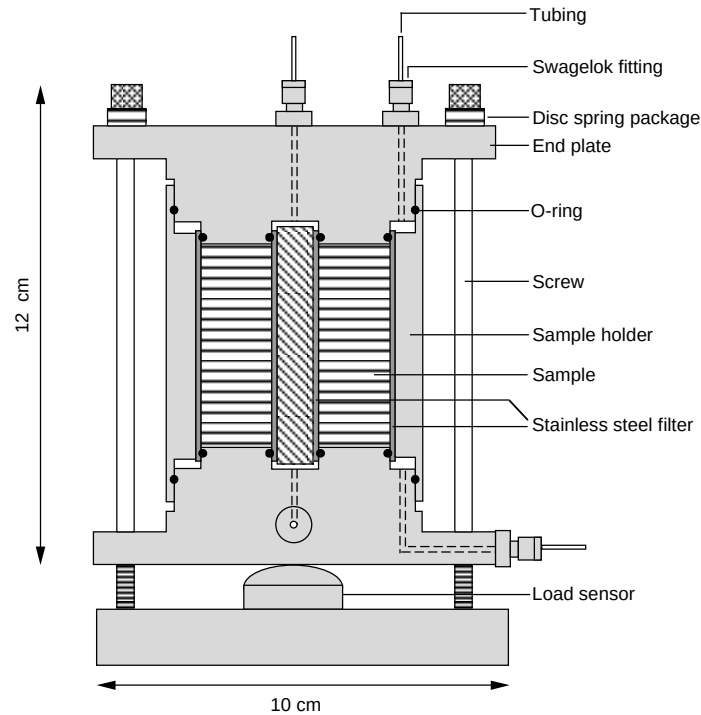


Figure 17: Cross section view of the PSI high-pressure through-diffusion cell for radial diffusion (PSI-HPDIF-RAD/1.0).

A well-defined load is applied to the cells by tightening the bolts. The cells are equipped with a load sensor (SENSY, Belgium) connected to a force indicator (FMD-6, Penko, the Netherlands) in order to monitor the pressure applied to the samples. The dead volumes (spiral groove, filter and tubings) are $4.8 \pm 0.5 \text{ cm}^3$ and $18.2 \pm 1.8 \text{ cm}^3$ at the inner and outer side of the cell, respectively.



Figure 18: Photograph of a cylindrical OPA sample and both stainless steel filters used for the radial diffusion experiment.

3.2.2 Samples and sample preparation

A bore core (Mont Terri, Benken) was placed in a Plexiglas™ cylinder and embedded in an epoxy resin (Epofix™, Struers GmbH). After the hardening of the resin, the cylinder was placed on a lathe and a cylindrical sample ($r_{\text{ext}} = 25.4 \text{ mm}$, $h = 52 \text{ mm}$) was prepared (Figure 18 and 19). The central hole ($r_{\text{int}} = 6.58 \text{ mm}$) in the sample was made by drilling.

Benken: a borecore originating from a depth between – 636.67 and – 636.82 m (15 cm) was used for the experiments. The core turned out to be broken in three pieces. The lower piece was used for the experiments, i.e. 636.76 – 636.82 m.

Mont Terri: a borecore MT/DI-A0/7 (3.02 – 3.09 m) was used for the experiments.

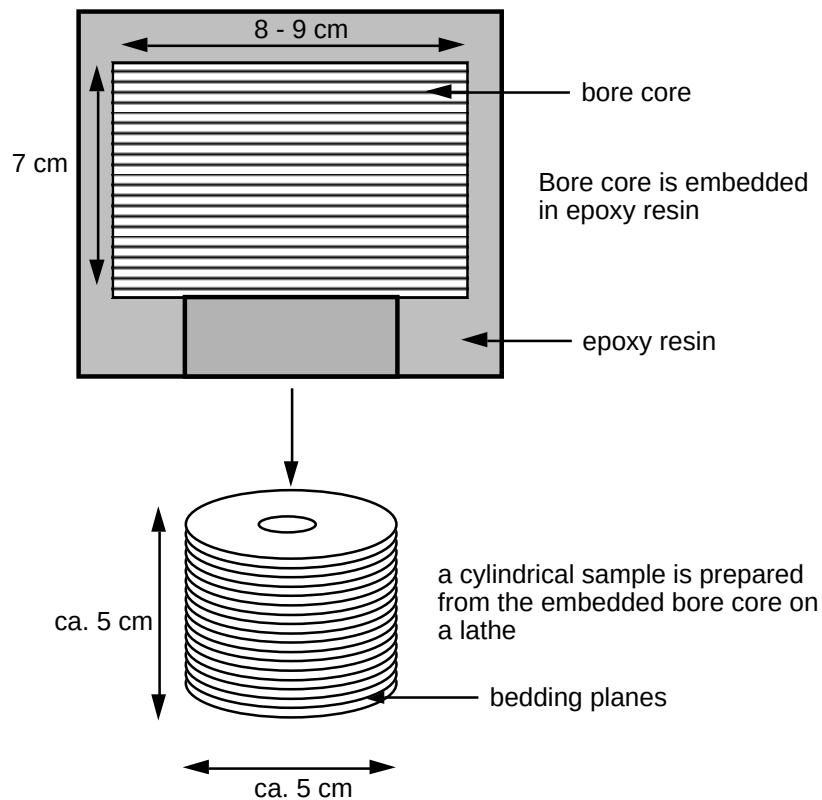


Figure 19: Preparation of a cylindrical sample with a central hole from a bore core.

3.2.3 Synthetic pore water

The composition of the pore water used is given in Table 1 of chapter 2.1.3

3.2.4 Experimental set-up

A similar experimental set-up was used as in the case of diffusion perpendicular to the bedding (see chapter 2.1.4). The planar diffusion cell was replaced by the radial diffusion cell (Figure 20). The volumes of the reservoirs used in the radial diffusion set-up were larger, i.e. 500 or 1000 cm³ for the large container and 50 or 100 cm³ for the small container. Table 11 gives an overview of the tracers used, as well as their activities (Bq·m⁻³) and concentrations [mol·dm⁻³].

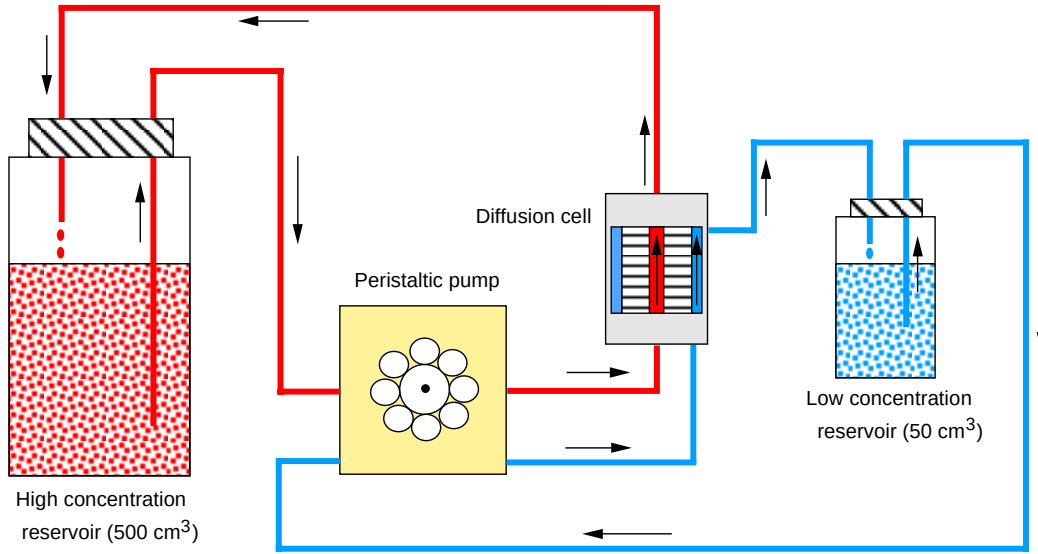


Figure 20: Experimental set-up of a through-diffusion experiment for diffusion parallel to the bedding.

3.2.5 Activity measurements

The activity measurements were performed in the same way as described in chapter 2.1.5.

3.2.6 Data processing

The same data processing procedure as described by Eqn. 2 – 5 and Eqn. 7 in chapter 2.1.6 was used. The average flux during Δt_i , $J_{r_{ext}}^{t_i}$, is calculated by:

$$J_{r_{ext}}^{t_i} = \frac{A_{dif}^{\Delta t_i}}{S_{r_{ext}} \cdot \Delta t_i} \quad (57)$$

where $S_{r_{ext}}$ [m²] represents the cross section area at the outer boundary, $r = r_{ext}$, of the sample.

Table 11: Overview of the tracers and their activity/concentration used in the diffusion studies parallel to the bedding. The activities given represent average values.

Sample	Activity x 10 ⁻⁶ [Bq m ⁻³]			
	HTO/1	HTO/2	³⁶ Cl ⁻	²² Na ⁺
BE-S	1200±83	1197±82	1046±70	979±65
BE-R	1143±77	1165±70	1059±65	906±55
MT-S	1207±71	1182±70	1070±62	975±56
MT-R	1130±77	1135±77	1054±64	918±56

BE: Benken samples; MT: Mont Terri samples; S: perpendicular; R: parallel

HTO/1 and HTO/2: 1.0·10⁻⁹ mol·dm⁻³; ³⁶Cl⁻: 2.4·10⁻⁵ mol·dm⁻³;

²²Na⁺: 2.0·10⁻¹⁰ mol·dm⁻³

3.2.7 Uncertainty estimation

The uncertainty estimation was done using the same procedure as described in chapter 2.1.7.

3.3 Diffusion perpendicular to the bedding

For comparison reasons, diffusion measurements perpendicular to the bedding were performed on samples from the same location and under similar overburden pressures. The equipment used was the high-pressure diffusion cell (PSI-HPDIF-PER/2.0) depicted in Figure 4 of chapter 2.1.1. The diffusion cells were equipped with a load sensor in order to monitor the pressure on the samples during the diffusion measurements. All experiments were performed in the same way as described in Chapter 2.

3.4 Theoretical background of radial through-diffusion

Consider a hollow cylindrical sample of height h with an internal radius r_{int} and an external radius r_{ext} . The concept behind the experiments is the same as in one-dimensional diffusion experiments (CHO et al., 1993), except for the fact that the high-concentration reservoir is the central cavity of the cylinder (from $r = 0$ to $r = r_{int}$) and breakthrough of the tracer occurs across the whole external surface ($r = r_{ext}$). The diffusion process has therefore a cylindrical geometry. The diffusion equation is given by:

$$\frac{\partial C}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D_a \frac{\partial C}{\partial r} \right) \quad (58)$$

where D_a is the apparent diffusion coefficient (see Eqn. 25) and C is the concentration of a given tracer in the rock. The initial and boundary conditions are given by:

$$\begin{aligned} C(r_{int} \leq r \leq r_{ext}, t = 0) &= 0 ; \\ C(0 \leq r < r_{int}, t > 0) &= C_0 ; \\ C(r > r_{ext}, t > 0) &= 0 ; \end{aligned}$$

and the tracer flow ($\text{mol} \cdot \text{s}^{-1}$ or $\text{Bq} \cdot \text{s}^{-1}$) at the external surface of the cylinder is equal to:

$$J(r_{ext}, t) = -2\pi h D_e \left(r \frac{\partial C}{\partial r} \right)_{r=r_{ext}} \quad (59)$$

The analytical solution of the cylindrical through-diffusion problem is (CRANK, 1975; p.84; 5.64):

$$\frac{A(r_{ext}, t)}{\pi \cdot C_0 \cdot \alpha \cdot h} = \frac{2(D_a \cdot t - \Lambda)}{\ln(r_{ext}/r_{int})} - 4 \sum_{n=1}^{\infty} \frac{J_0(r_{int}\beta_n) J_0(r_{ext}\beta_n) \exp(-D_a \beta_n^2 t)}{\beta_n^2 (J_0^2(r_{int}\beta_n) - J_0^2(r_{ext}\beta_n))} \quad (60)$$

where $A(r_{ext}, t)$ is the total amount [mol] or activity [Bq] of tracer diffused through the sample at time t . α is the rock capacity factor (see Eqn. 26). Λ [m^2] in Eqn. 60 is given by:

$$\Lambda = \frac{r_{int}^2 - r_{ext}^2 + (r_{int}^2 + r_{ext}^2) \ln(r_{ext}/r_{int})}{4 \ln(r_{ext}/r_{int})} \quad (61)$$

The $J_0(x)$ terms are Bessel functions of the first kind of order zero and the β_n terms are the positive roots of:

$$U_0(r_{int}\beta_n) = J_0(r_{int}\beta_n) Y_0(r_{ext}\beta_n) - J_0(r_{ext}\beta_n) Y_0(r_{int}\beta_n) = 0 \quad (62)$$

The $Y_0(x)$ terms are Bessel functions of the second kind of order zero. The first five positive roots $r_{int}\beta_n$ of Eqn. 62 for different ratios r_{ext}/r_{int} , are tabulated in CRANK (1975, Table 5.3, p. 380). At steady state, when $t \rightarrow \infty$ the solution (Eqn. 60) reduces to:

$$A(r_{ext}, t) = \frac{2\pi \cdot C_0 \cdot h}{\ln(r_{ext}/r_{int})} (D_e \cdot t - \alpha \cdot \Lambda) \quad (63)$$

hence the total amount of tracer diffused through becomes a linear function of time:

$$A(r_{ext}, t) = a \cdot t - b \quad (64)$$

with:

$$a = \frac{2\pi \cdot C_0 \cdot h \cdot D_e}{\ln(r_{ext}/r_{int})} \quad (65)$$

and

$$b = \frac{2\pi \cdot C_0 \cdot h \cdot \alpha \cdot \Lambda}{\ln(r_{ext}/r_{int})} \quad (66)$$

The concentration profile of the tracer in the rock at steady state is given by:

$$C_r = C_0 \left(1 - \frac{\ln \frac{r_{int}}{r}}{\ln \frac{r_{int}}{r_{ext}}} \right) \quad (67)$$

A typical concentration profile at steady state as described by Eqn. 67 is depicted in Figure 16.

3.5 Results and discussion

3.5.1 Diffusion of HTO

Figure 21 (A-D) shows the flux and total diffused activity for the diffusion of HTO parallel and perpendicular to the bedding in the two samples (series 1). The curves clearly show two stages: a transient phase in which the flux increases and a steady state phase in which the flux is constant as a function of time and the total diffused activity increases linearly with time.

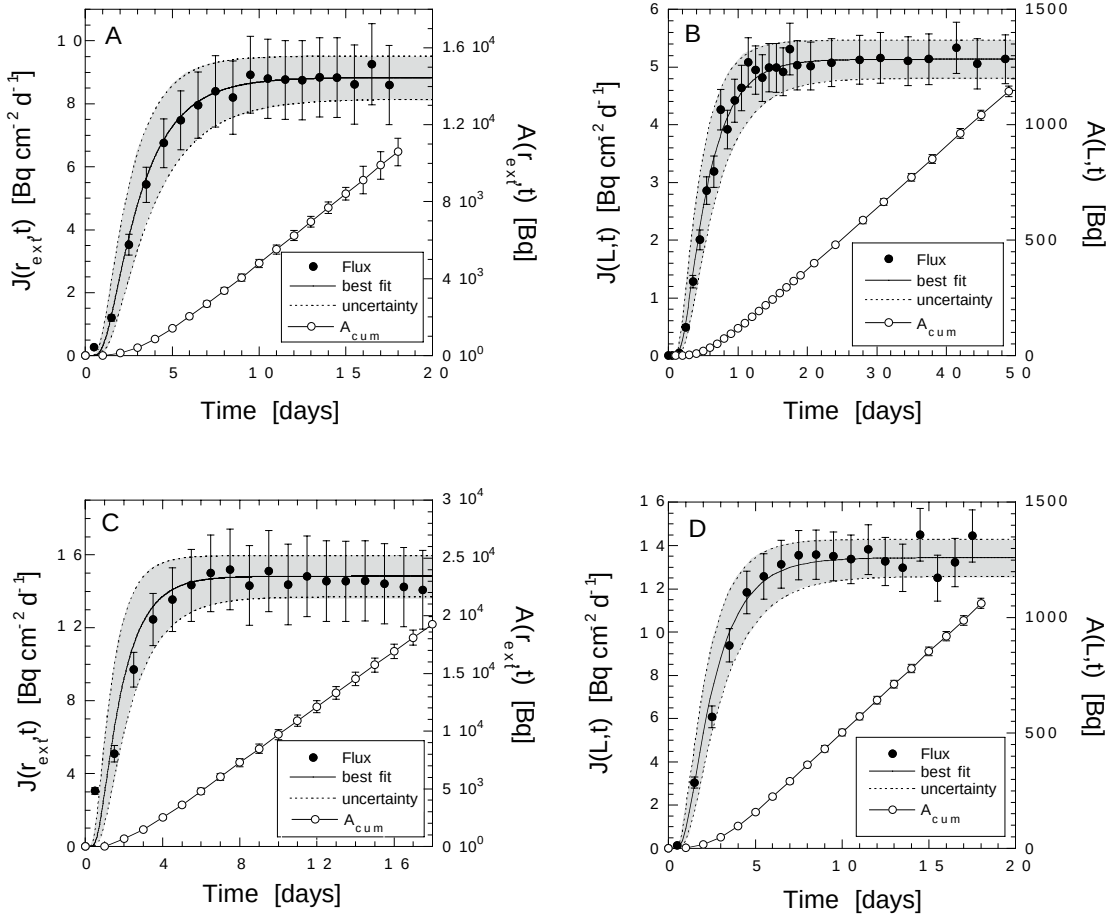


Figure 21: Fluxes and total diffused activity of HTO through Opalinus Clay (series 1). Diffusion parallel to the bedding (A: Benken, C: Mont Terri). Diffusion perpendicular to the bedding (B: Benken, D; Mont Terri). The solid curves for the fluxes were calculated using the analytical solution (Eqn. 60 diffusion parallel, Eqn. 31 diffusion perpendicular) using the parameters in Table 12. The shaded areas represent the uncertainty on the calculated curves.

Unlike the cumulative $A(L,t)$ - or $A(r_{ext},t)$ -plot, the flux clearly shows when the system is in steady state. The latter phase was used to deduce a diffusion coefficient for the tracer and the porosity accessible to the tracer. The slope of the curve ($A(L,t)$ vs time or $A(r_{ext})$ vs time) gives the effective diffusion coefficient by applying Eqn. 65 for diffusion parallel to the bedding and Eqn. 34 for diffusion perpendicular to the bedding. The porosity can be calculated from the intercept of an asymptote extrapolated from the steady-state region of the curve to the Y-axis. The concentration of HTO at the high concentration boundary decreased slightly ($<10\%$) during the experiment. This decrease,

however, is acceptable and the boundary condition of constant concentration can be assumed to be fulfilled.

The quality of the parameters (D_e and α) was tested by using them as input parameters for the full analytical solution (Eqn. 60 for parallel diffusion or Eqn. 31 for perpendicular diffusion) and calculating the whole diffusion curve ($J(r_{ext}, t)$ vs. time or $J(L, t)$ vs time). In all cases it was possible to reproduce the flux in both the initial and the final steady state stages of the experiments (solid flux-curves in Fig. 21). An overview of the results for HTO is given in Table 12. As can be seen from the data from series 1 and series 2, the diffusion measurements are reproducible. Compared with values for diffusion perpendicular to the bedding, the effective diffusion coefficients for diffusion parallel to the bedding are larger by a factor between 4 (MT) and 6 (BE). The larger value for the Benken sample indicates a greater degree of preferential orientation of the platelets which might be due to the higher overburden pressure (MEADE, 1964). The porosity values, ε , are similar.

Table 12: Overview of effective diffusion coefficients and diffusion accessible porosities for the diffusion of HTO into Opalinus clay parallel and perpendicular to the bedding

Sample	D_e (parallel) [m ² ·s ⁻¹]	α (parallel) [-]	D_e (perpend.) [m ² ·s ⁻¹]	α (perpend.) [-]
BE-1	(3.10±0.25)×10 ⁻¹¹	0.15±0.02	(5.42±0.34)×10 ⁻¹²	0.14±0.02
BE-2	(3.20±0.23)×10 ⁻¹¹	0.13±0.02	(5.41±0.35)×10 ⁻¹²	0.13±0.02
MT-1	(5.39±0.43)×10 ⁻¹¹	0.15±0.03	(1.42±0.09)×10 ⁻¹¹	0.17±0.02
MT-2	(5.42±0.38)×10 ⁻¹¹	0.17±0.02	(1.37±0.08)×10 ⁻¹¹	0.14±0.02

1 = series 1; 2 = series 2 (repetition of the measurement on the same sample)

Because D_w , ε and δ are independent of the direction of diffusion, the different D_e values must be due to differences in tortuosity. The path lengthening (tortuosity) is smaller in the direction parallel to the bedding so that D_e is expected to be larger for diffusion parallel to the bedding, as found experimentally.

3.5.2 Diffusion of ³⁶Cl⁻

Figure 22 (A-D) shows the flux and total diffused activity for the diffusion of ³⁶Cl⁻ parallel and perpendicular to the bedding in the two samples. The values of D_e and α are summarised in Table 13.

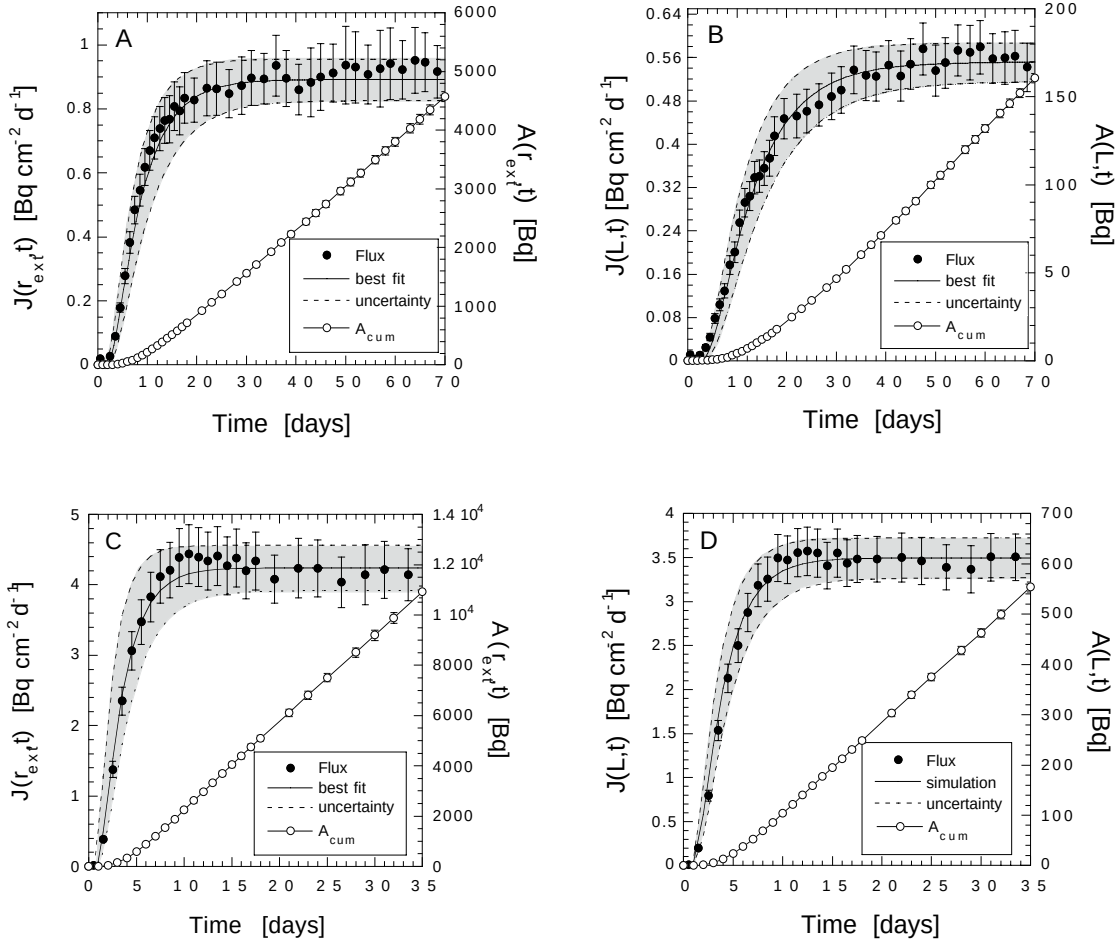


Figure 22: Fluxes and total diffused activity of ^{36}Cl through Opalinus Clay. Diffusion parallel to the bedding (A: Benken, C: Mont Terri). Diffusion perpendicular to the bedding (B: Benken, D; Mont Terri). The solid curves for the fluxes were calculated using analytical solution (Eqn. 60 diffusion parallel, Eqn. 31 diffusion perpendicular) using the parameters in Table 13. The shaded areas represent the uncertainty on the calculated curves.

A similar picture as for HTO is obtained, i.e. diffusion parallel is faster than diffusion perpendicular to the bedding. The rock capacity factor is independent of the direction.

The value of D_e for diffusion parallel to the bedding is larger by a factor 5.0 for the Benken samples and 3.9 for the Mont Terri samples. This is comparable with the observations made for HTO.

Table 13: Overview of effective diffusion coefficients and diffusion accessible porosities for the diffusion of $^{36}\text{Cl}^-$ into Opalinus clay parallel and perpendicular to the bedding.

Sample	D_e (parallel) [m ² ·s ⁻¹]	α (parallel) [-]	D_e (perpend.) [m ² ·s ⁻¹]	α (perpend.) [-]
BE-1	(3.38±0.25)×10 ⁻¹²	0.045±0.005	(6.74±0.42)×10 ⁻¹³	0.041±0.003
MT-1	(1.59±0.11)×10 ⁻¹¹	0.082±0.007	(4.13±0.25)×10 ⁻¹²	0.075±0.006

3.5.3 Diffusion of $^{22}\text{Na}^+$

Figure 23 (A-D) shows the results (flux and total diffused activity vs time) for the diffusion of $^{22}\text{Na}^+$. The values of the effective diffusion coefficient and rock capacity factor are summarized in Table 14. As was already observed for HTO and $^{36}\text{Cl}^-$ also in the case of $^{22}\text{Na}^+$ the diffusion parallel to the bedding is larger than diffusion perpendicular to the bedding. The rock capacity factor is independent of the sample orientation.

Table 14: Overview of effective diffusion coefficients and diffusion accessible porosities for the diffusion of $^{22}\text{Na}^+$ into Opalinus clay parallel and perpendicular to the bedding.

Sample	D_e (parallel) [m ² ·s ⁻¹]	α (parallel) [-]	D_e (perpend.) [m ² ·s ⁻¹]	α (perpend.) [-]
BE-1	(7.75±0.55)×10 ⁻¹¹	0.89±0.08	(1.33±0.08)×10 ⁻¹¹	1.11±0.11
MT-1	(7.20±0.51)×10 ⁻¹¹	0.62±0.05	(1.52±0.08)×10 ⁻¹¹	0.48±0.04

The value of D_e for diffusion parallel to the bedding is larger by a factor 5.8 for the Benken samples and 4.7 for the Mont Terri samples. This is comparable with the observations made for HTO and $^{36}\text{Cl}^-$.

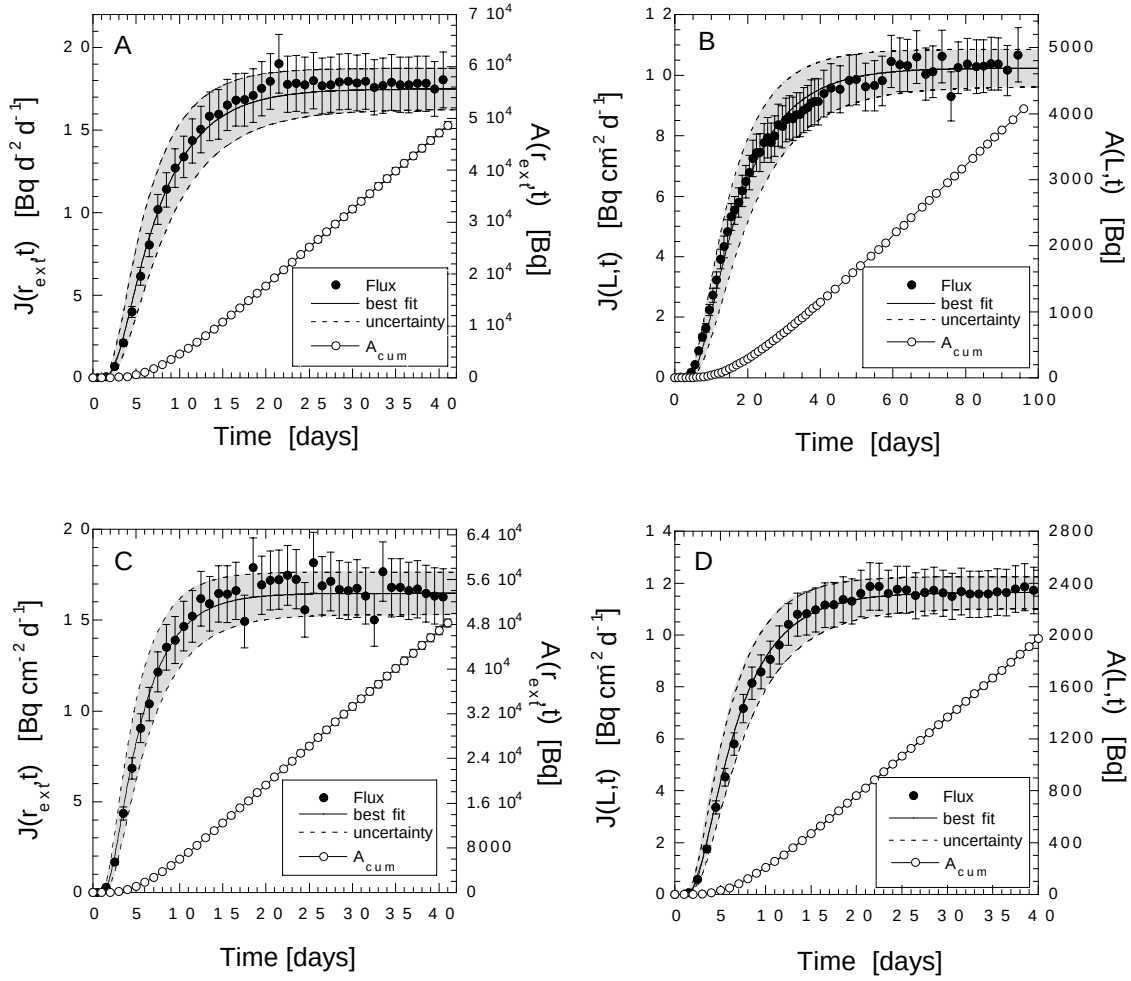


Figure 23: Fluxes and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay. Diffusion parallel to the bedding (A: Benken, C: Mont Terri). Diffusion perpendicular to the bedding (B: Benken, D: Mont Terri). The solid curves for the fluxes were calculated using analytical solution (Eqn. 60 diffusion parallel, Eqn. 31 diffusion perpendicular) using the parameters in Table 14. The shaded areas represent the uncertainty on the calculated curves.

3.5.4 Summary of results

Figure 24 summarises the results for the three radionuclides measured in this study. Figure 24A depicts the effective diffusion coefficients parallel to the bedding against those perpendicular to the bedding. The diagonal line represents the case for isotropic diffusion, i.e. the diffusion coefficients are independent of the direction of diffusion.

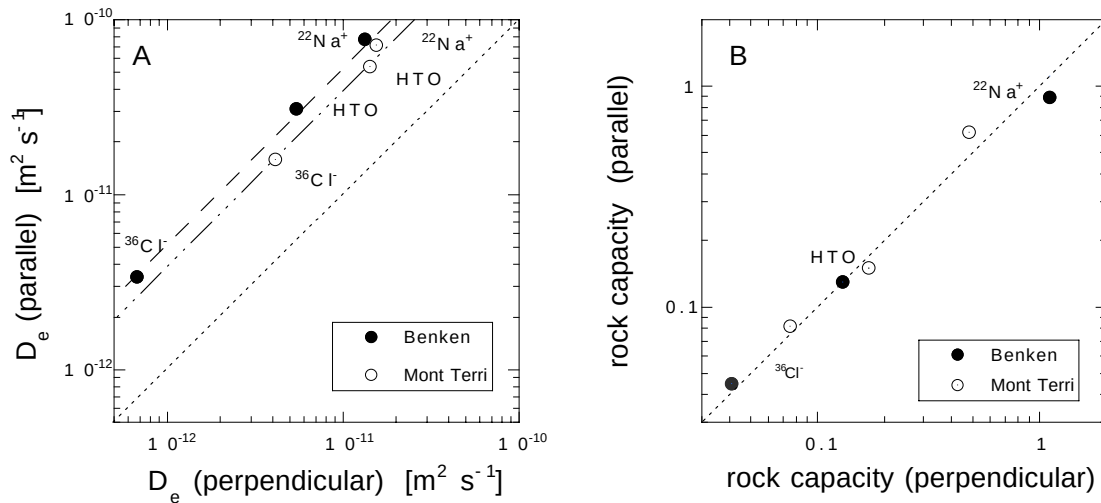


Figure 24: Effective diffusion coefficients (A) and rock capacity factors (B) for the two diffusion directions. The diagonal lines ($x=y$) represent isotropic properties.

As can be seen from this plot, all three radionuclides deviate from this diagonal line. It can further be seen from the plot that the three radionuclides lie more or less on the same line, meaning that the ratio of the diffusion coefficients parallel to the bedding to those perpendicular to the bedding is constant and independent on the type of radionuclide. The ratio for Benken is larger than the ratio for Mont Terri, indicating that the anisotropy for the Benken samples is larger than the anisotropy for the Mont Terri samples. Figure 24B is a similar plot for the rock capacity factors. Unlike the diffusion coefficient, the rock capacity is independent on the direction of diffusion, hence is isotropic.

A more detailed explanation for the factor 4 to 6 cannot yet be given. However, for porous media with pores smaller than the dimension of the clay particles, the ratio of the particle length, l , to the particle width, d , seems to play a central role, as shown in a very simplified sketch of the microstructure of a layered argillaceous rock (Figure 25).

For a sample having dimensions $L \times L$, the pathway, P , for diffusion along the orientation of the platelets is L . The pathway for diffusion perpendicular to the orientation is given by:

$$P \approx L + (n - 1) \cdot l / 2 \quad (68)$$

where n = number of clay layers of thickness d and length l . When the pore diameter is much smaller than the thickness of the clay layer, then $L = n \times d$.

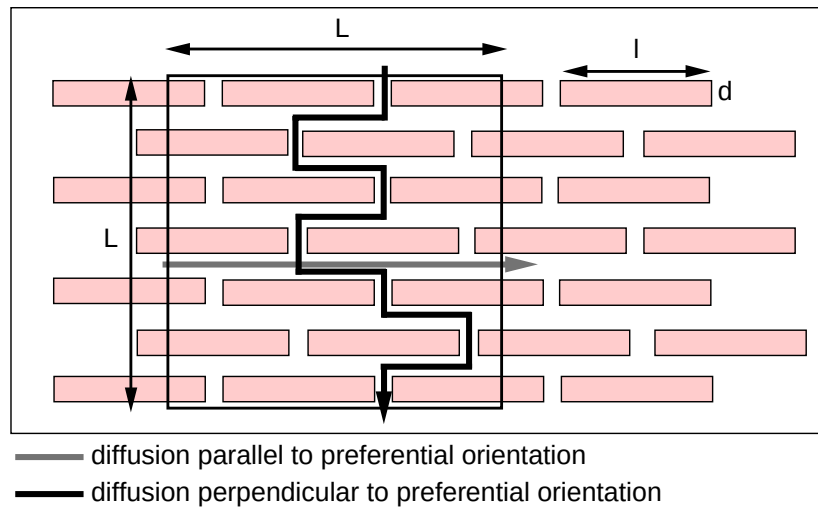


Figure 25: Simplified sketch of the microstructure of a layered argillaceous rock.

4 EFFECT OF TEMPERATURE ON DIFFUSION

4.1 Introduction

The temperature in the Opalinus Clay layer (OPA) at the Benken site is between 30 °C and 40 °C (Figure 26). So far, all diffusion measurements on OPA have been performed at room temperature, i.e. 20 ± 2 °C. It is well known that molecular diffusion is temperature dependent. Studies of the temperature dependence of the self-diffusion coefficient of H₂O in bulk liquid water show that D can be described by the Arrhenius equation (EISENBERG & KAUFMANN, 1969):

$$D = A \cdot e^{-\frac{E_A}{R \cdot T}} \quad (69)$$

with:

D = diffusion coefficient ($\text{m}^2 \cdot \text{s}^{-1}$)

A = (empirical) Arrhenius parameter ($\text{m}^2 \cdot \text{s}^{-1}$), also called pre-exponential factor

E_A = activation energy ($\text{kJ} \cdot \text{mol}^{-1}$)

R = gas constant ($8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$)

T = absolute temperature (K)

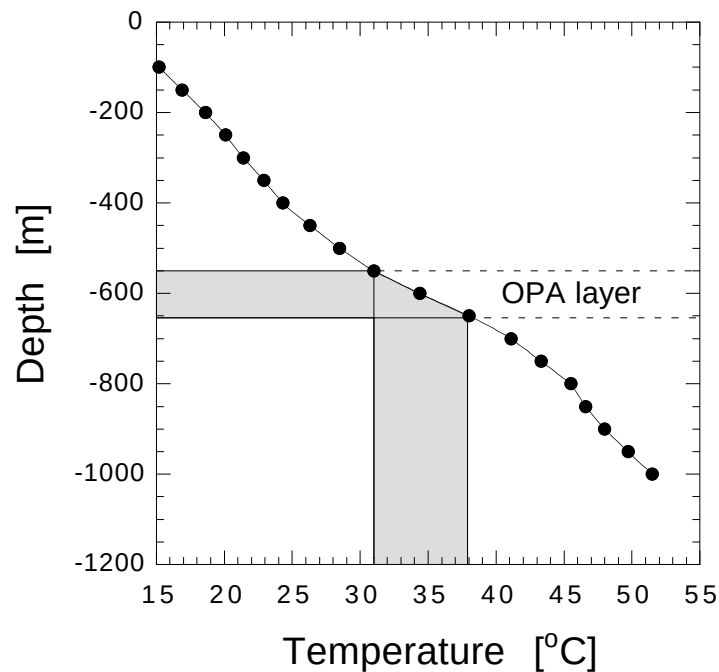


Figure 26: Temperature profile at the Benken site (NAGRA, 2001)

WANG et al. (1953) found that over the temperature range 1.1 - 55 °C, E_A is 19.2 ± 0.4 kJ·mol⁻¹ for HDO and HTO tracers in H₂O and 18.4 ± 1.3 kJ·mol⁻¹ for a H₂¹⁸O tracer. The activation energy derived from measurements with HTO in H₂O over a temperature range between 1 - 45 °C (MILLS, 1973) is 18.8 kJ·mol⁻¹. Water is a structural liquid (MILLS, 1973) and the energy required to break hydrogen bonds between the molecules is reflected in the activation energy of the self-diffusion.

The main question now is whether diffusion coefficients for OPA at higher temperature can be calculated from measurements at room temperature by using the Arrhenius relationship (Eqn. 69) with an activation energy similar to the one for diffusion processes in bulk liquid water. Because diffusion in compacted systems takes place in the confined pore water, the problem can be reformulated to the question of the structure of water in confined systems. If water in confined systems has the same structure as water in bulk liquid water, it can be expected that the temperature dependence of the diffusion in OPA is of the form as given by Eqn. 69 and that the activation energy for diffusion processes in such confined systems is similar to that of diffusion processes in bulk liquid water. A different structure would result in a different activation energy.

So far no information on the activation energy of the self-diffusion of water in compacted clay systems is available. This study focusses on the temperature dependence of the diffusion of HTO in Opalinus clay. The main emphasis of the study lies on the experimental technique for measuring the activation energy.

4.2 Materials and methods

4.2.1 Experimental set-up

The experimental set-up used to study the effect of temperature on diffusion is identical with the one used for diffusion perpendicular to the bedding (chapter 2.1.4). The temperature was controlled by keeping the diffusion cells in a water bath (see Figure 27).

4.2.2 Samples

The samples used were samples from the deep borehole Benken (- 636.67 and - 636.82 m) and from Mont Terri (DI-A). The preparation of the samples was described elsewhere (chapter 2.1.2).

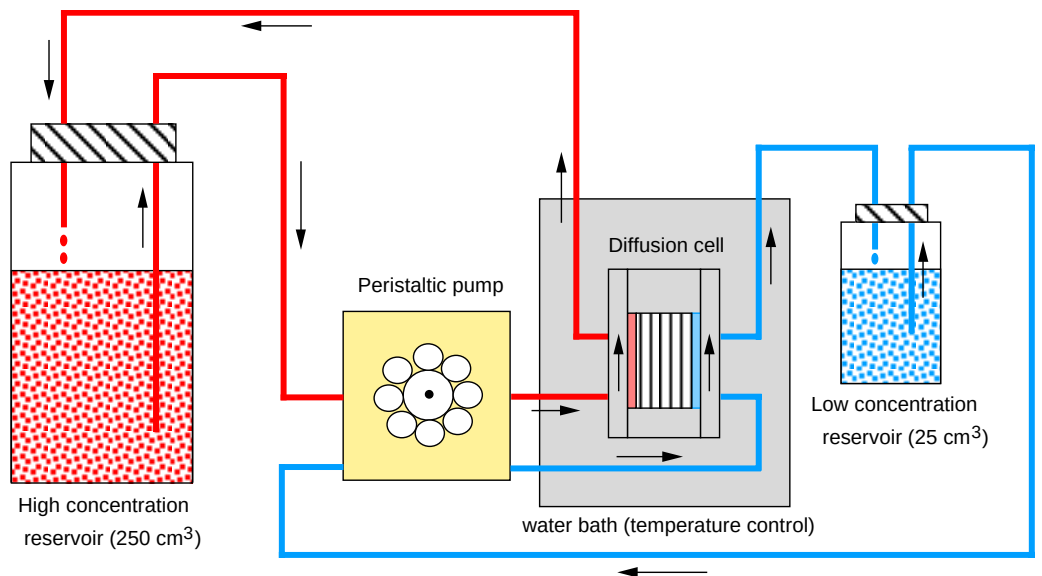


Figure 27: Experimental set-up for studying the effect of temperature on diffusion.

4.2.3 Theoretical background

Because molecular diffusion is a direct consequence of the Brownian motion of dissolved molecules, temperature has a significant effect on diffusion ($D \propto kT$). An increase in temperature results in an increase in the value of the diffusion coefficient (LI & GREGORY, 1974). A temperature change will therefore result in a change in flux and total diffused mass.

The temperature dependence of diffusion can be studied by the through-diffusion technique in two ways. In the step approach the whole diffusion curve (transient and steady state phase) is used. Diffusion experiments have to be performed at different temperatures. Either different samples, each at a given temperature, or one sample at different temperatures can be used. In the case where only one sample is used, a through-diffusion measurement has to be followed by an out-diffusion experiment in order to wash out the tracer present in the rock. After the out-diffusion step, the temperature can be changed and through-diffusion can be re-started. In both cases, similar results will be obtained. Figure 28 shows typical results for a series of through-diffusion measurements performed at different temperatures. The higher the temperature, the larger the slope of the asymptotes. From the slope of the asymptote, the diffusion coefficient for the given temperature can be calculated. The intercept gives the rock capacity factor (= diffusion accessible porosity for non-sorbing tracers). Because the porosity is independent on the temperature, the intercept on the Y-axis is also independent on temperature.

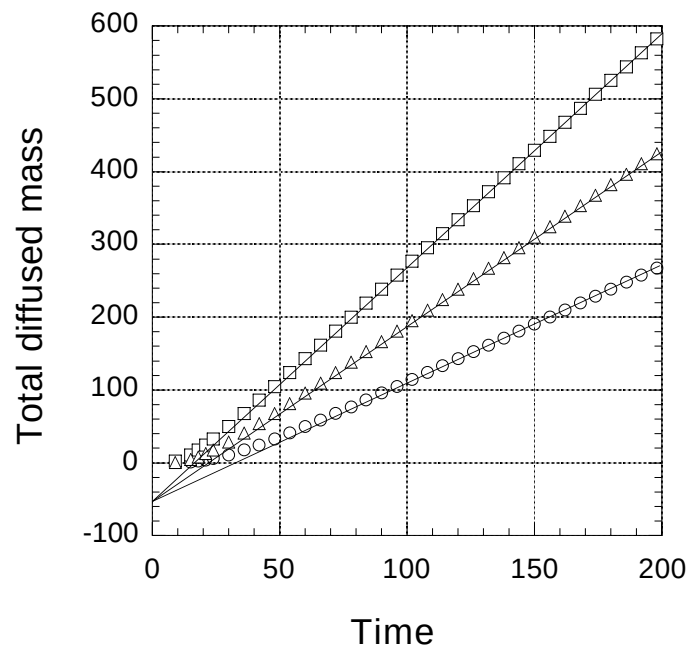


Figure 28: Schematic representation of the effect of temperature on the total diffused mass in a through-diffusion experiment. Each curve represents a diffusion experiment performed at a different temperature. The slope depends on temperature. The intercept is independent on temperature.

In the case of the continuous method only the steady state phase is used. A diffusion experiment is started at a given temperature. As soon as the diffusion experiment is in the steady state phase, the temperature is changed. This will cause a short transient phase, its duration depending on the heat conductivity of sample and sample holder, followed by a steady state phase. This transient phase will usually last no longer than a few hours. As soon as the temperature of the sample has reached a constant value, diffusion will again be in steady state because the concentration gradient in the rock is independent of temperature and depends only on the concentration of the tracer at both boundaries. The total accumulated mass curve (Figure 29) now shows different linear regions. The slope of the asymptotes again can be used directly to calculate the diffusion coefficient. The intercept, however, also varies with temperature and has no longer a physical meaning and cannot be used any longer for calculating the porosity. Only the intercept of the asymptote of the first part of the curve, i.e. for the initial temperature, can be used for determination of the diffusion accessible porosity.

The advantages of the last method are the short experimental time and the fact that the same sample is used for the whole temperature range.

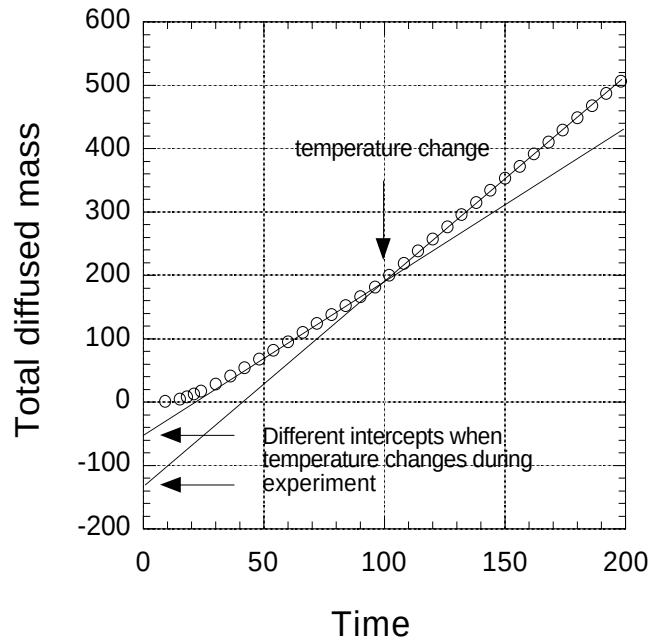


Figure 29: Schematic representation of the effect of temperature on the total diffused activity in a through-diffusion experiment with varying temperature. The temperature is increased in the steady state phase from T_1 to T_2 .

4.2.4 Diffusion experiment

The diffusion cells were loaded with cylindrical samples of Opalinus clay from Benken (B) and Mont Terri (M), sandwiched between two stainless steel filters. The cells were closed and a confining pressure of 3 and 5 MPa was put on the samples of Mont Terri and Benken, respectively. The containers were filled with synthetic pore water and circulated for two weeks. The cells were placed in a water bath to keep the temperature at the desired values. After two weeks the solutions were replaced by fresh ones and the solutions in the large containers were labeled with HTO. The concentration of HTO was $\sim 1200 \pm 90 \text{ Bq}\cdot\text{cm}^{-3}$. The solutions in the small containers were regularly replaced after a given time interval Δt to keep the concentration of the tracer in this compartment as low as possible, i.e. $< 1\%$ of the concentration in the high-concentration compartment. As soon as the diffusion was in the steady state phase, the experiment was stopped and the activity washed out. Hereafter the temperature was increased and diffusion started again. This procedure is further denoted as the “step method (S)”. In a second series, the temperature was gradually increased as soon as the diffusion was in the steady state phase. This procedure is further called the “continuous method (C)”. All experiments were performed in duplicate (BS1, BS2; BC1, BC2; MS1, MS2; MC1, MC2).

4.3 Results and discussion

Figure 30 shows the cumulative activity for the diffusion of HTO through OPA from Benken (BS1) as measured by the step method. For $t \rightarrow \infty$ the cumulative activity becomes a linear function of time. The slope of the asymptote increases with temperature, indicating that the diffusion coefficient increases with increasing temperature. The intercept, however, is independent of the temperature, indicating that the porosity is independent of temperature. Similar observations were also made for the other samples.

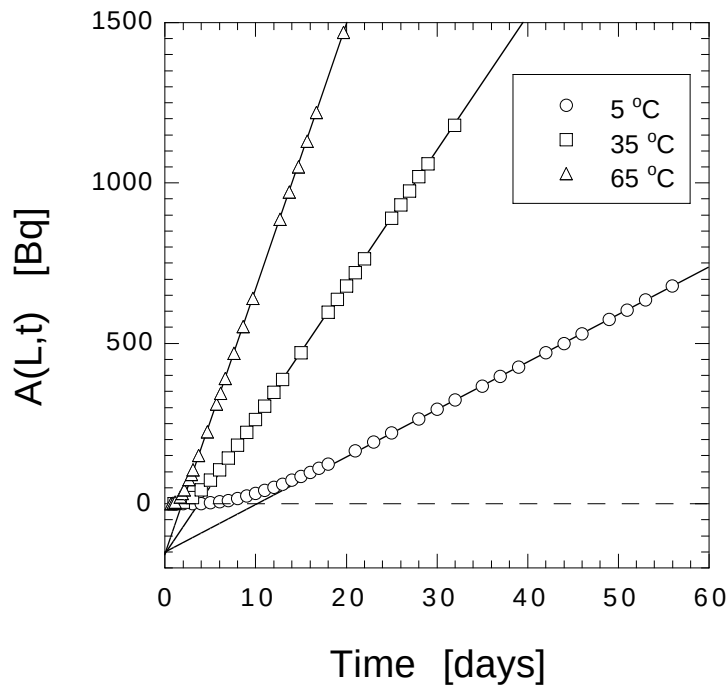


Figure 30: Total diffused activity of HTO as a function of time for Opalinus Clay from Benken (BS1) for three different temperatures. The data were measured using the step method (S).

In the case of the continuous method, the accumulated activity vs time curve shows different linear parts with increasing slope (Figure 31B). Each part reflects a different temperature. The slopes of the asymptotes are proportional to the diffusion coefficient. The intercepts with the Y-axis, however, have no longer a physical meaning, i.e. no porosity values can be extracted from the intercept. The effective diffusion coefficients for HTO in OPA at different temperatures are summarized in Tables 15 and 16.

It is clear from the data that the two techniques used give similar values for the diffusion coefficients. The advantage of the step method is that also a porosity value for each

temperature can be extracted. In the case of the continuous method, only a value for the porosity at the starting temperature can be obtained. The diffusion coefficient increases with increasing temperature whereas the porosity value is independent of temperature. The temperature dependence of the diffusion coefficient is shown in Figure 32 and 33 for the diffusion in Benken and Mont Terri OPA, respectively.

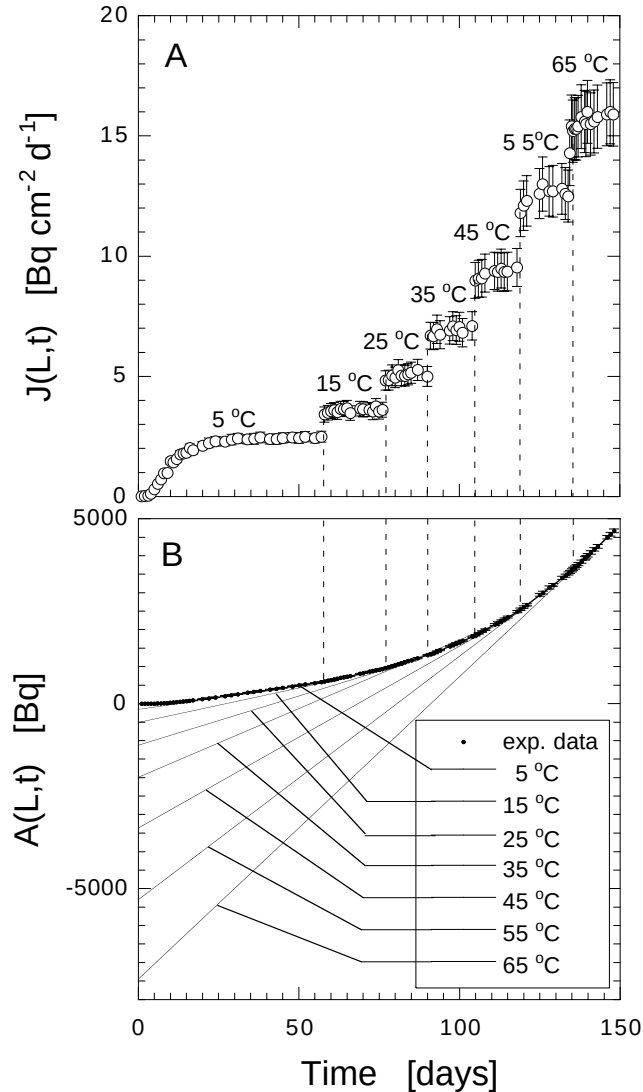


Figure 31: A) Flux of HTO as a function of time and temperature for Opalinus Clay from Benken (BC1) for six different temperatures; B) Total diffused activity of HTO as a function of time. The data were accumulated using the continuous method (C).

As can be seen, the temperature dependence is of an Arrhenius type (Eqn. 69). The average activation energy of the diffusion of HTO in Opalinus Clay (Benken: $E_A = 22 \text{ kJ} \cdot \text{mol}^{-1}$ and MT: $E_A = 20 \text{ kJ} \cdot \text{mol}^{-1}$) is larger than that of the diffusion in bulk liquid water

($E_A = 18 \text{ kJ}\cdot\text{mol}^{-1}$). The larger value indicates that the properties of the confined water differ from those of bulk water. This is due to the interaction of water with clay surfaces and/or with the loading cations (hydration water), which is different from the hydrogen bonds between water molecules.

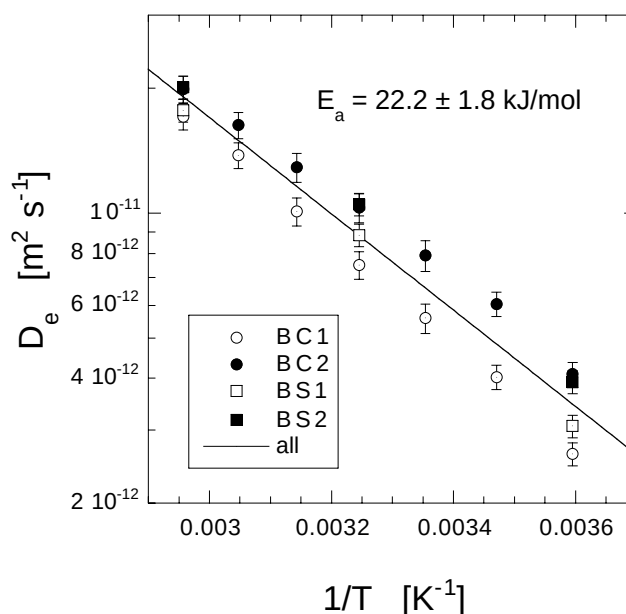


Figure 32: Effective diffusion coefficients as a function of the reciprocal temperature (Arrhenius plot) for the diffusion of HTO in Benken OPA.

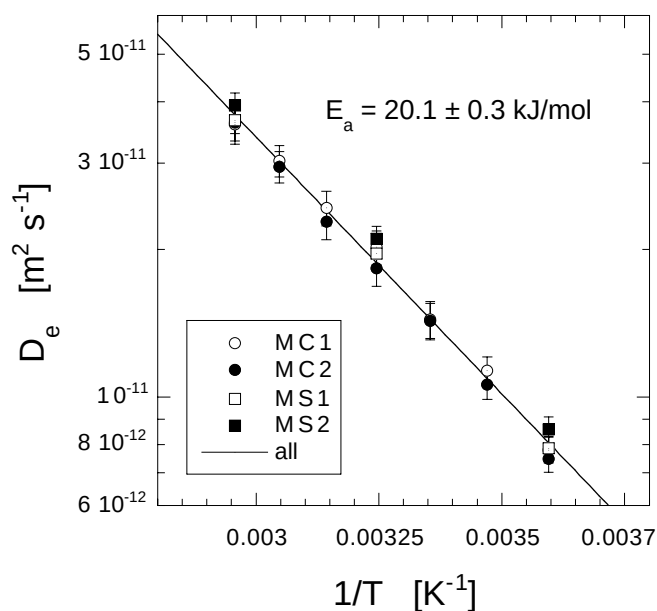


Figure 33: Effective diffusion coefficients as a function of the reciprocal temperature (Arrhenius plot) for the diffusion of HTO in Mont Terri OPA.

Table 15.1: Effective diffusion coefficients for HTO in Opalinus Clay from Benken for different temperatures as measured with two different methods: the step method (S) and the continuous method (C).

T	BS1	BS2	BC1	BC2
5	$(3.07 \pm 0.19) \times 10^{-12}$	$(3.92 \pm 0.25) \times 10^{-12}$	$(2.63 \pm 0.17) \times 10^{-12}$	$(4.10 \pm 0.27) \times 10^{-12}$
15	nm	nm	$(4.03 \pm 0.27) \times 10^{-12}$	$(6.04 \pm 0.41) \times 10^{-12}$
25	nm	nm	$(5.59 \pm 0.46) \times 10^{-12}$	$(7.91 \pm 0.67) \times 10^{-12}$
35	$(8.85 \pm 0.55) \times 10^{-12}$	$(1.05 \pm 0.65) \times 10^{-11}$	$(7.50 \pm 0.58) \times 10^{-12}$	$(1.03 \pm 0.08) \times 10^{-11}$
45	nm	nm	$(1.01 \pm 0.08) \times 10^{-11}$	$(1.29 \pm 0.10) \times 10^{-11}$
55	nm	nm	$(1.38 \pm 0.10) \times 10^{-11}$	$(1.63 \pm 0.12) \times 10^{-11}$
65	$(1.77 \pm 0.11) \times 10^{-11}$	$(2.01 \pm 0.13) \times 10^{-11}$	$(1.71 \pm 0.12) \times 10^{-11}$	$(1.99 \pm 0.15) \times 10^{-11}$
E_A	22.9 ± 1.5 kJ/mol	21.4 ± 1.3 kJ/mol	24.3 ± 0.4 kJ/mol	20.2 ± 0.5 kJ/mol

nm: not measured

Table 15.2: Diffusion accessible porosity values for HTO in Opalinus Clay from Benken for different temperatures measured with two different methods: the step method (S) and the continuous method (C).

T	BS1	BS2	BC1	BC2
5	0.13 ± 0.01	0.15 ± 0.02	0.12 ± 0.01	0.15 ± 0.02
15	nm	nm	-	-
25	nm	nm	-	-
35	0.14 ± 0.01	0.15 ± 0.02	-	-
45	nm	nm	-	-
55	nm	nm	-	-
65	0.15 ± 0.02	0.15 ± 0.02	-	-
	0.14 ± 0.01	0.15 ± 0.02	0.12 ± 0.01	0.15 ± 0.02

nm: not measured

Table 16.1: Effective diffusion coefficients for HTO in Opalinus Clay from Mont Terri at different temperatures measured with two different methods: the step method (S) and the continuous method (C).

T	MS1	MS2	MC1	MC2
5	$(7.85 \pm 0.47) \times 10^{-12}$	$(8.58 \pm 0.52) \times 10^{-12}$	$(7.80 \pm 0.47) \times 10^{-12}$	$(7.48 \pm 0.45) \times 10^{-12}$
15	nm	nm	$(1.13 \pm 0.08) \times 10^{-11}$	$(1.06 \pm 0.07) \times 10^{-11}$
25	nm	nm	$(1.44 \pm 0.12) \times 10^{-11}$	$(1.43 \pm 0.12) \times 10^{-11}$
35	$(1.96 \pm 0.12) \times 10^{-11}$	$(2.10 \pm 0.13) \times 10^{-11}$	$(2.02 \pm 0.16) \times 10^{-11}$	$(1.83 \pm 0.15) \times 10^{-11}$
45	nm	nm	$(2.43 \pm 0.20) \times 10^{-11}$	$(2.28 \pm 0.19) \times 10^{-11}$
55	nm	nm	$(3.03 \pm 0.22) \times 10^{-11}$	$(2.95 \pm 0.22) \times 10^{-11}$
65	$(3.67 \pm 0.22) \times 10^{-11}$	$(3.93 \pm 0.24) \times 10^{-11}$	$(3.66 \pm 0.27) \times 10^{-11}$	$(3.63 \pm 0.35) \times 10^{-11}$
E_a	20.2 ± 1.0 kJ/mol	19.9 ± 0.9 kJ/mol	19.9 ± 0.6 kJ/mol	20.3 ± 0.3 kJ/mol

nm: not measured

Table 16.2: Diffusion accessible porosity values for HTO in Opalinus Clay from Mont Terri at different temperatures measured with two different methods: the step method (S) and the continuous method (C).

T	MS1	MS2	MC1	MC2
5	0.10 ± 0.01	0.17 ± 0.02	0.09 ± 0.01	0.13 ± 0.01
15	nm	nm	-	-
25	nm	nm	-	-
35	0.12 ± 0.02	0.20 ± 0.02	-	-
45	nm	nm	-	-
55	nm	nm	-	-
65	0.12 ± 0.01	0.19 ± 0.02	-	-
	0.12 ± 0.02	0.19 ± 0.02	0.09 ± 0.01	0.13 ± 0.01

nm: not measured

When the diffusion coefficient at a given temperature T_1 is known ($D_e^{T_1}$), its value at a temperature T_2 can be calculated by the following equation:

$$D_e^{T_2} = D_e^{T_1} \cdot e^{\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)} \quad (70)$$

For a temperature of 40 °C, the diffusion coefficients of HTO in Opalinus Clay from Benken will be higher by a factor of 1.7 – 1.8 as compared with 22 °C. This is shown in Figure 34. It is clear that the use of an activation energy of 18 kJ·mol⁻¹ results in an underestimation of the D_e at higher temperatures.

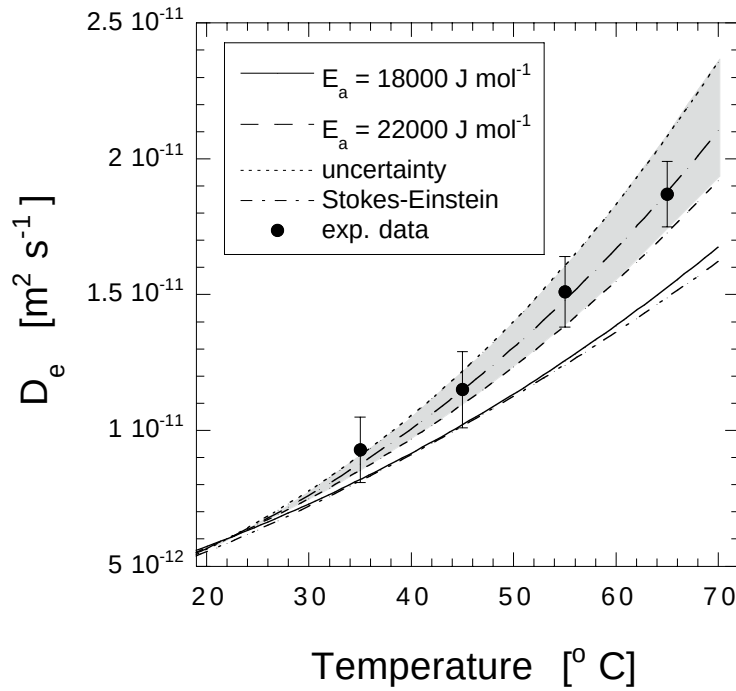


Figure 34: Experimental and calculated temperature dependence of the effective diffusion coefficient for the diffusion of HTO in OPA from Benken.

Instead of the Arrhenius equation, also the Einstein-Stokes relationship can be used for estimating the effective diffusion coefficients at other temperatures:

$$D_i = \frac{k \cdot T}{6\pi \cdot r_{si} \cdot \eta} \quad (71)$$

with:

k = Boltzmann constant [kg·m²·s⁻²·K⁻¹]

r_{si} = hydrodynamic radius of particle I [m]

T = temperature [K]

η = viscosity of the medium [kg·m⁻¹·s⁻¹]

When the diffusion coefficient at a given temperature T_0 is known, the diffusion coefficient at other temperatures T_i can be calculated using the Stokes-Einstein equation. For the diffusion in OPA Eqn. 71 can be rearranged:

$$\left(\frac{D_e^0 \cdot \eta_0}{T_0} \right) = \left(\frac{D_e^i \cdot \eta_i}{T_i} \right) \quad (72)$$

or

$$D_e^i = D_e^0 \cdot \frac{\eta_0}{\eta_i} \cdot \frac{T_i}{T_0} \quad (73)$$

Eqn.73 was applied to the diffusion of HTO in OPA. A reference temperature of 22 °C was chosen. At this temperature $D_e = 6.0 \times 10^{-12} \text{ m}^2 \cdot \text{s}^{-1}$ and $\eta = 1.002 \text{ cp}$. The values for η_i at different temperatures were calculated by (CRC 1979, p. F-51):

$$\log_{10} \frac{\eta_i}{\eta_{293}} = \frac{1.3272 \cdot (293 - T_i) - 0.001053 \cdot (T_i - 293)^2}{(T_i - 168)} \quad (74)$$

The results in Figure 34 show that Eqn. 73 does not represent the temperature dependence of D_e observed in the experiments. This, however, does not mean that the Stokes-Einstein relationship is not applicable to diffusion in OPA. It again indicates that the confined water in OPA has properties different from bulk liquid water. Moreover, the applicability of the Stokes-Einstein relationship can only be validated when the viscosity of the confined water in OPA and its temperature dependence is known.

5 HETEROGENEITY OF OPALINUS CLAY FROM THE DEEP BOREHOLE IN BENKEN WITH RESPECT TO DIFFUSION

5.1 Introduction

Opalinus Clay is not homogeneous along its depth profile (depth: -539 to - 652 m). Whether the diffusion characteristics of OPA vary along the depth is unknown and therefore diffusion on samples originating from different depths was investigated. Both the diffusion of HTO parallel and perpendicular to the bedding was measured.

5.2 Materials and Methods

The methodology for measuring through-diffusion is extensively described in previous chapters (2 and 3). The samples used were Opalinus Clay samples from the deep borehole in Benken and originate from the following depths (Table 17):

Table 17: Origin of the OPA samples used to explore the heterogeneity of OPA with respect to diffusion characteristics.

Sample	Depth [m]	Diffusion mode
BE-S0	-563.74 to -563.76	perpendicular
BE-S1	-589.48 to -589.50	perpendicular
BE-S2	-651.03 to -651.19	perpendicular
BE-R3	-563.61 to -563.68	parallel
BE-R4	-589.40 to -584.48	parallel

5.3 Results and discussion

Figures A11-1 to A11-5 show the results of the through-diffusion of HTO measured on the samples from different depths. The parameters extracted from the diffusion measurements are summarised in Table 18 together with results obtained in earlier experiments on samples originating from a depth of - 636 m.

It is clear from Table 18 that the effective diffusion coefficients of HTO for diffusion perpendicular to the bedding decrease with increasing depth. However, the difference between samples from the top and the bottom of the OPA layer is not larger than a factor of 1.5. For diffusion parallel to the bedding, no significant difference between the top and bottom of the OPA layer was found. It can further be observed that the rock capacity

factor (or porosity) has similar values along the whole depth profile. Based on these first results on relatively few samples, it can be concluded that - with respect to diffusion properties - the Opalinus Clay layer in the Zürcher Weinland is very homogeneous.

Table 18: Effective diffusion coefficients and rock capacity factors for the diffusion of HTO through Opalinus Clay from Benken at different depths.

Depth [m]	D_e (perp) [m ² s ⁻¹]	α (perp)	D_e (parallel) [m ² s ⁻¹]	α (parallel)
-564	$(8.78 \pm 0.54) \times 10^{-12}$	0.16 ± 0.02	$(3.31 \pm 0.23) \times 10^{-11}$	0.17 ± 0.02
-589	$(7.26 \pm 0.44) \times 10^{-12}$	0.16 ± 0.02	$(3.31 \pm 0.23) \times 10^{-11}$	0.17 ± 0.02
-636	$(5.42 \pm 0.34) \times 10^{-12}$	0.14 ± 0.02	$(3.10 \pm 0.25) \times 10^{-11}$	0.15 ± 0.02
-651	$(5.83 \pm 0.37) \times 10^{-12}$	0.17 ± 0.02	n.m.	n.m.

n.m.: not measured because not enough material available

The results of the out-diffusion experiments for diffusion perpendicular to the bedding are depicted in Figure A12-1.1 – A12-1.3. The expected curves in these figures were calculated using Eqn. 53 and 54 and the parameters summarised in Table 18. As can be seen, the predicted and experimental data are in good agreement. Through-diffusion and out-diffusion occur by the same diffusion process, characterised by the same value for the diffusion coefficient.

6 CONCLUSIONS

The through- and out-diffusion of HTO, $^{36}\text{Cl}^-$ and $^{125}\text{I}^-$ and $^{22}\text{Na}^+$ were studied in OPA samples from Mont Terri and Benken as a function of confining pressure. Only a small effect of pressure on diffusion could be observed, i.e. a decrease of the effective diffusion coefficients with increasing confining pressure. Porosity values were hardly affected at all.

The diffusion of anions is slower than the diffusion of HTO. Also the diffusion-accessible porosity is smaller for anions than for HTO. This can be explained by an anion exclusion effect. The anion exclusion effect is larger in Benken OPA than in Mont Terri OPA because the ionic strength of the pore water is larger in the latter case, resulting in a stronger shielding of the surface charges. In the case of $^{22}\text{Na}^+$ the values for the effective diffusion coefficient are equal or even larger than those for HTO. In the case of Mont Terri OPA, the values for the effective diffusion coefficients were similar to those of HTO. In the case of Benken OPA, the effective diffusion coefficients of $^{22}\text{Na}^+$ were larger by a factor of 2.2. This phenomenon cannot be explained so far, but is comparable to experimental data from Andra on Callovo-Oxfordian claystones from the Meuse/Haute Sarne site.

Diffusion of all radiotracers was slower in OPA from Benken than in OPA from Mont Terri, indicating that the pore diameter of OPA originating from Benken is smaller than that of Mont Terri OPA. In the case of anions, the difference is larger than for HTO. This is due to the effect of anion exclusion superimposed on the pore diameter effect. In the case of $^{22}\text{Na}^+$ the effect is smaller than for HTO.

Whereas HTO and $^{36}\text{Cl}^-$ can be considered as non-sorbing tracers, $^{125}\text{I}^-$ and $^{22}\text{Na}^+$ clearly show an interaction with OPA. The mechanism of the interaction (sorption) between $^{125}\text{I}^-$ and OPA is so far unclear and needs further investigation. In the case of $^{22}\text{Na}^+$ the sorption can be explained by an ion exchange reaction.

Out-diffusion data of the given radionuclides were in good agreement with most of the through-diffusion data in the case of HTO and $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$. Both processes could be described by a single porous medium model with the same diffusion coefficients and rock capacity factors. In the case of $^{125}\text{I}^-$, the out-diffusion could not be predicted using a single porous medium model with the parameter values extracted from through-diffusion experiments. This observation suggests that additional (still unknown) processes (sorption/desorption with slow kinetics?) are taking place.

The effective diffusion coefficients for diffusion parallel to the bedding are a factor of 4 - 6 larger than the effective diffusion coefficients for diffusion perpendicular to the bedding (Table 19). This is due to the anisotropic tortuosity of the Opalinus Clay. The rock capacity factor, however, is independent of the diffusion direction. The anisotropic

diffusion is more pronounced for samples originating from the Benken deep borehole compared to those from Mont Terri, indicating that the clay platelets might be more preferentially orientated in the Benken samples.

Table 19: Overview of the “best - estimate” values for the effective diffusion coefficients and rock capacity factors for the diffusion of HTO, $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$ in Opalinus clay parallel and perpendicular to the bedding.

Sample	Isotope	D_e (parallel) ($\text{m}^2\cdot\text{s}^{-1}$)	α (parallel)	D_e (perpend.) ($\text{m}^2\cdot\text{s}^{-1}$)	α (perpend.)
BE/1	HTO	$(3.1\pm 0.3)\times 10^{-11}$	0.15 ± 0.02	$(5.4\pm 0.4)\times 10^{-12}$	0.14 ± 0.02
BE/2	HTO	$(3.2\pm 0.2)\times 10^{-11}$	0.13 ± 0.02	$(5.4\pm 0.4)\times 10^{-12}$	0.13 ± 0.02
MT/1	HTO	$(5.4\pm 0.4)\times 10^{-11}$	0.15 ± 0.03	$(1.4\pm 0.1)\times 10^{-11}$	0.17 ± 0.02
MT/2	HTO	$(5.4\pm 0.4)\times 10^{-11}$	0.17 ± 0.02	$(1.4\pm 0.1)\times 10^{-11}$	0.14 ± 0.02
¹ MT	HDO	$(5.1 - 8.4)\times 10^{-11}$	$0.17 - 0.24$	n.m	n.m
¹ MT	H ₂ O ¹⁸	$(5.6 - 10)\times 10^{-11}$	$0.19 - 0.25$	n.m	n.m
BE/1	$^{36}\text{Cl}^-$	$(3.4\pm 0.3)\times 10^{-12}$	0.05 ± 0.01	$(6.7\pm 0.4)\times 10^{-13}$	0.04 ± 0.01
MT/1	$^{36}\text{Cl}^-$	$(1.6\pm 0.1)\times 10^{-11}$	0.08 ± 0.01	$(4.1\pm 0.3)\times 10^{-12}$	0.08 ± 0.01
¹ MT	Cl^-	$(1.8 - 5.6)\times 10^{-11}$	$0.10 - 0.15$	n.m	n.m
¹ MT	Br^-	$(1.7 - 4.5)\times 10^{-11}$	$0.10 - 0.15$	n.m	n.m
BE/1	$^{22}\text{Na}^+$	$(7.8\pm 0.6)\times 10^{-11}$	0.89 ± 0.08	$(1.3\pm 0.1)\times 10^{-11}$	1.11 ± 0.11
MT/1	$^{22}\text{Na}^+$	$(7.2\pm 0.5)\times 10^{-11}$	0.62 ± 0.05	$(1.5\pm 0.1)\times 10^{-11}$	0.48 ± 0.04

n.m: not measured

¹data from VAN DER KAMP & VAN STEMPTVOORT (1998); no confining pressure applied

MT: Mont Terri (applied confining pressure: 7MPa); BE: Benken (applied confining pressure: 14MPa)

The temperature dependence of the diffusion of HTO in Opalinus Clay was studied using two different techniques, i.e. a step method and a continuous method. Both techniques gave similar results. The dependence of the diffusion coefficient on temperature was of an Arrhenius type. The activation energy of the self-diffusion of HTO in OPA is $22 \text{ kJ}\cdot\text{mol}^{-1}$ (Benken) and $20 \text{ kJ}\cdot\text{mol}^{-1}$ (Mont Terri), and is higher than that of the self-diffusion of HTO in bulk liquid water ($18 \text{ kJ}\cdot\text{mol}^{-1}$). This indicates that the structure of the confined water in OPA is somewhat different from bulk liquid water. The diffusion coefficient of HTO perpendicular to bedding in OPA from Benken at the relevant

temperature of 30-40 °C is $\sim 10^{-11} \text{ m}^2 \cdot \text{s}^{-1}$. This is a factor of ~ 2 higher than the value measured so far at room temperature.

The effective diffusion coefficients perpendicular to bedding of HTO decrease with increasing depth of the Opalinus Clay in the Zürcher Weinland (deep borehole Benken). The difference between the top and bottom of the clay layer, however, is not larger than a factor of 1.5. The rock capacity factors show no significant variation with depth. These observations indicate that the Opalinus Clay layer is very homogeneous with respect to its diffusion properties.

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8 ACKNOWLEDGEMENTS

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APPENDIX 1: THROUGH-DIFFUSION OF HTO (series 1)

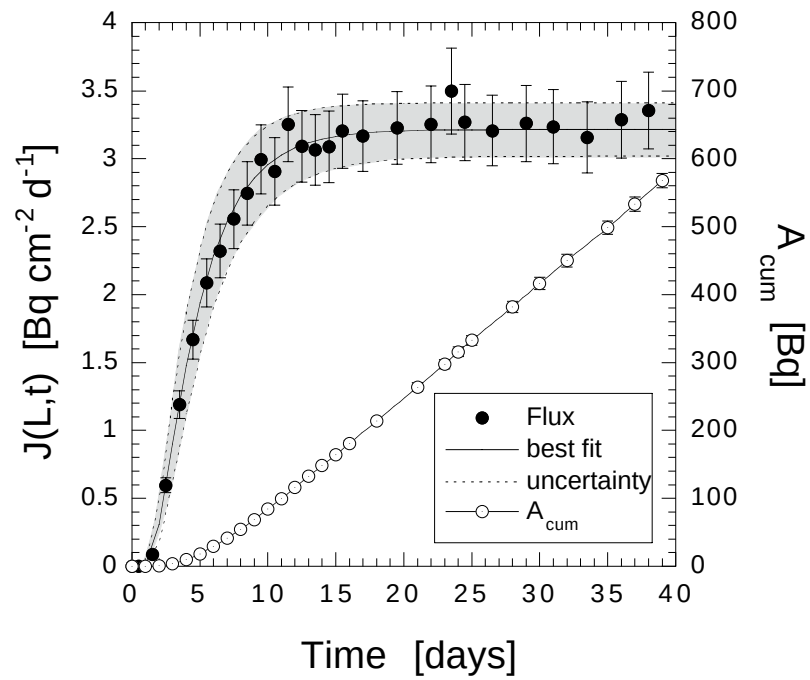


Figure A1-1.1: Flux ($J(L,t)$) and total diffused activity (A_{cum}) of HTO through Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

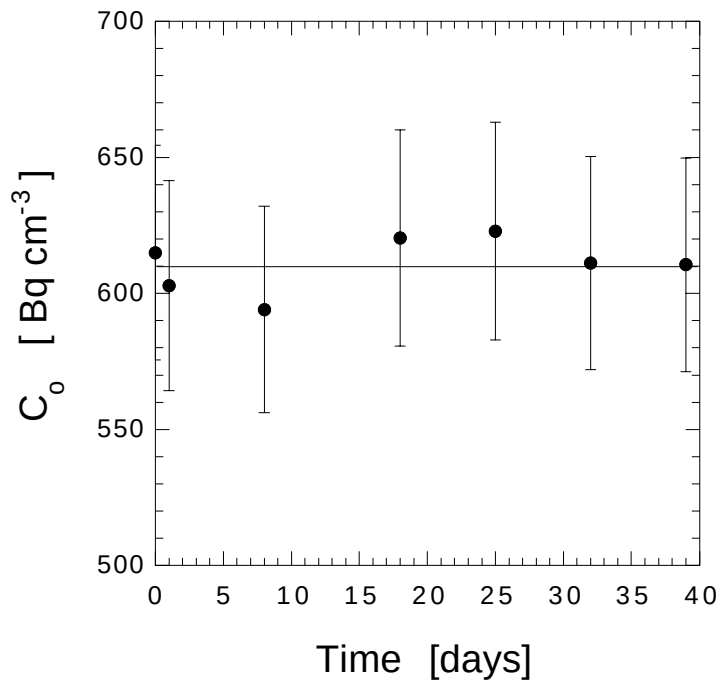


Figure A1-1.2: Concentration C_o of HTO in the high-concentration container as a function of time (BE-2a).

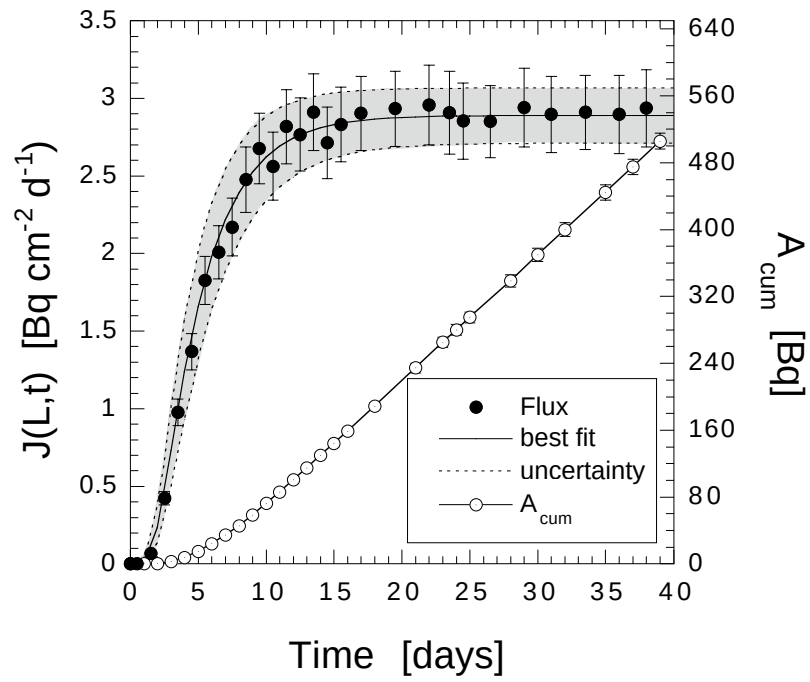


Figure A1-2.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

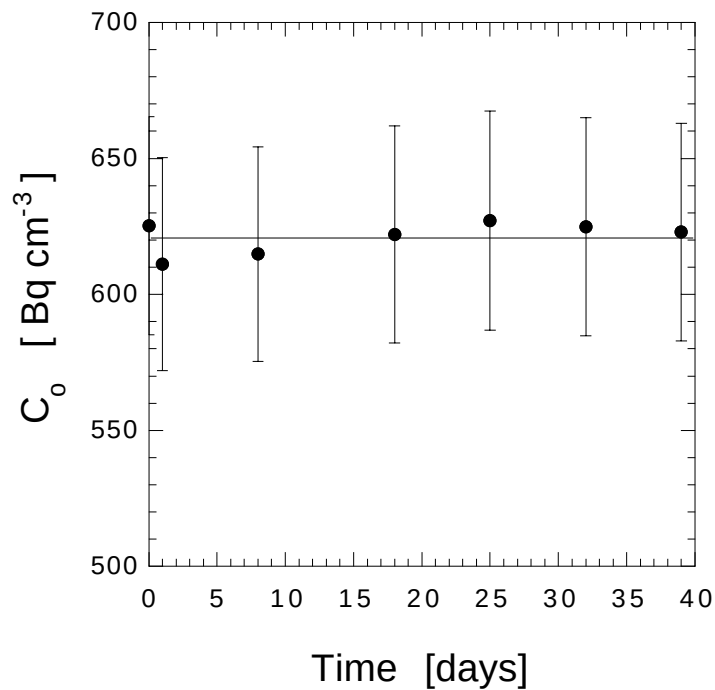


Figure A1-2.2: Concentration C_o of HTO in the high-concentration container as a function of time (BE-3a).

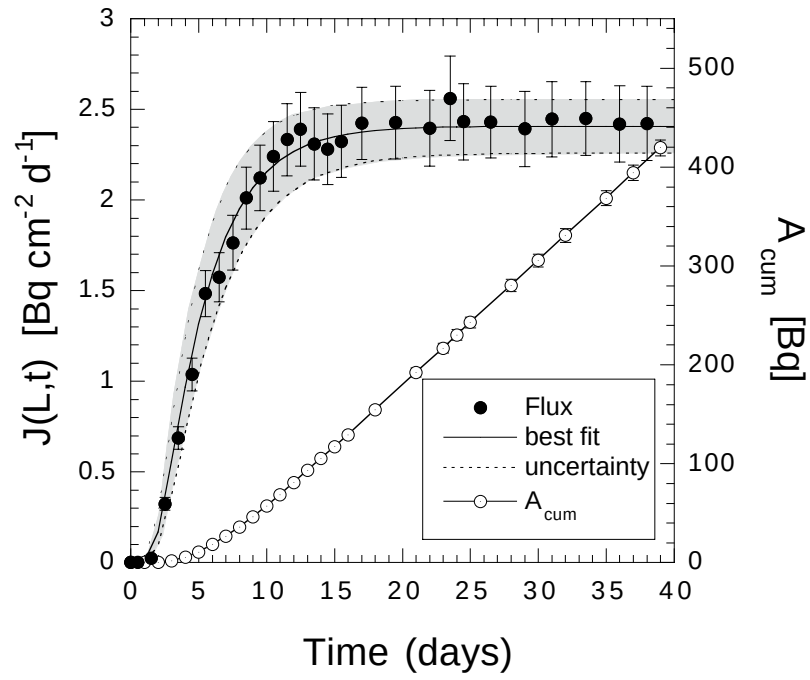


Figure A1-3.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

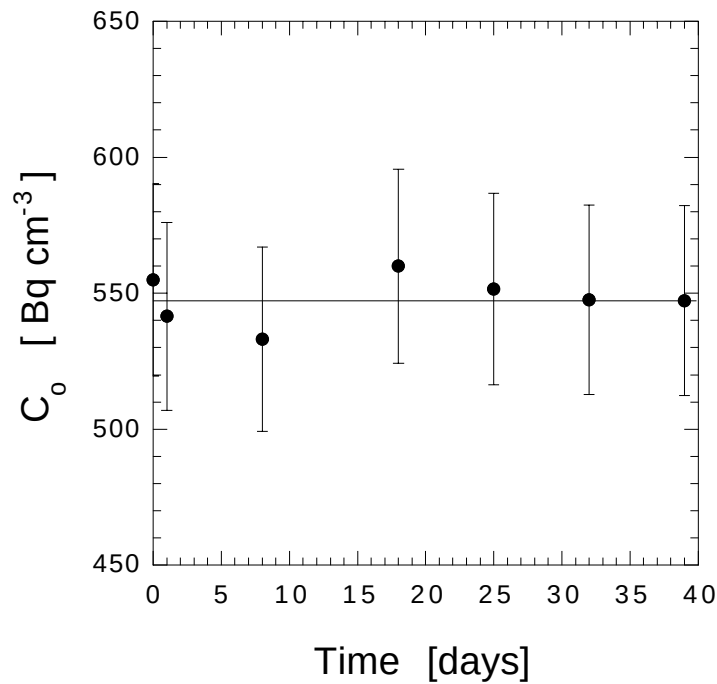


Figure A1-3.2: Concentration C_o of HTO in the high-concentration container as a function of time (BE-4a).

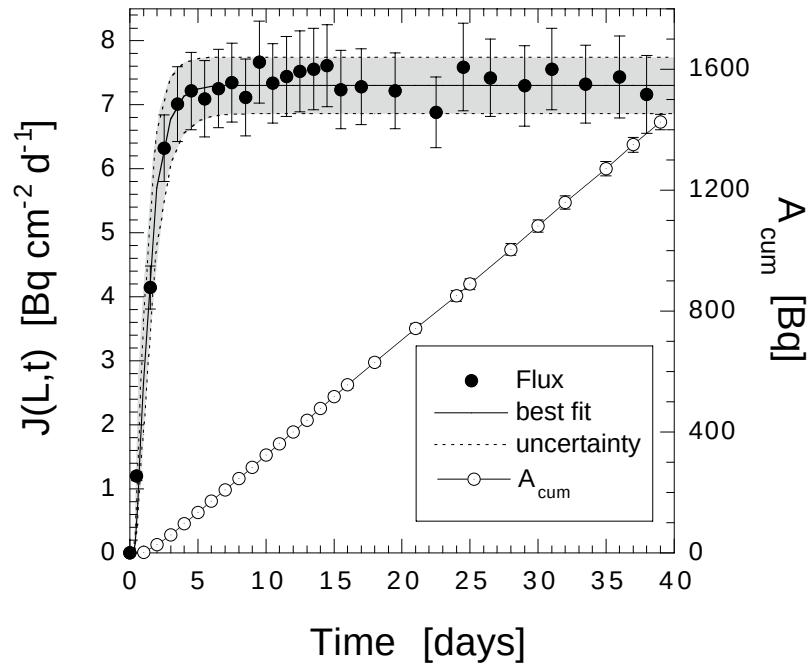


Figure A1-4.1: Flux and total diffused activity of HTO through Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

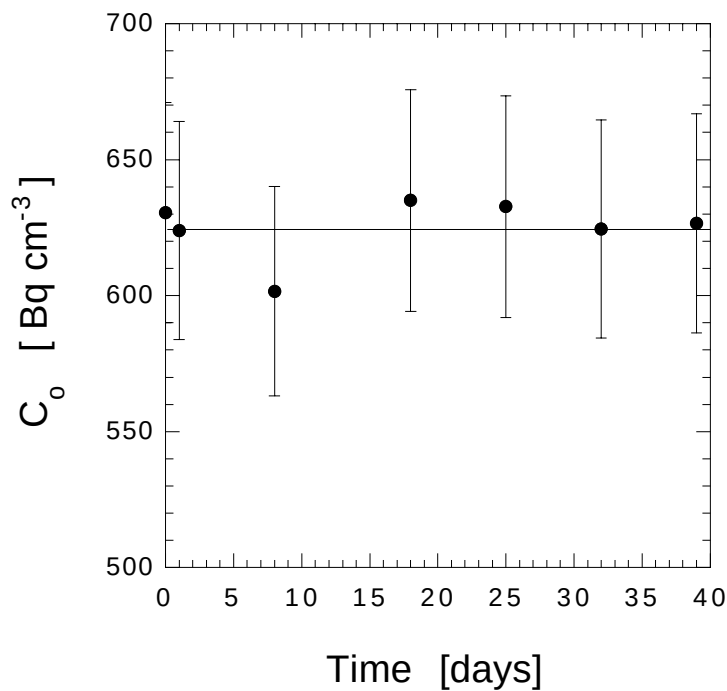


Figure A1-4.2: Concentration C_o of HTO in the high-concentration container as a function of time (MT-1a).

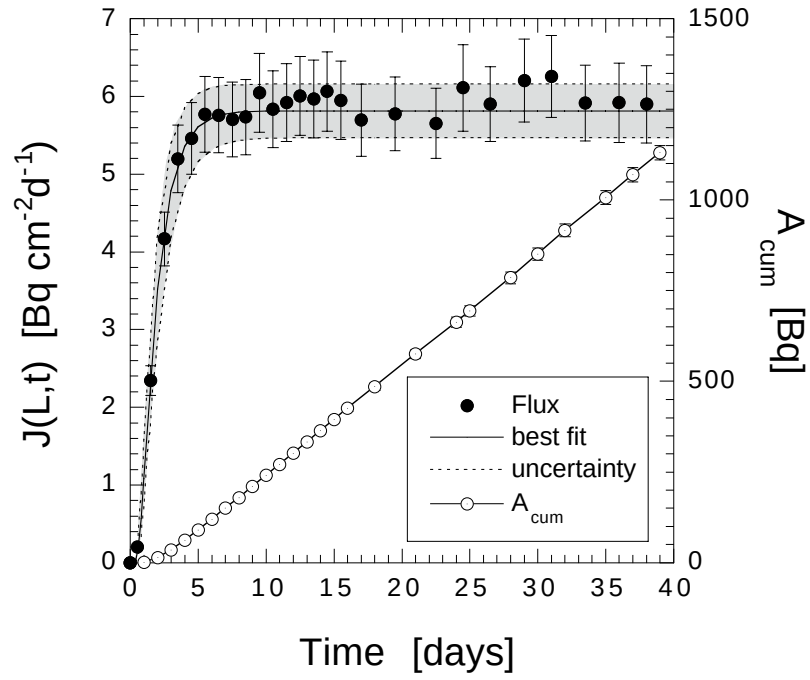


Figure A1-5.1: Flux and total diffused activity of HTO through Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

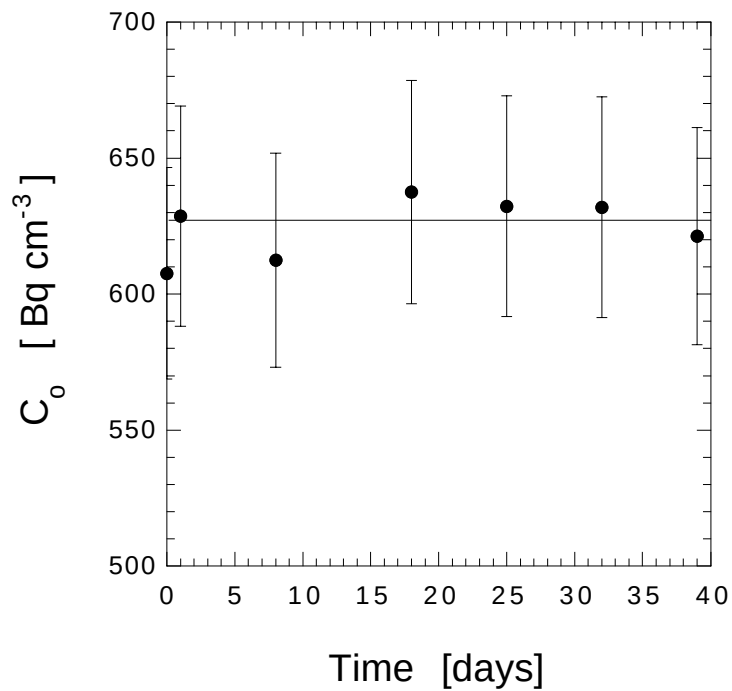


Figure A1-5.2: Concentration C_0 of HTO in the high-concentration container as a function of time (MT-3a).

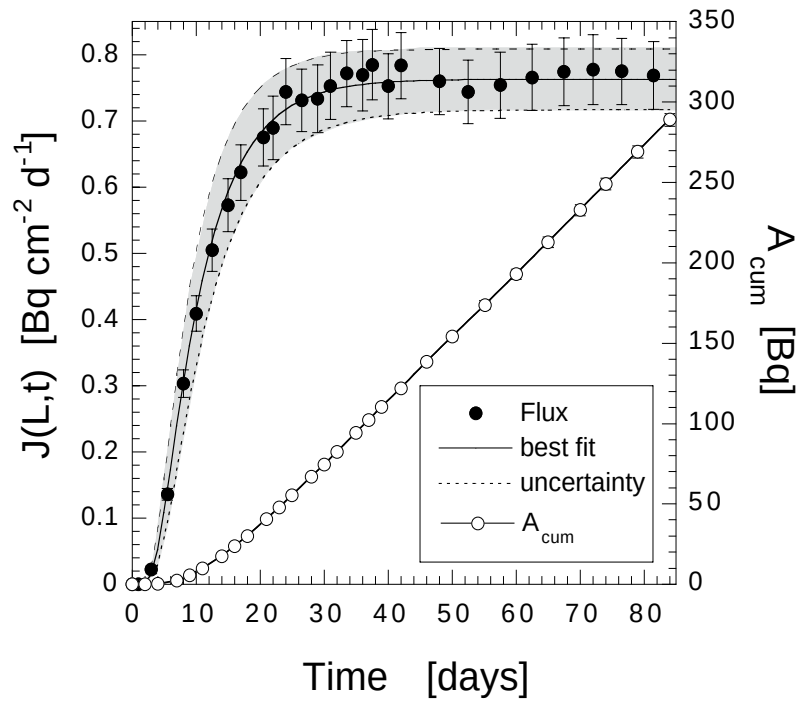
APPENDIX 2: THROUGH-DIFFUSION OF $^{36}\text{Cl}^-$ 

Figure A2-1.1: Flux and total diffused activity of $^{36}\text{Cl}^-$ through Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

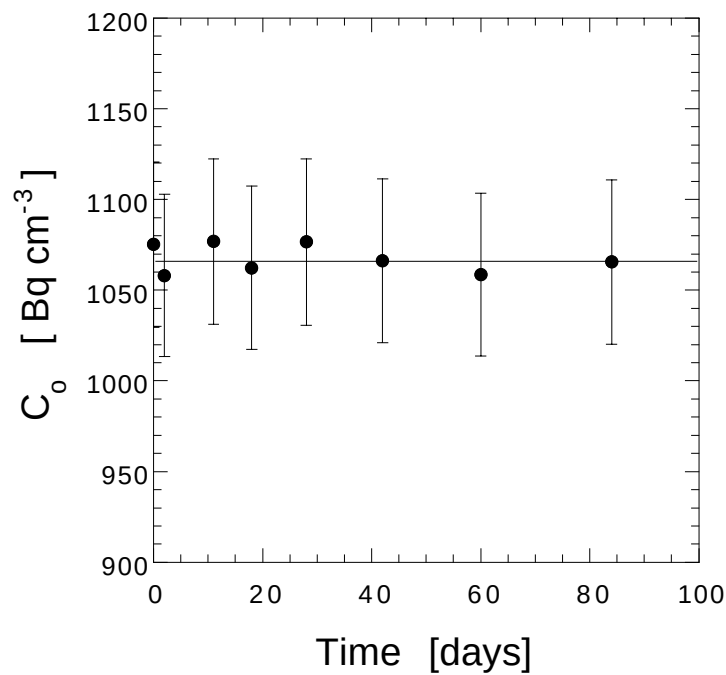


Figure A2-1.2: Concentration C_o of $^{36}\text{Cl}^-$ in the high-concentration container as a function of time (BE-2a).

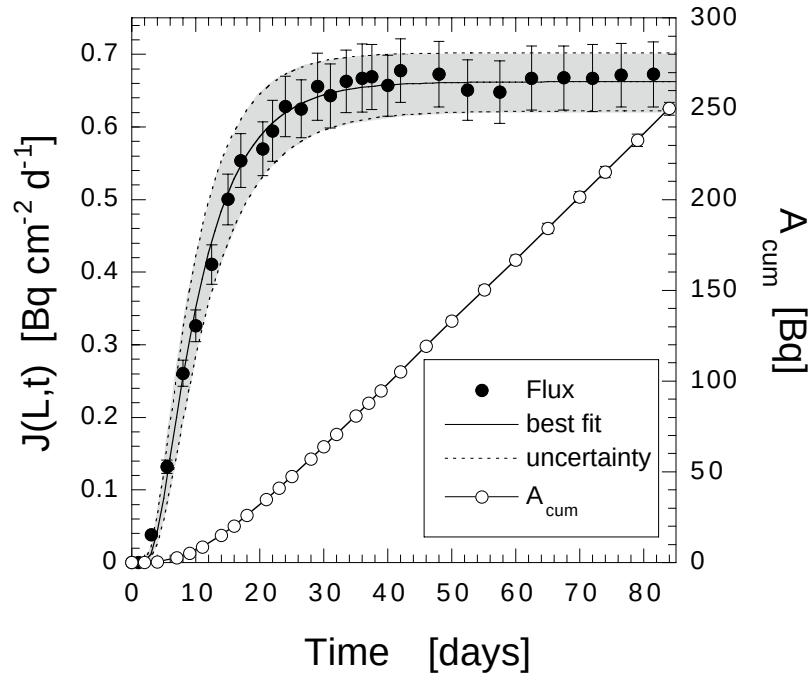


Figure A2-2.1: Flux and total diffused activity of ^{36}Cl through Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

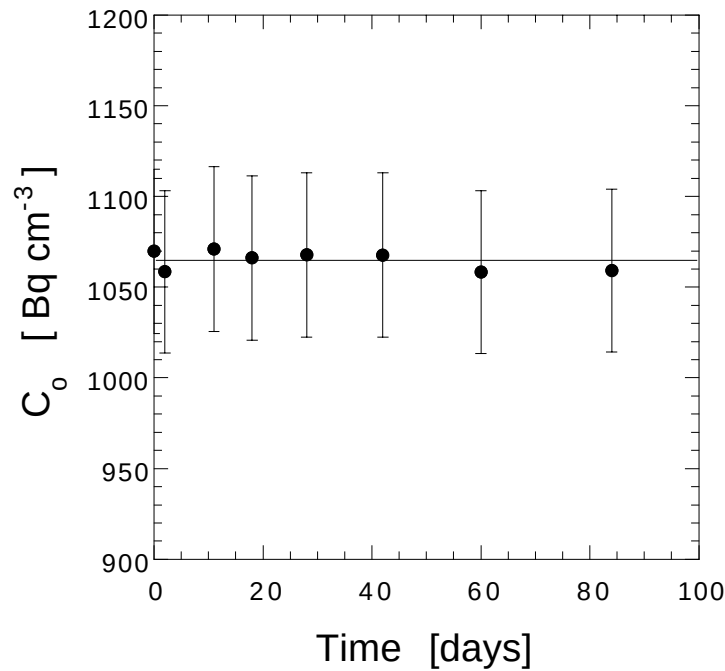


Figure A2-2.2: Concentration C_0 of ^{36}Cl in the high-concentration container as a function of time (BE-3a).

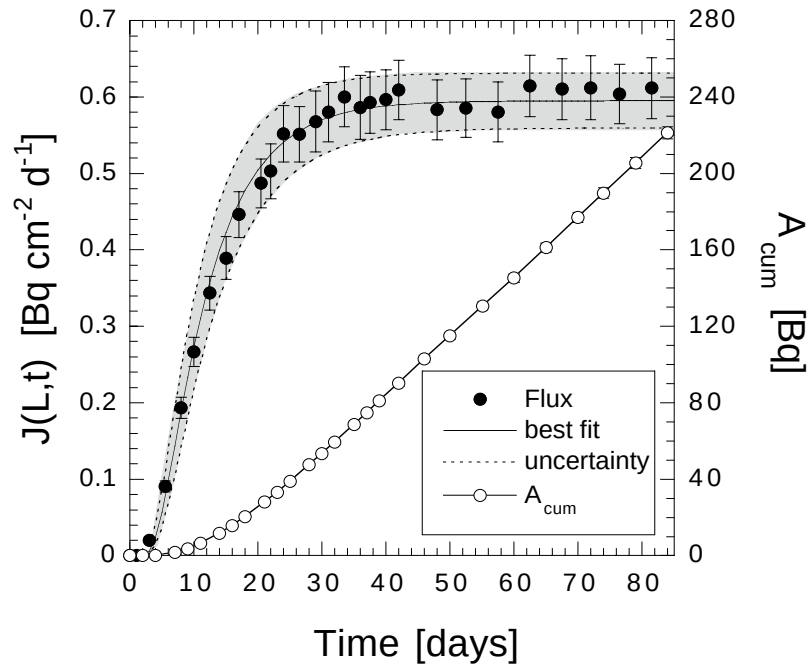


Figure A2-3.1: Flux and total diffused activity of $^{36}\text{Cl}^-$ through Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

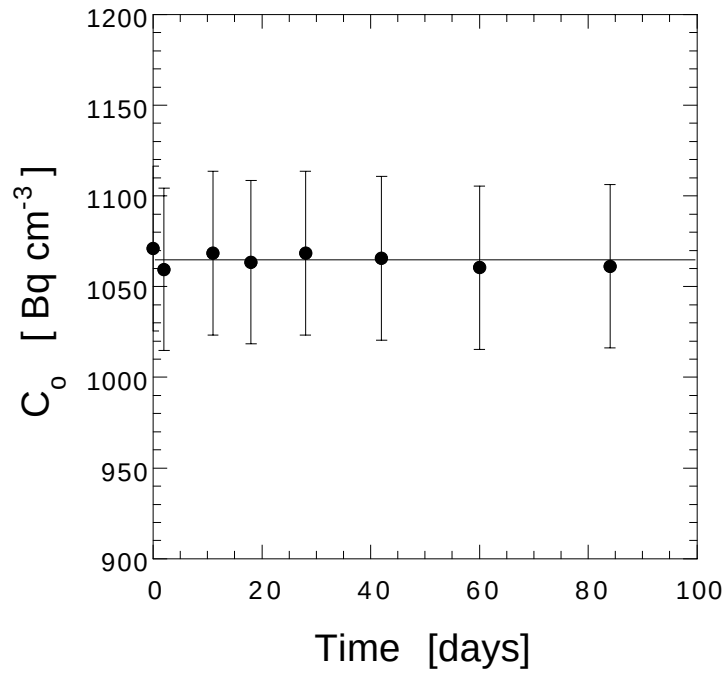


Figure A2-3.2: Concentration C_o of $^{36}\text{Cl}^-$ in the high-concentration container as a function of time (BE-4a).

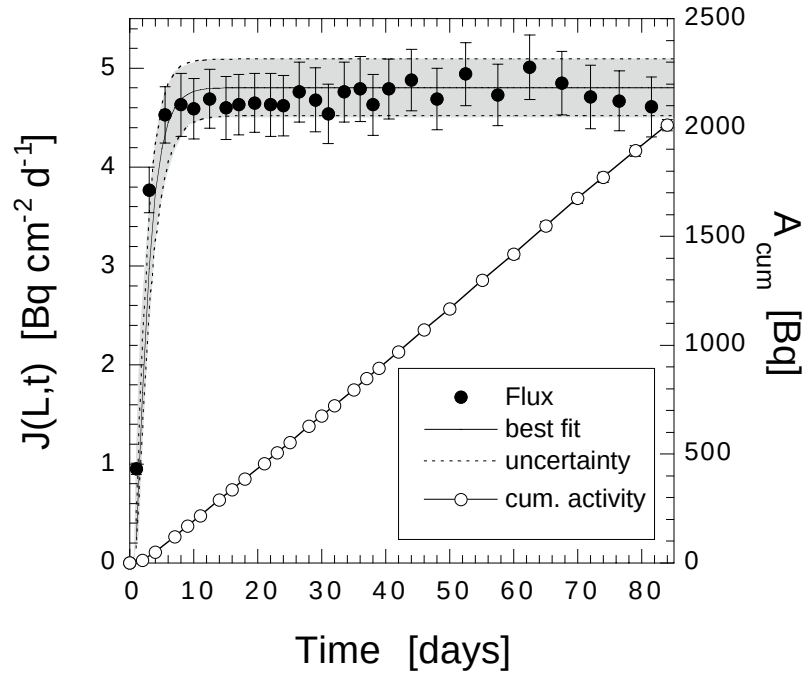


Figure A2-4.1: Flux and total diffused activity of $^{36}\text{Cl}^-$ through Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

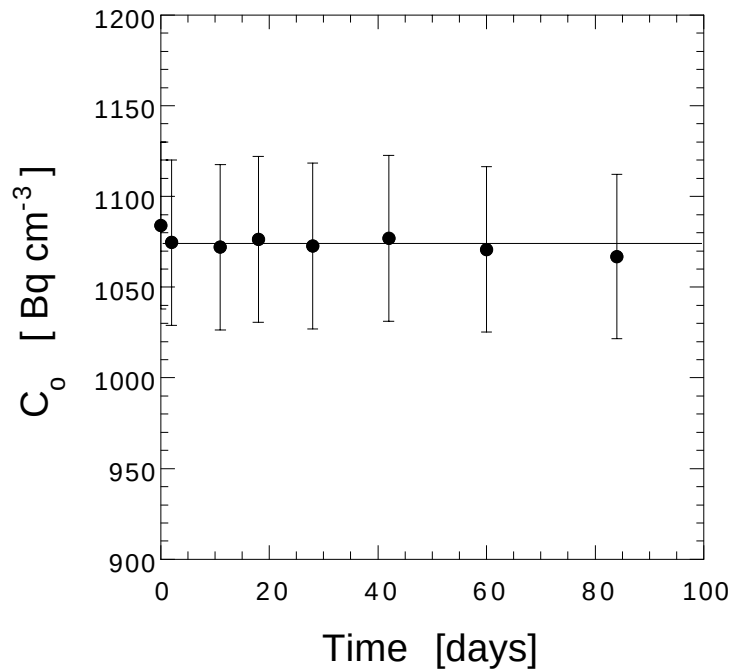


Figure A2-4.2: Concentration C_o of $^{36}\text{Cl}^-$ in the high-concentration container as a function of time (MT-1a).

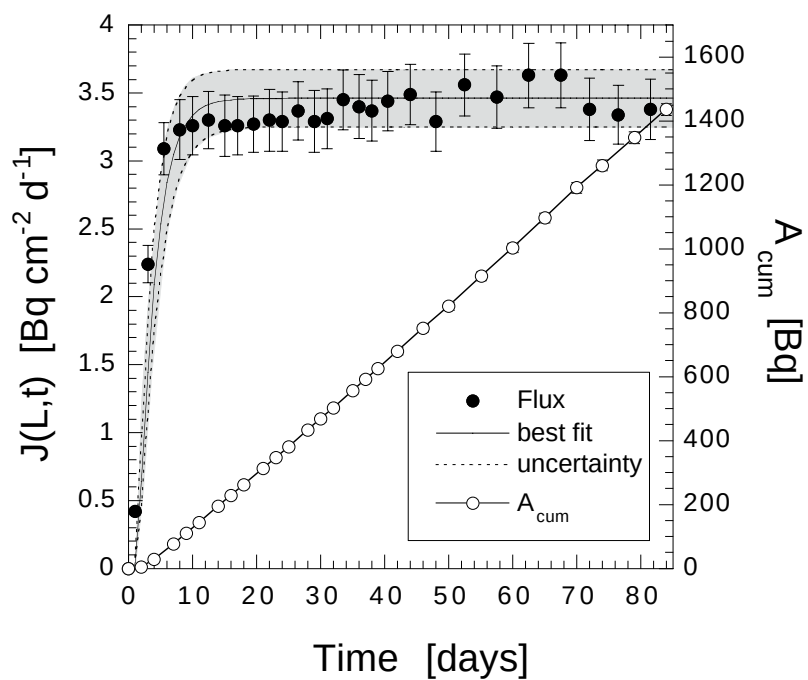


Figure A2-5.1: Flux and total diffused activity of $^{36}\text{Cl}^-$ through Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

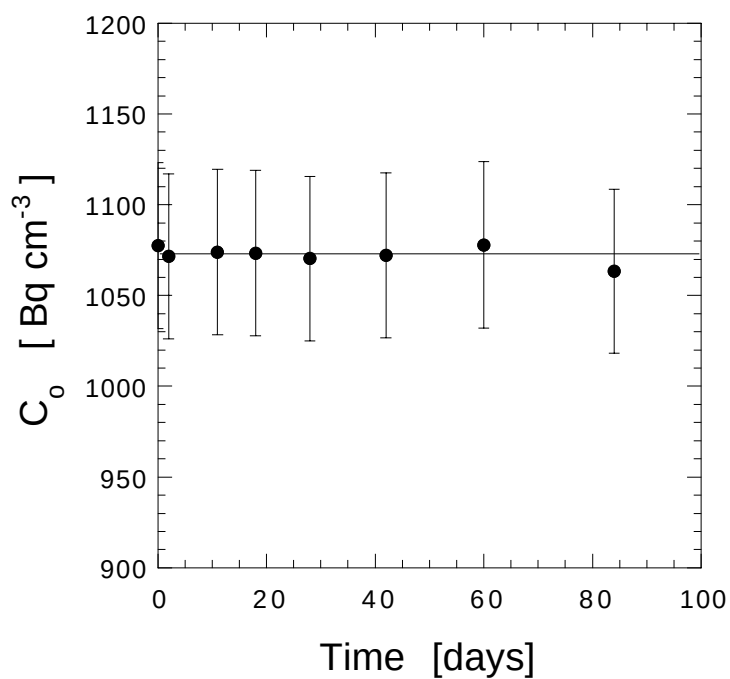


Figure A2-5.2: Concentration C_o of $^{36}\text{Cl}^-$ in the high-concentration container as a function of time (MT-3a).

APPENDIX 3: THROUGH-DIFFUSION OF $^{125}\text{I}^-$

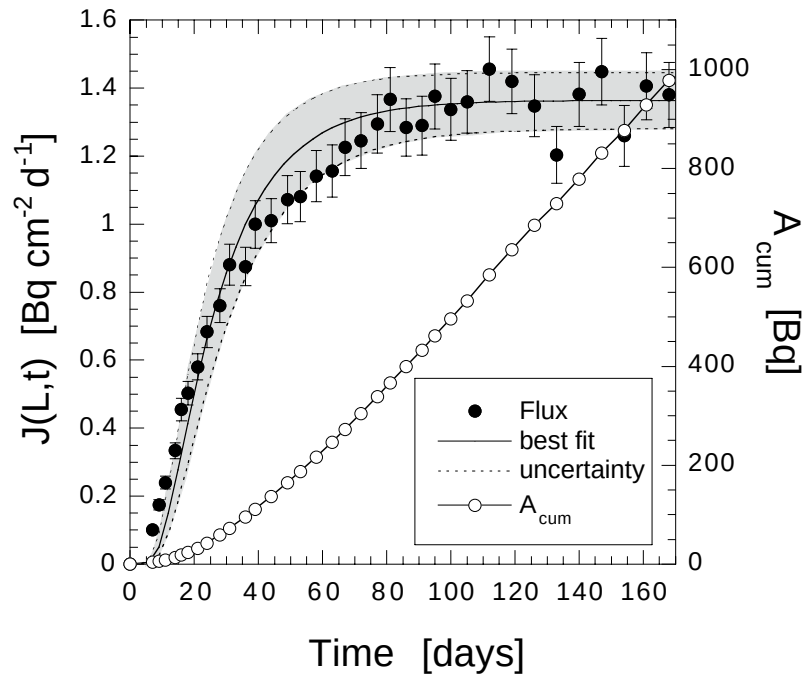


Figure A3-1.1: Flux and total diffused activity of $^{125}\text{I}^-$ through Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

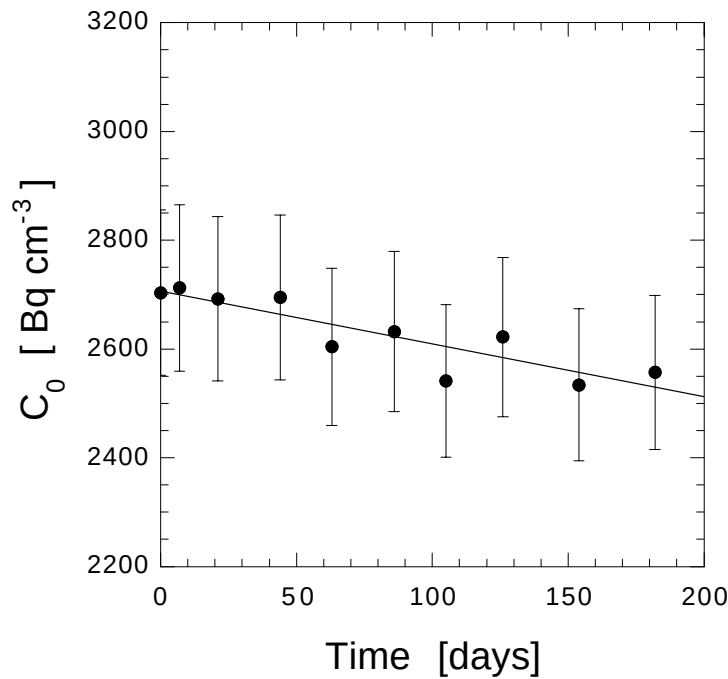


Figure A3-1.2: Concentration C_0 of $^{125}\text{I}^-$ in the high-concentration container as a function of time (BE-2a).

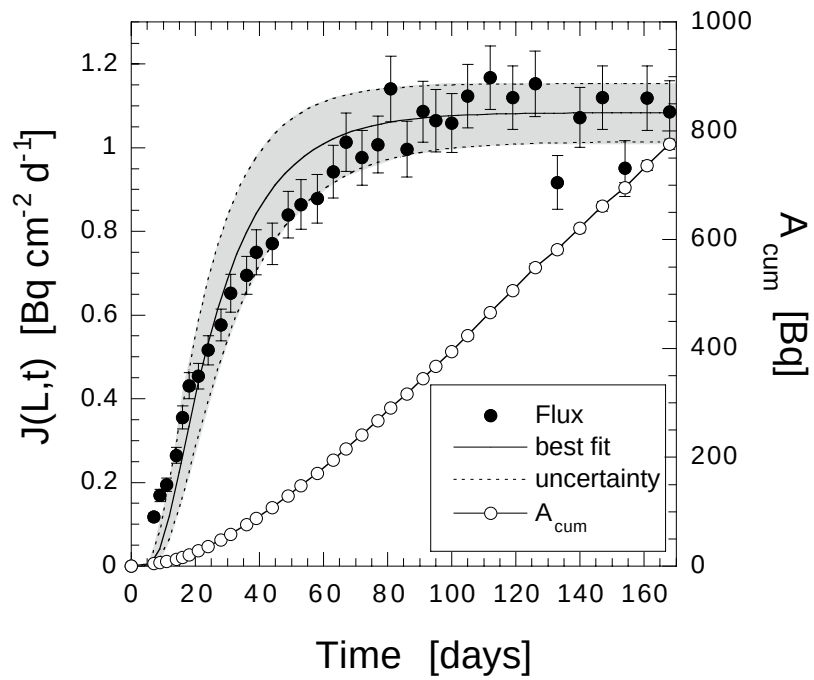


Figure A3-2.1: Flux and total diffused activity of ^{125}I through Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

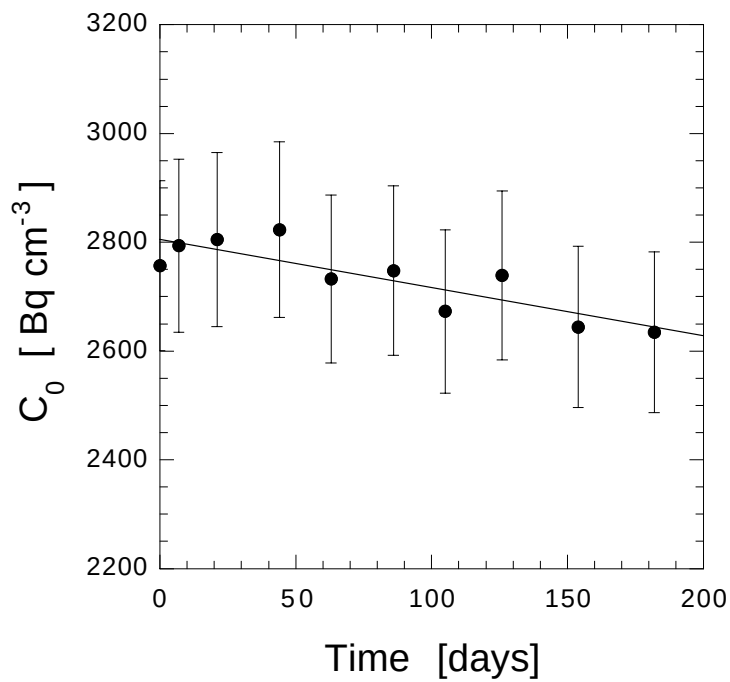


Figure A3-2.2: Concentration C_0 of ^{125}I in the high-concentration container as a function of time (BE-3a).

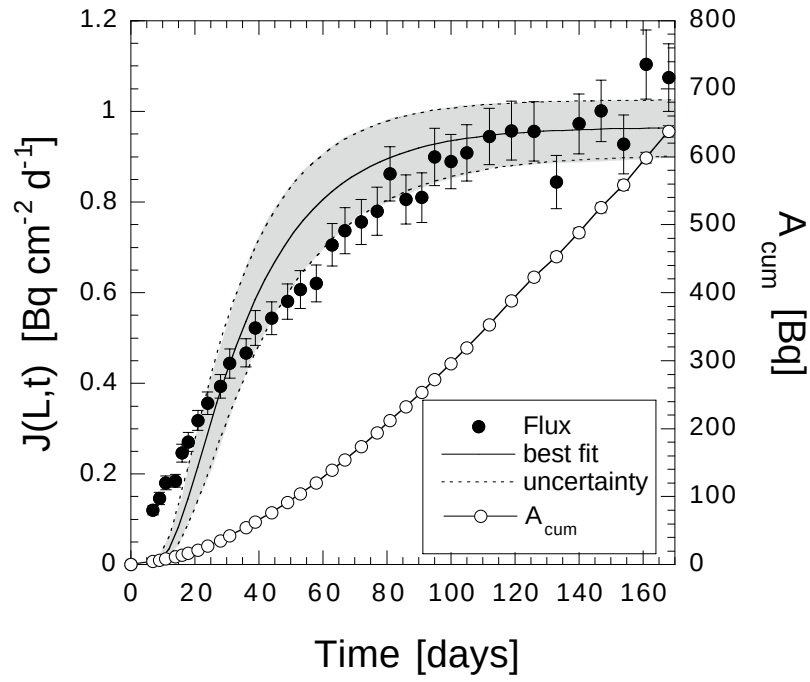


Figure A3-3.1: Flux and total diffused activity of ^{125}I through Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

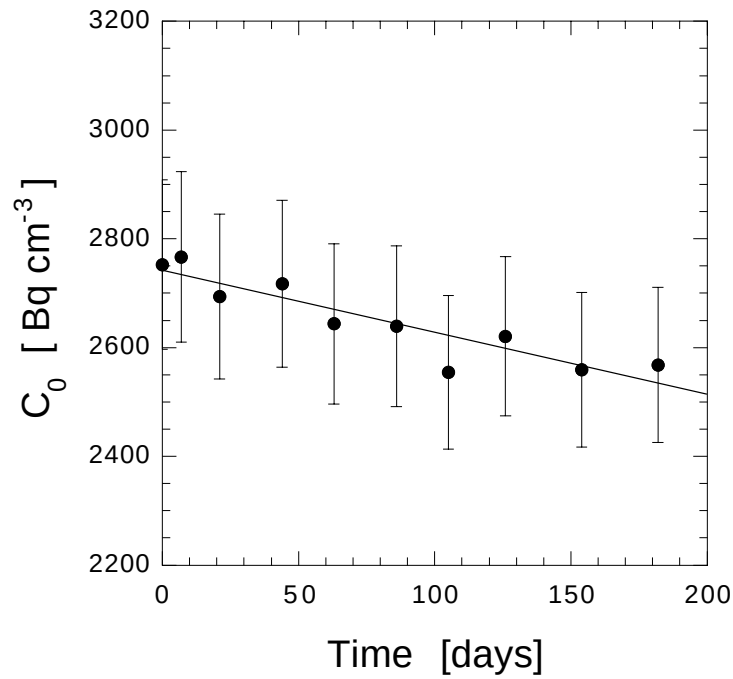


Figure A3-3.2: Concentration C_0 of ^{125}I in the high-concentration container as a function of time (BE-4a).

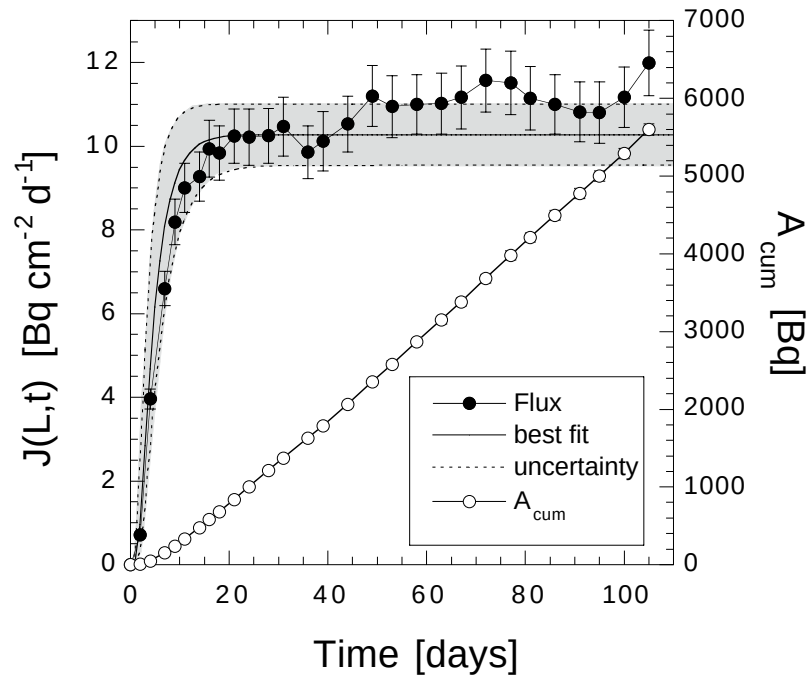


Figure A3-4.1: Flux and total diffused activity of ^{125}I through Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

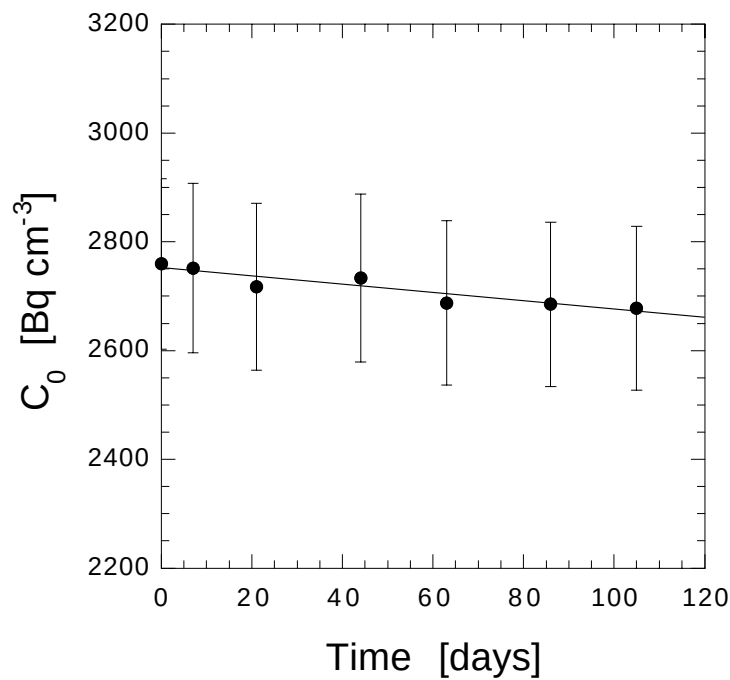


Figure A3-4.2: Concentration C_0 of ^{125}I in the high-concentration container as a function of time (MT-1a).

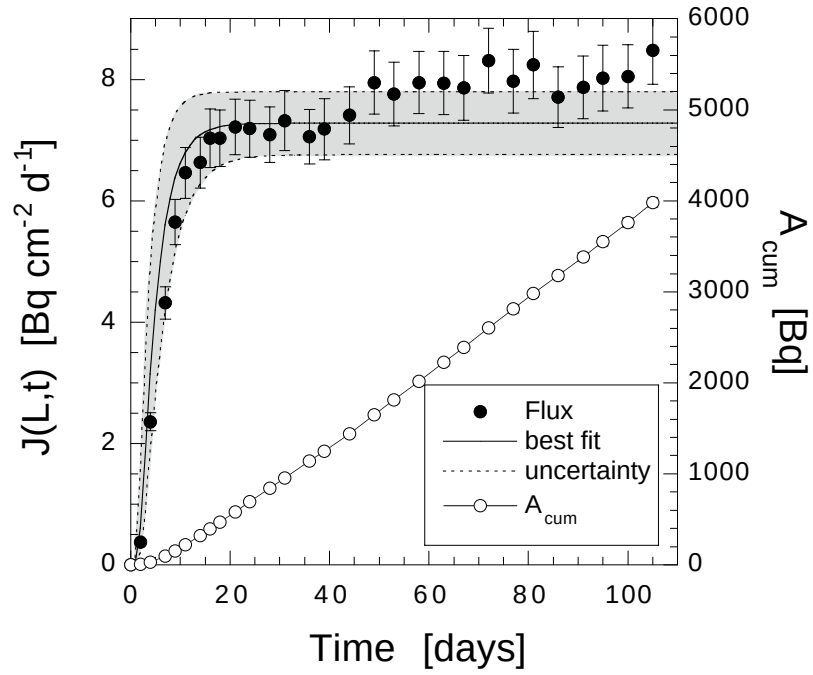


Figure A3-5.1: Flux and total diffused activity of ^{125}I through Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

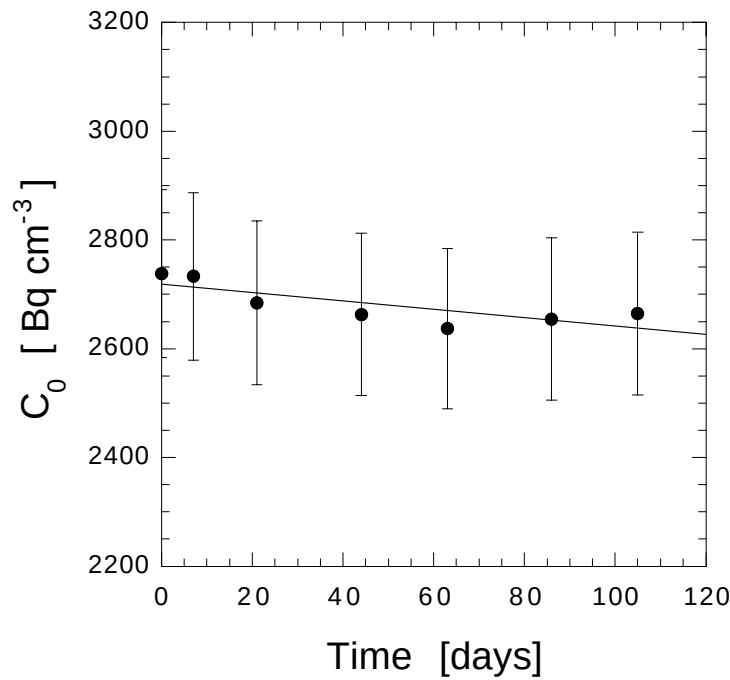


Figure A3-5.2: Concentration C_0 of ^{125}I in the high-concentration container as a function of time (MT-3a).

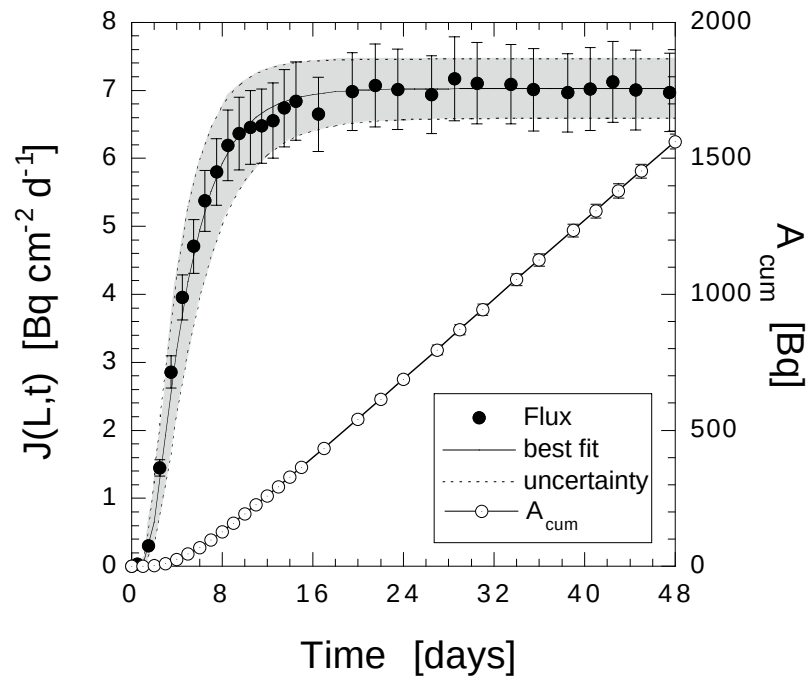
APPENDIX 4: THROUGH-DIFFUSION OF HTO (series 2)

Figure A4-1.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

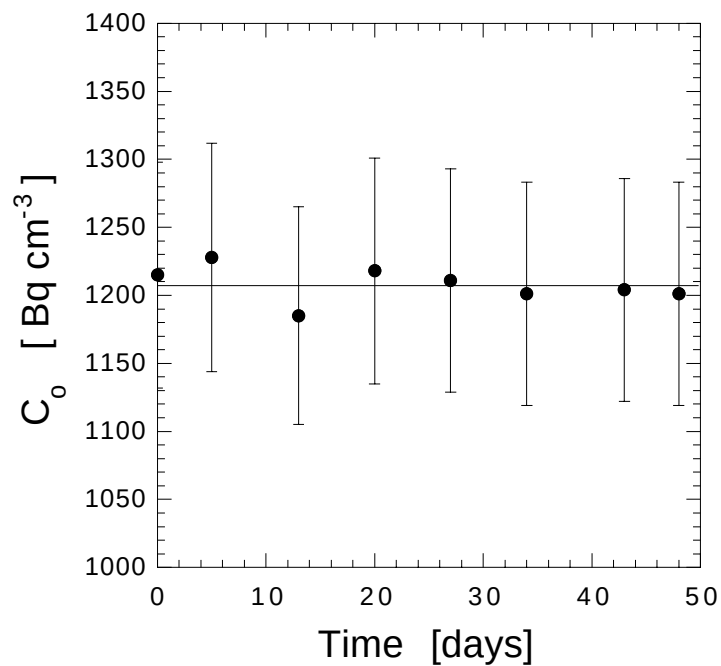


Figure A4-1.2: Concentration C_o of HTO in the high-concentration container as a function of time (BE-2a).

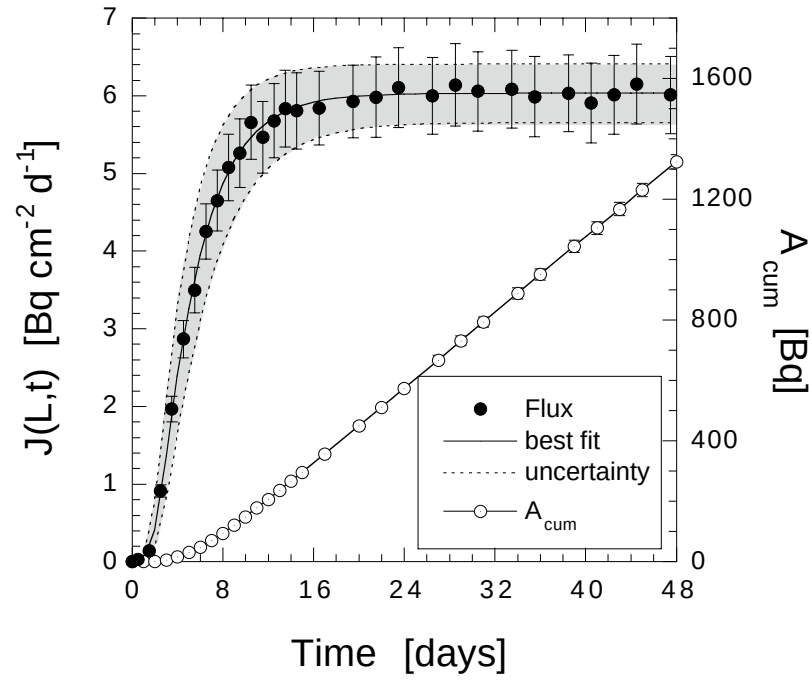


Figure A4-2.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

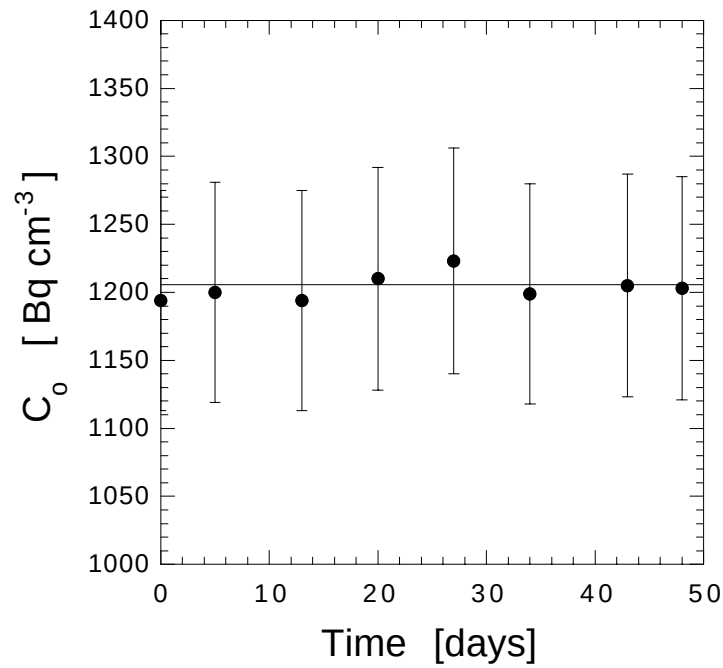


Figure A4-2.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-3a).

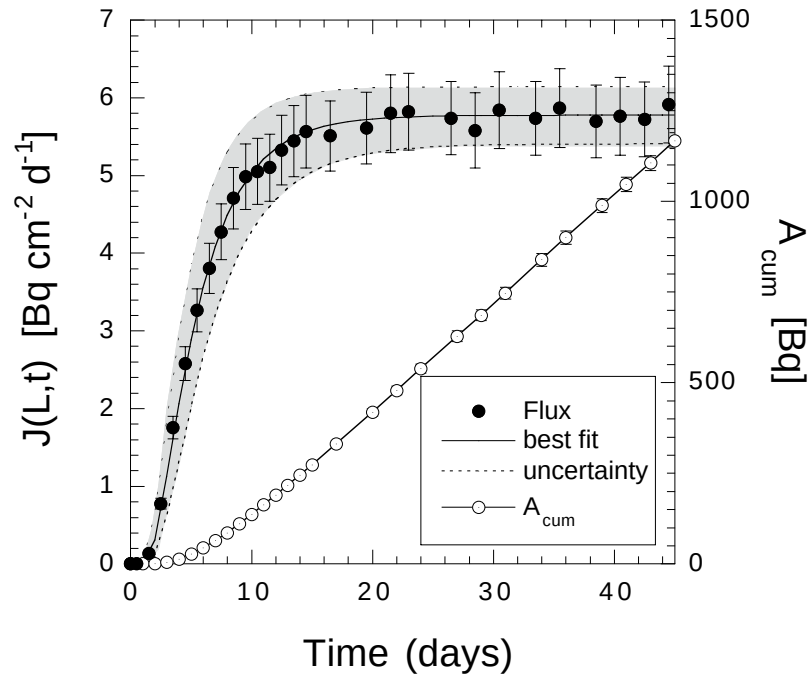


Figure A4-3.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

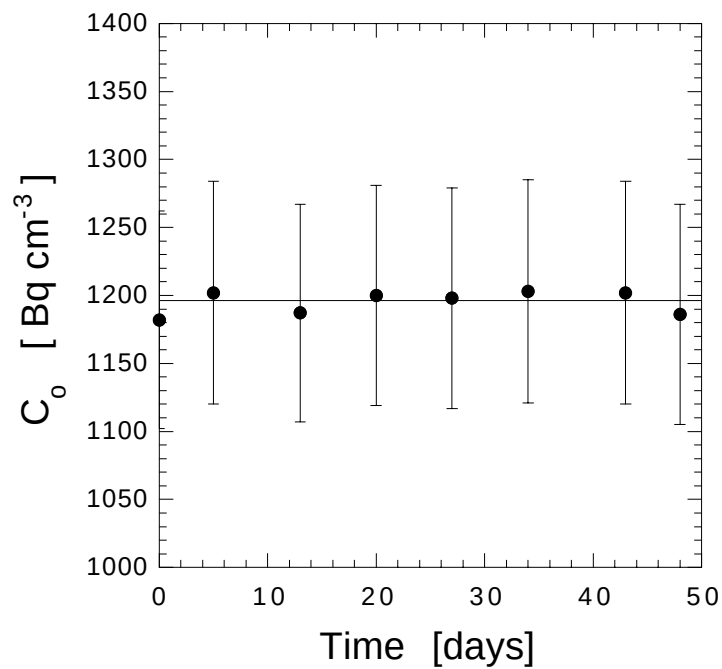


Figure A4-3.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-4a).

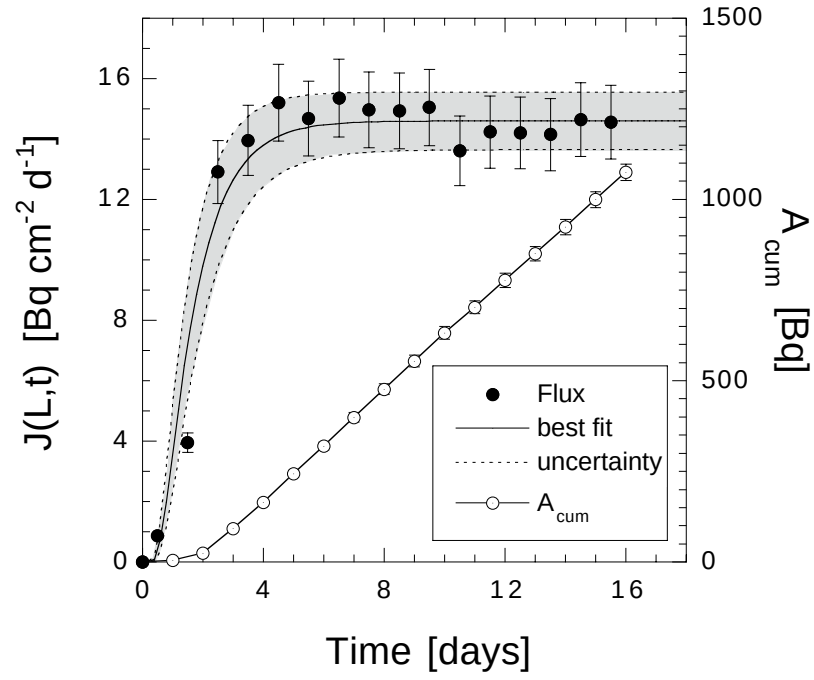


Figure A4-4.1: Flux and total diffused activity of HTO through Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

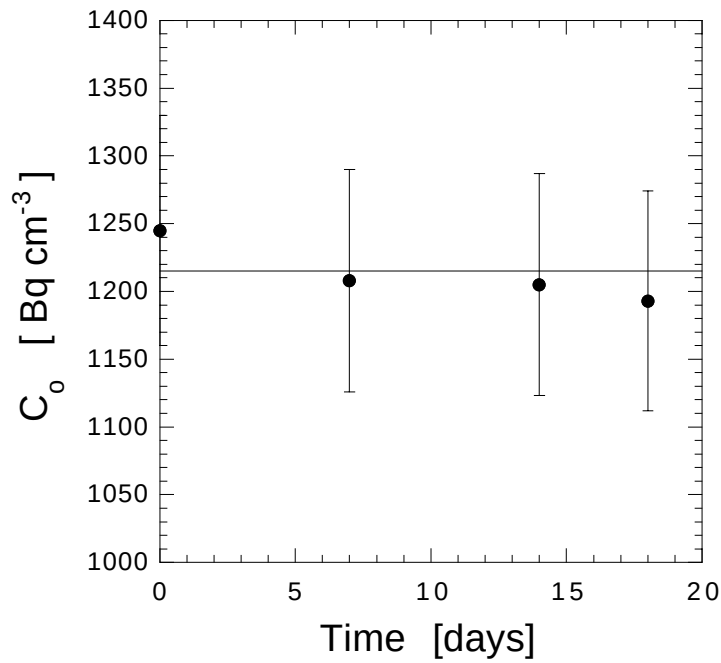


Figure A4-4.2: Concentration C_o of HTO in the high-concentration container as a function of time (MT-1a).

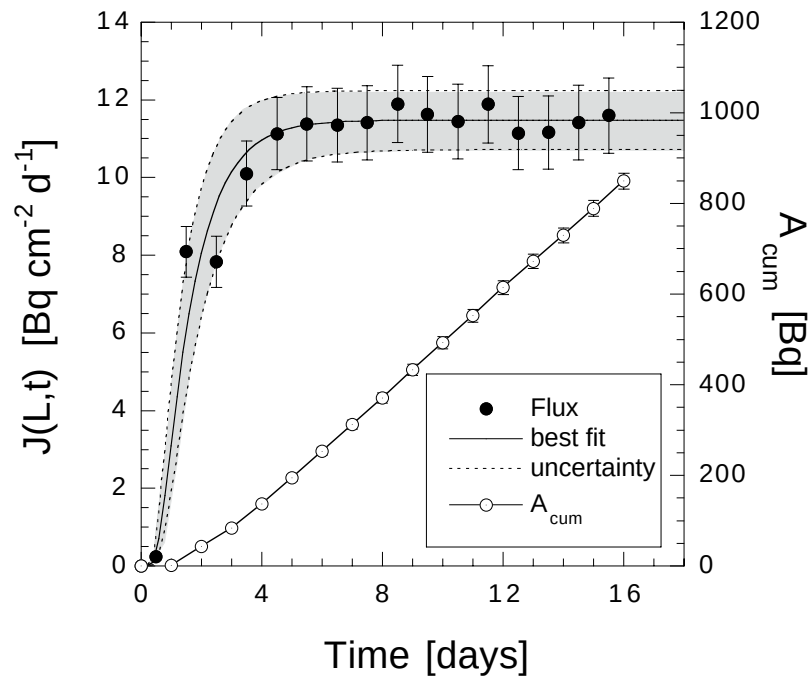


Figure A4-5.1: Flux and total diffused activity of HTO through Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

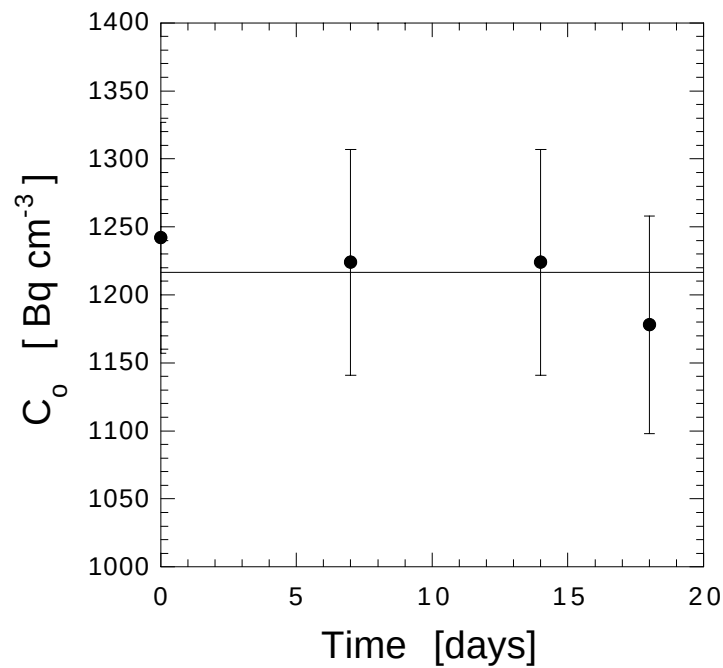


Figure A4-5.2: Concentration C_o of HTO in the high-concentration container as a function of time (MT-3a).

APPENDIX 5: THROUGH-DIFFUSION OF $^{22}\text{Na}^+$

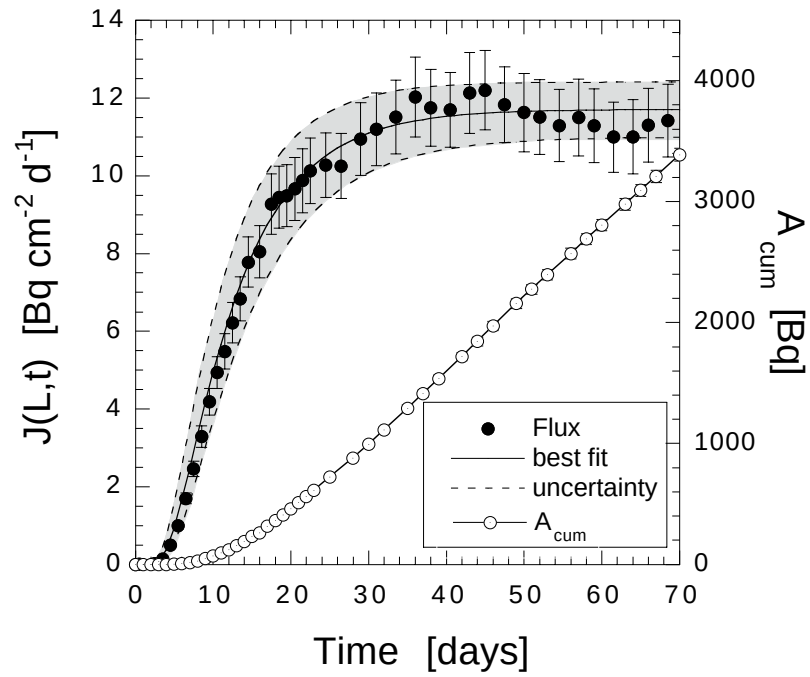


Figure A5-1.1: Flux and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

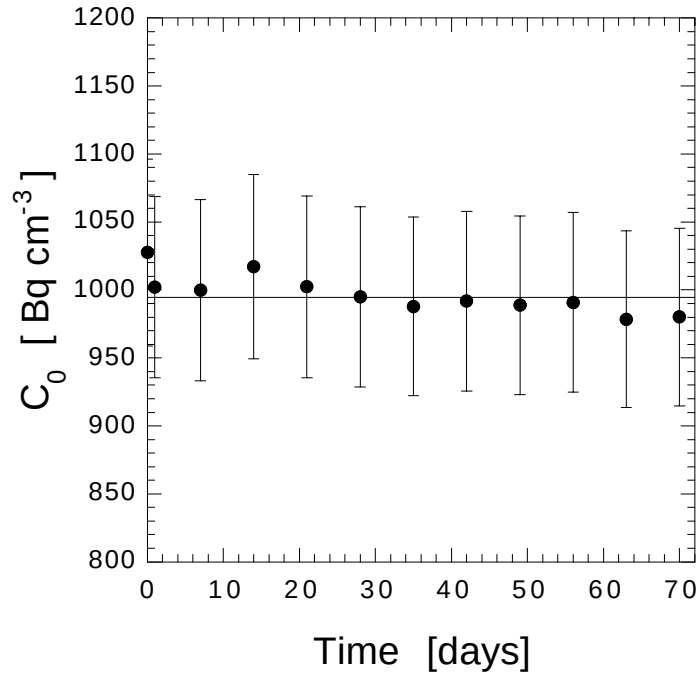


Figure A5-1.2: Concentration C_0 of $^{22}\text{Na}^+$ in the high-concentration container as a function of time (BE-2a).

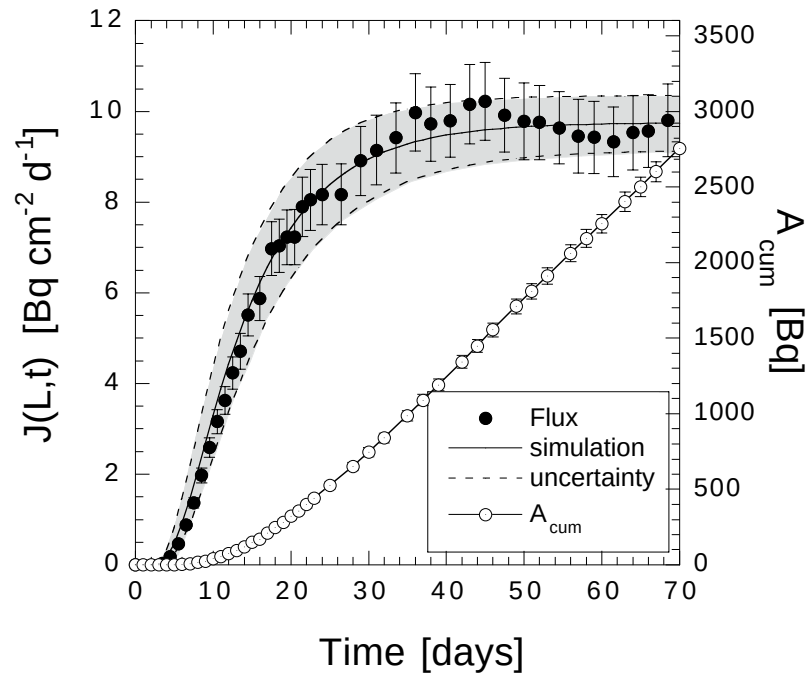


Figure A5-2.1: Flux and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

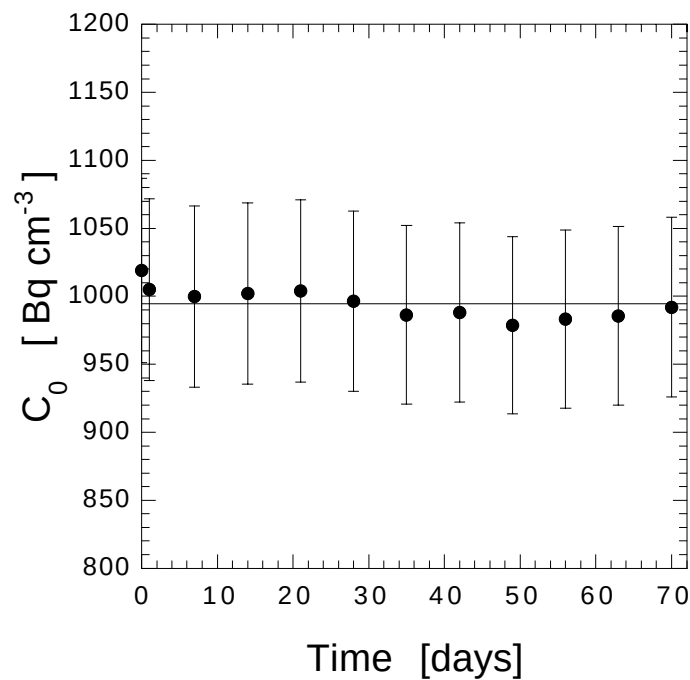


Figure A5-2.2: Concentration C_0 of $^{22}\text{Na}^+$ in the high-concentration container as a function of time (BE-3a).

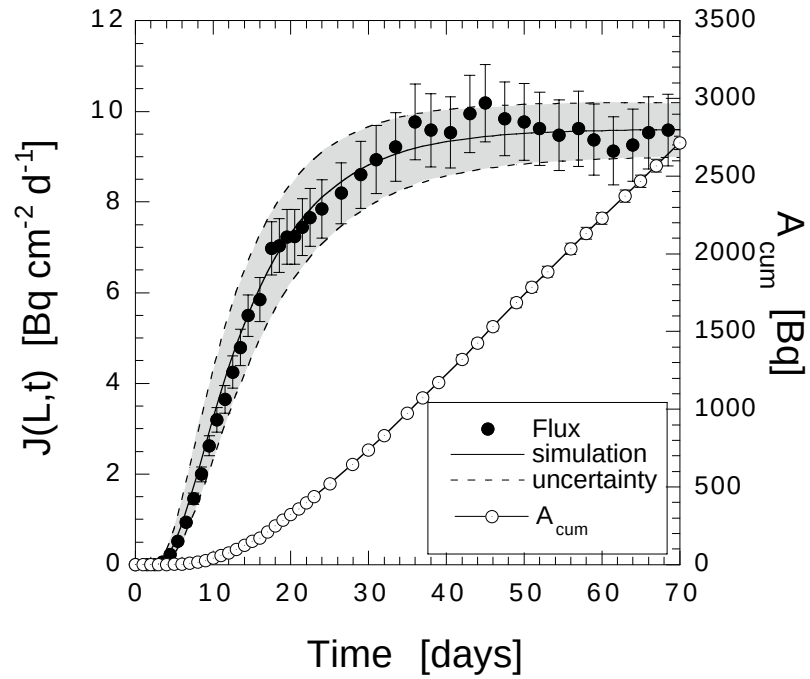


Figure A5-3.1: Flux and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay from Benken at a confining pressure of 14 MPa (BE-4a).

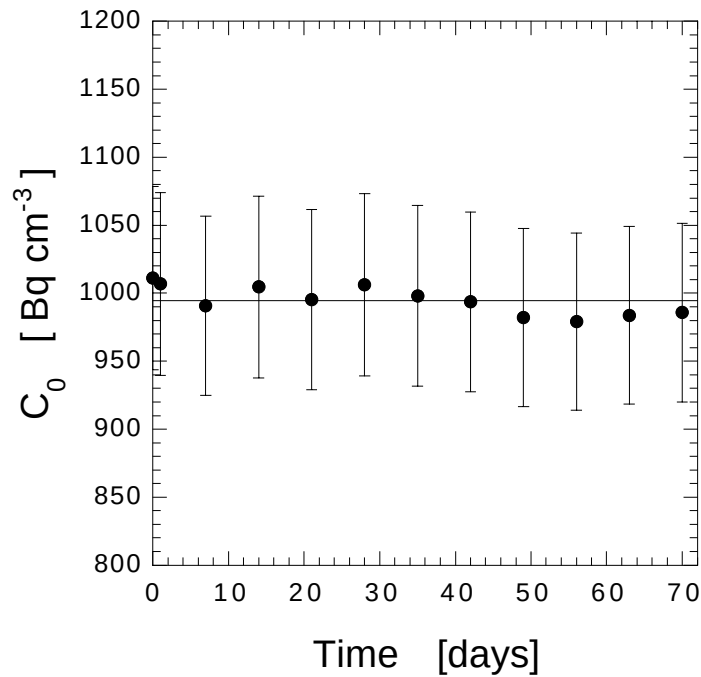


Figure A5-3.2: Concentration C_0 of $^{22}\text{Na}^+$ in the high-concentration container as a function of time (BE-4a).

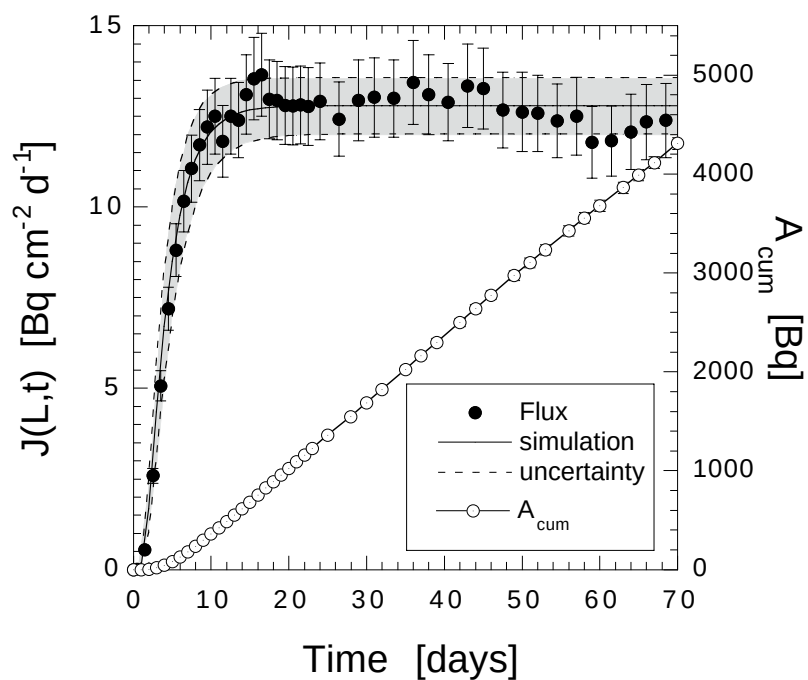


Figure A5-4.1: Flux and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

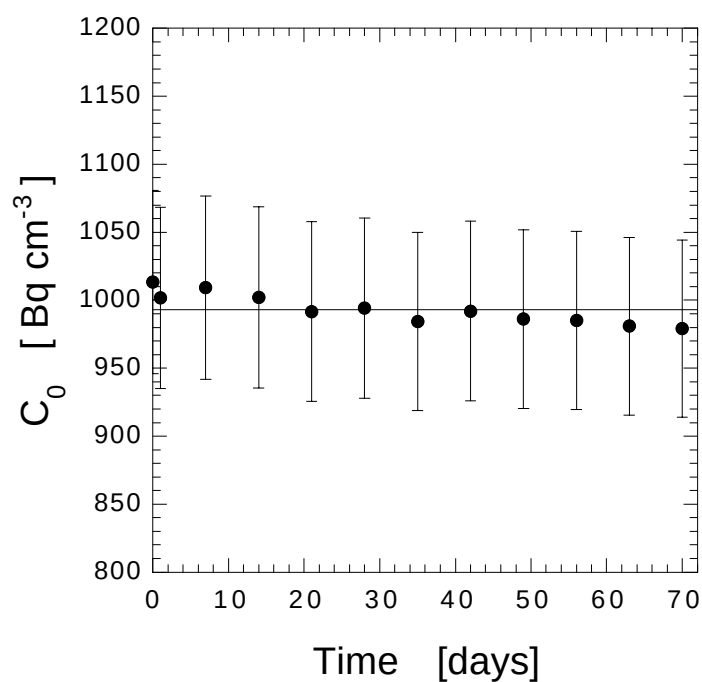


Figure A5-4.2: Concentration C_0 of $^{22}\text{Na}^+$ in the high-concentration container as a function of time (MT-1a).

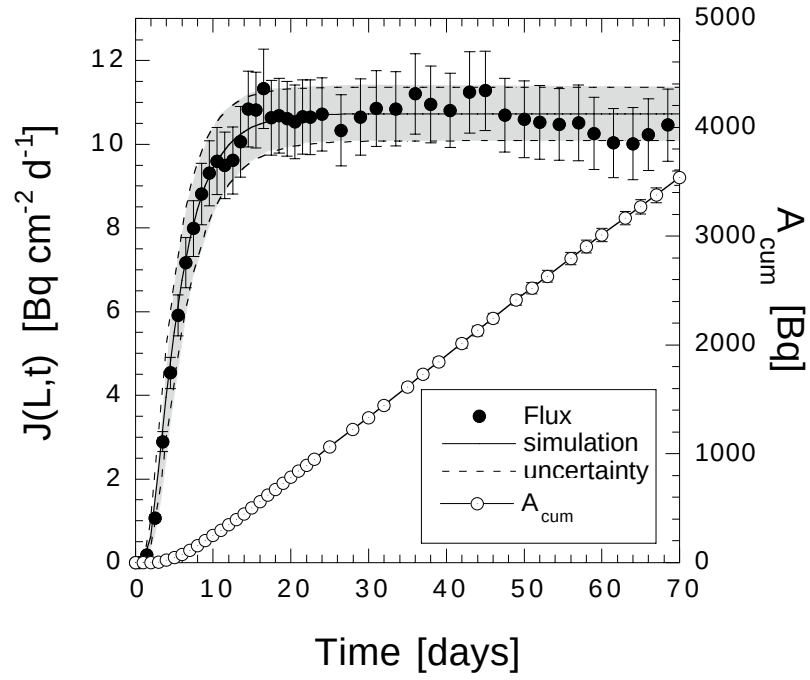


Figure A5-5.1: Flux and total diffused activity of $^{22}\text{Na}^+$ through Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

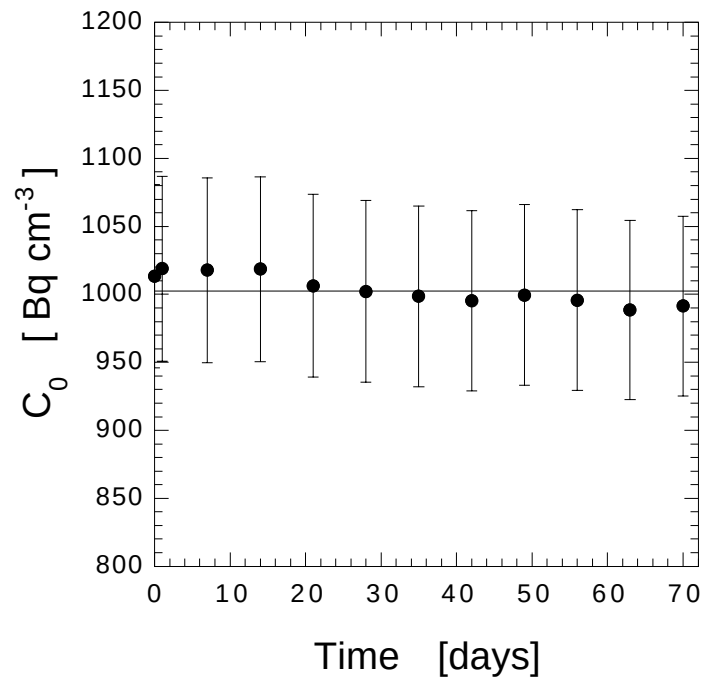


Figure A5-5.2: Concentration C_0 of $^{22}\text{Na}^+$ in the high-concentration container as a function of time (MT-3a).

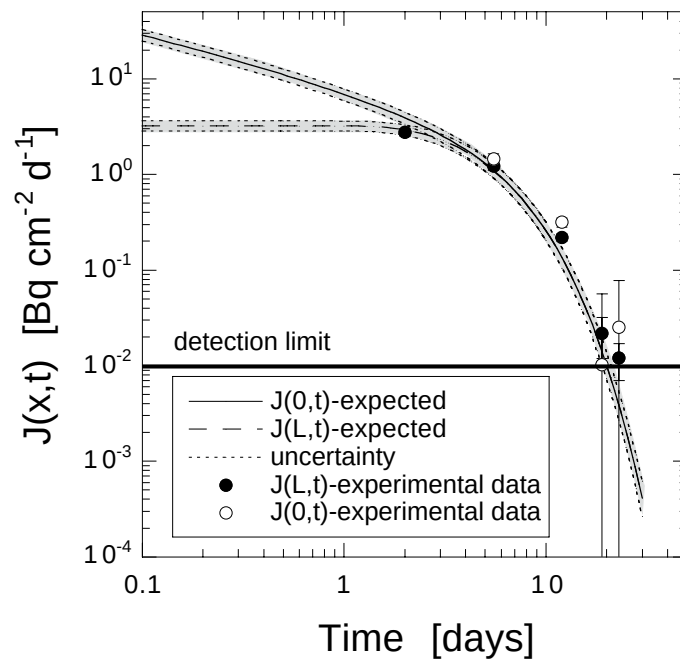
APPENDIX 6: OUT-DIFFUSION OF HTO (series 1)

Figure A6-1.1: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

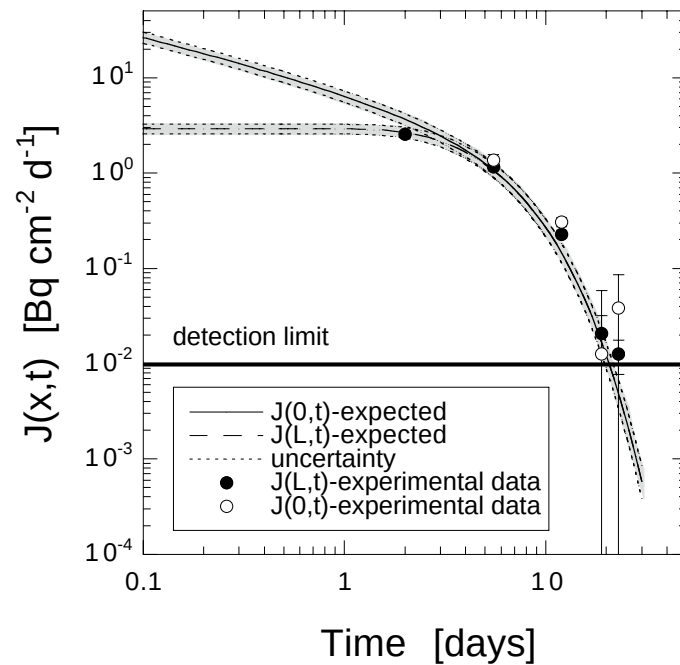


Figure A6-1.2: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

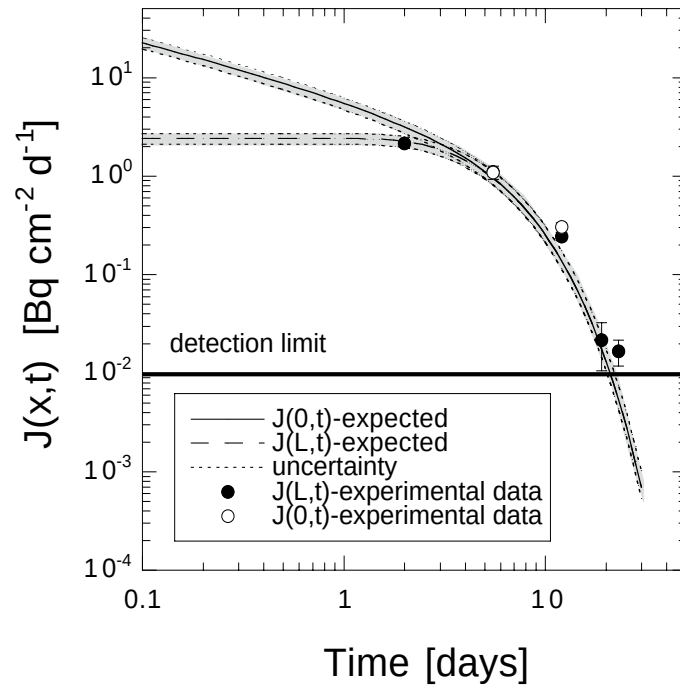


Figure A6-1.3: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

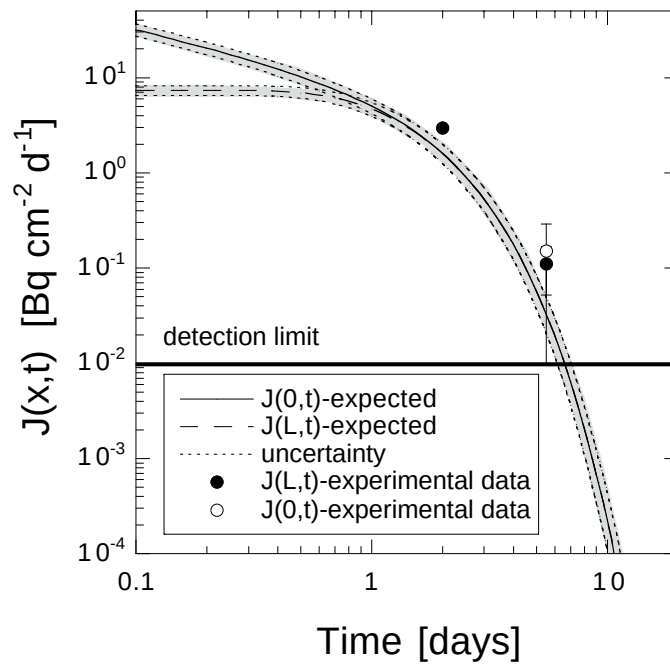


Figure A6-1.4: Out-diffusion of HTO in Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

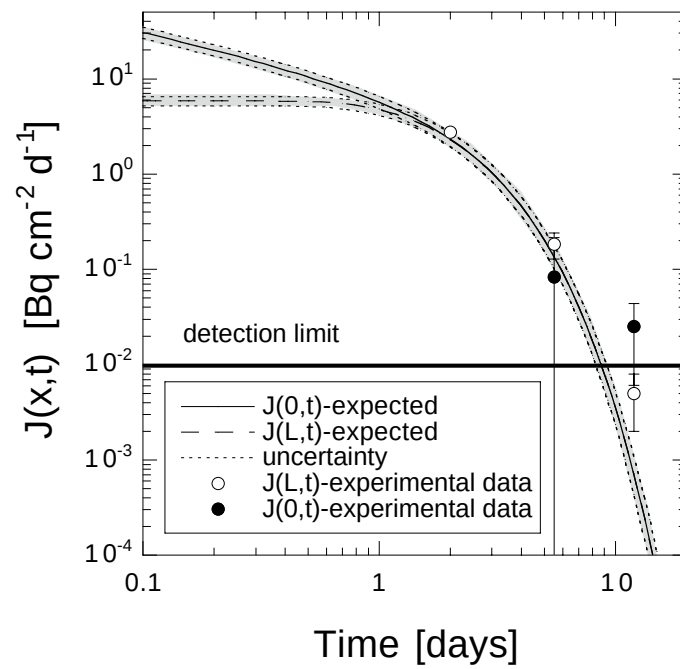


Figure A6-1.5: Out-diffusion of HTO in Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

APPENDIX 7: OUT-DIFFUSION OF $^{36}\text{Cl}^-$

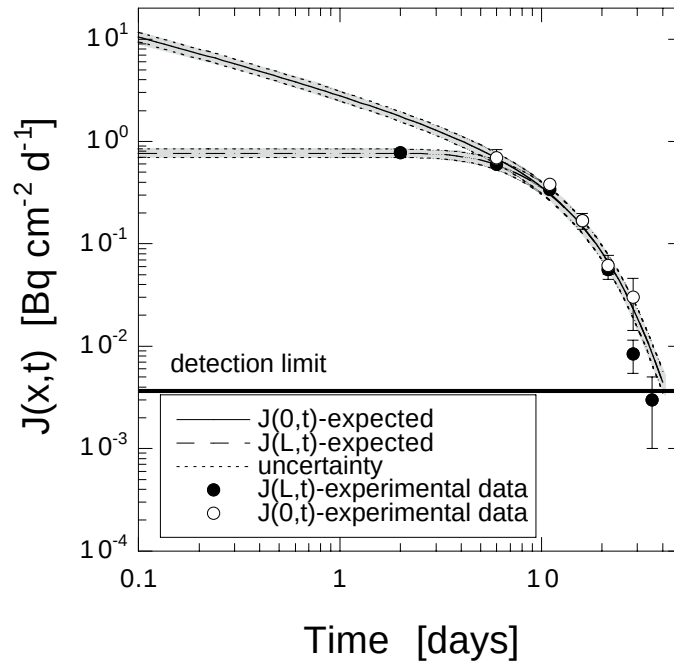


Figure A7-1.1: Out-diffusion of $^{36}\text{Cl}^-$ in Opalinus Clay from Benken at a confining of 4 MPa (BE-2a).

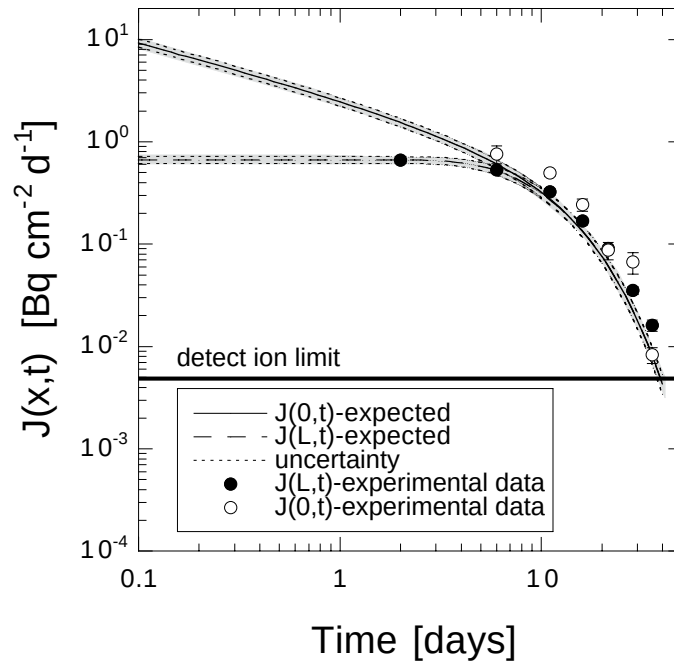


Figure A7-1.2: Out-diffusion of $^{36}\text{Cl}^-$ in Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

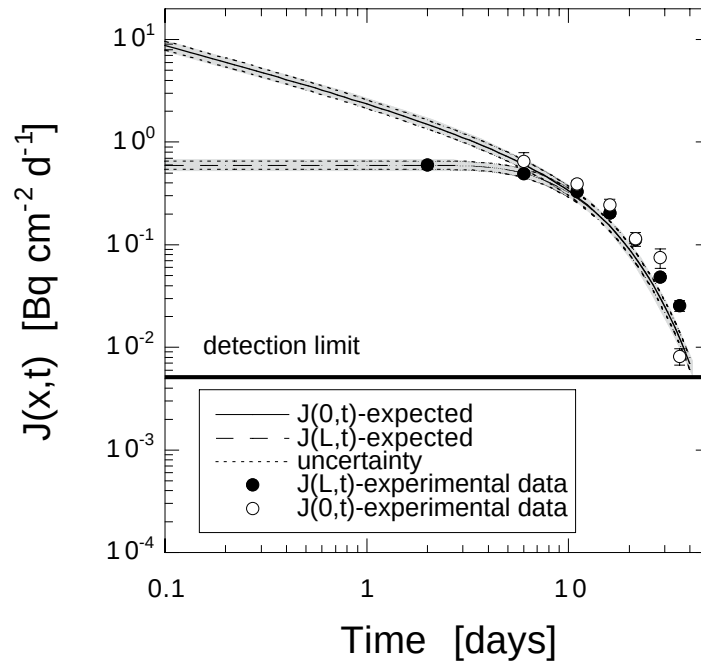


Figure A7-1.3: Out-diffusion of $^{36}\text{Cl}^-$ in Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

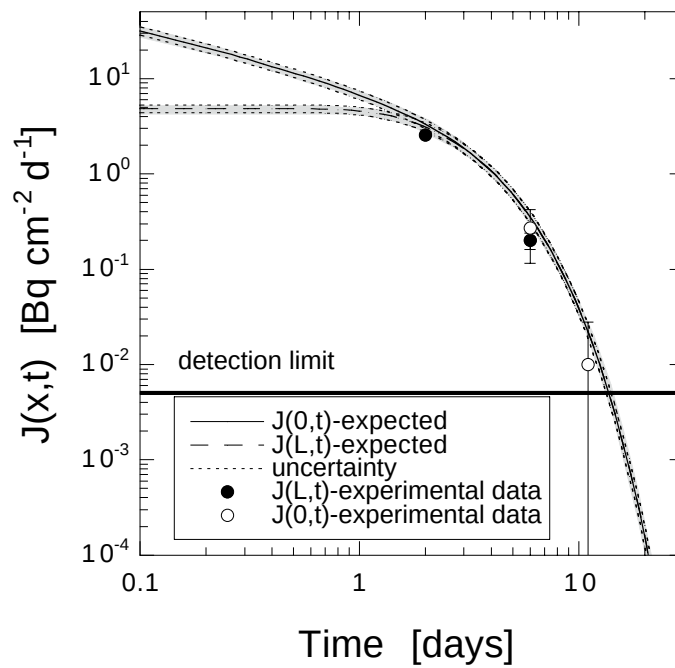


Figure A7-1.4: Out-diffusion of $^{36}\text{Cl}^-$ in Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

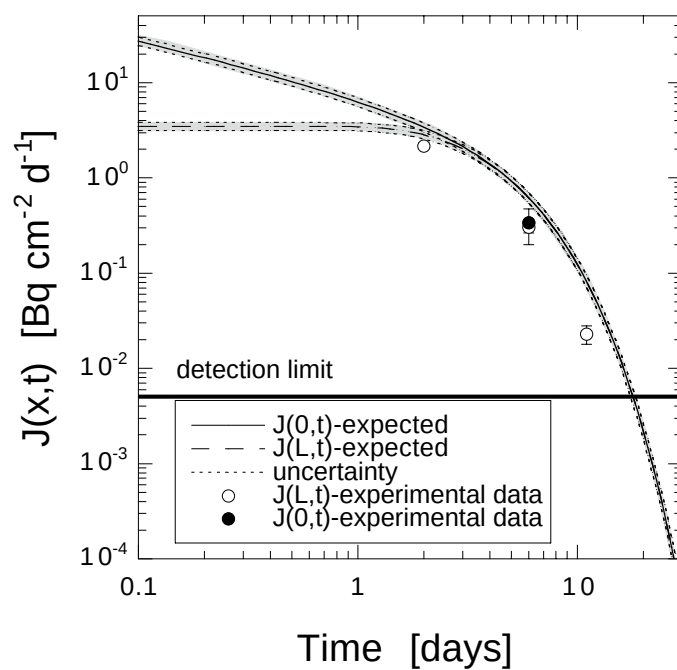


Figure A7-1.5: Out-diffusion of $^{36}\text{Cl}^-$ in Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

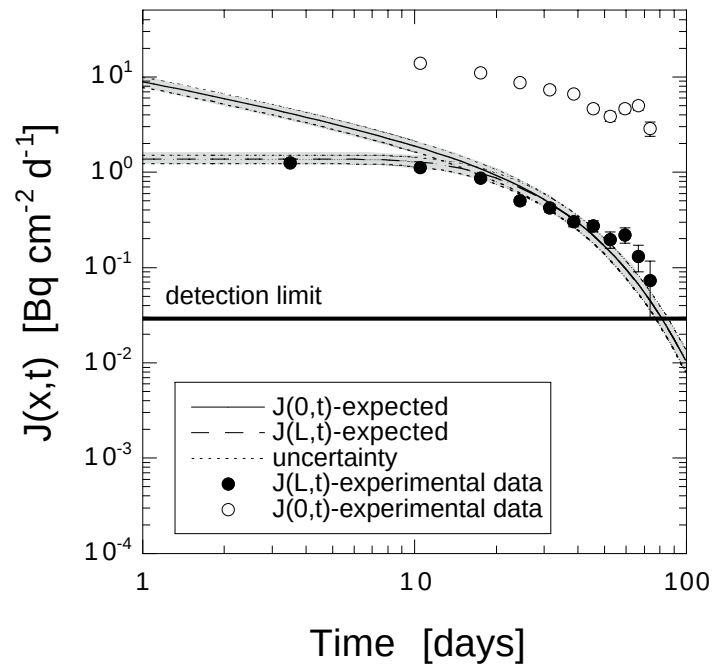
APPENDIX 8: OUT-DIFFUSION OF ^{125}I 

Figure A8-1.1: Out-diffusion of ^{125}I in Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

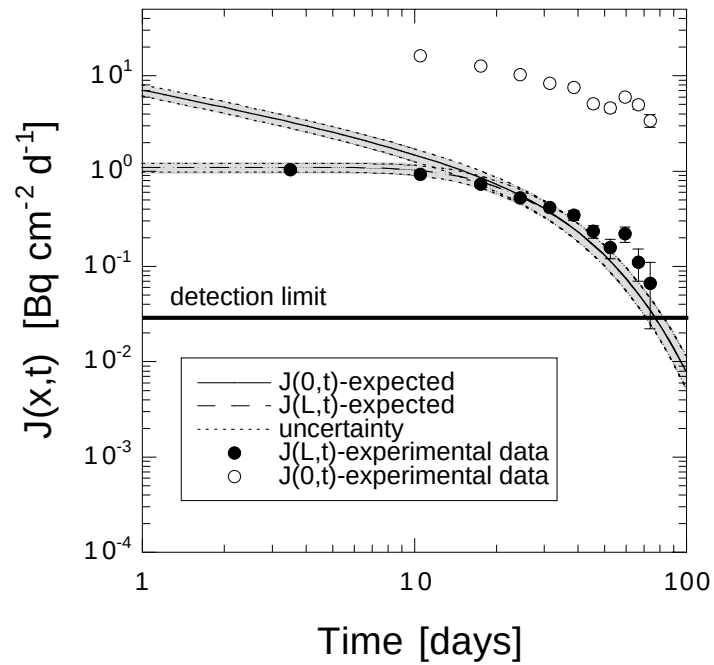


Figure A8-1.2: Out-diffusion of ^{125}I in Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

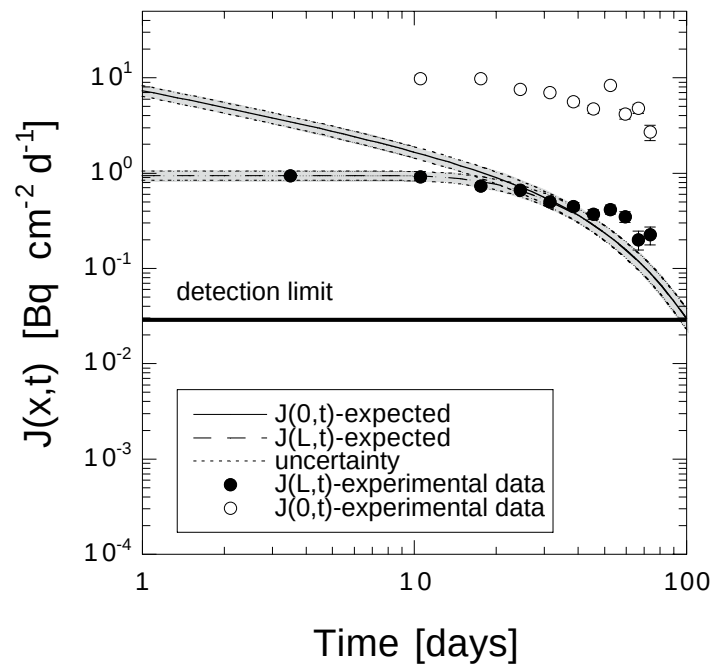


Figure A8-1.3: Out-diffusion of ^{125}I in Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

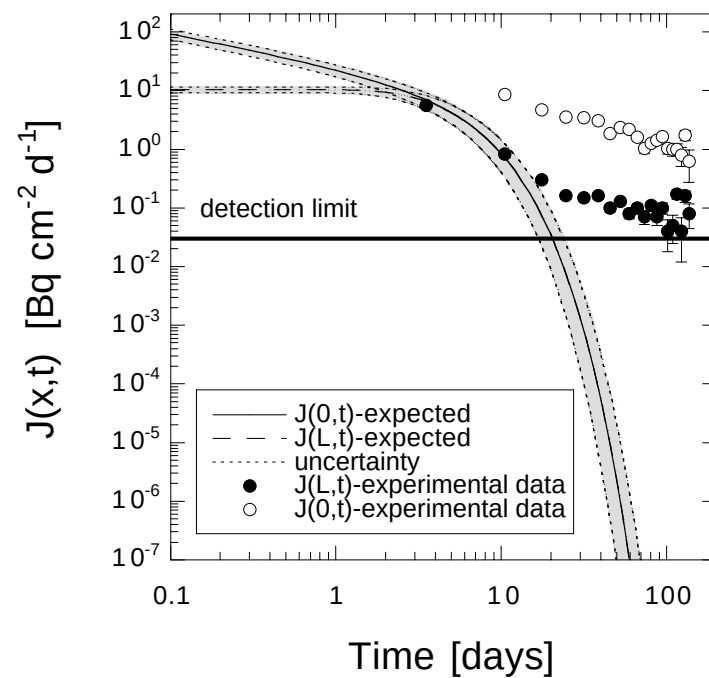


Figure A8-1.4: Out-diffusion of ^{125}I in Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

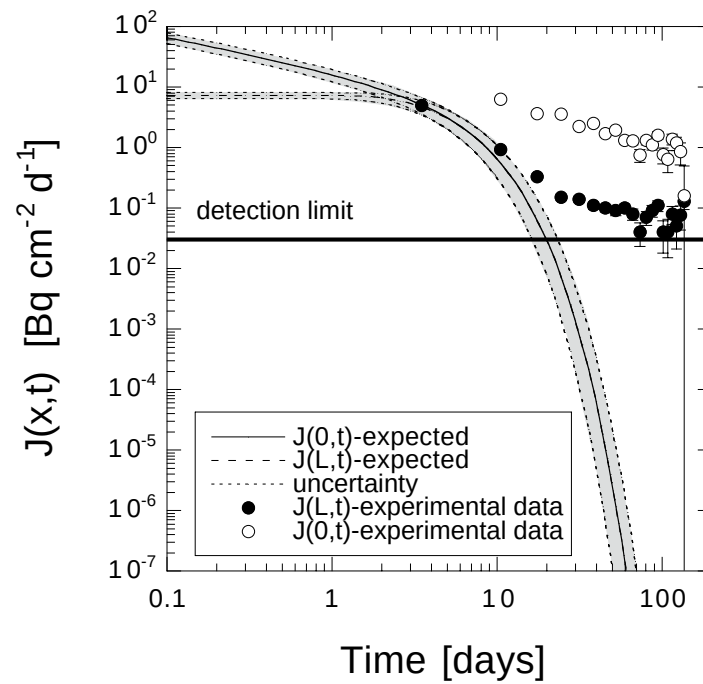


Figure A8-1.5: Out-diffusion of $^{125}\text{I}^-$ in Opalinus Clay from Benken at a confining pressure of 5 MPa (MT-3a).

APPENDIX 9: OUT-DIFFUSION OF HTO (series 2)

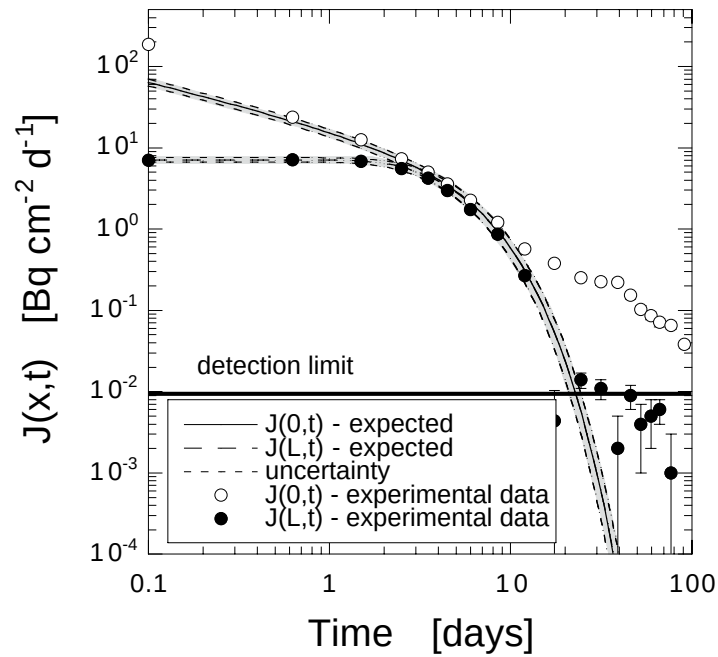


Figure A9-1.1: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

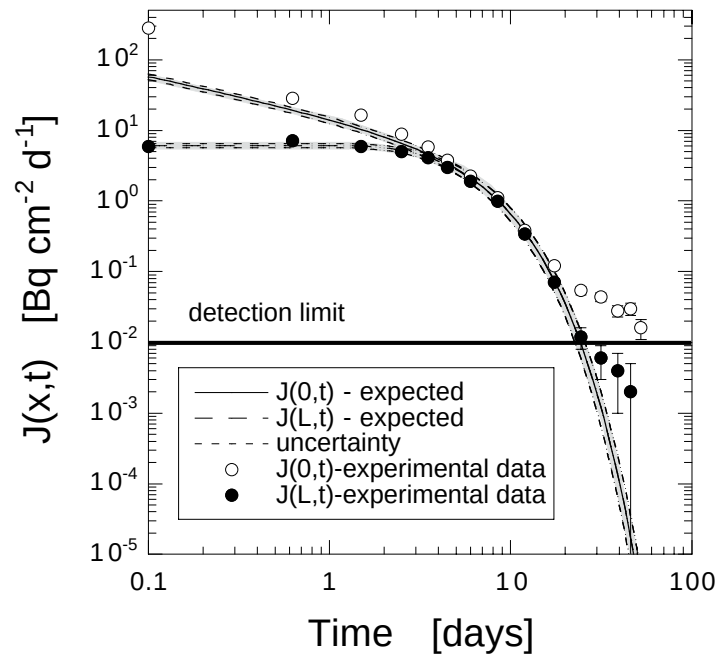


Figure A9-1.2: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

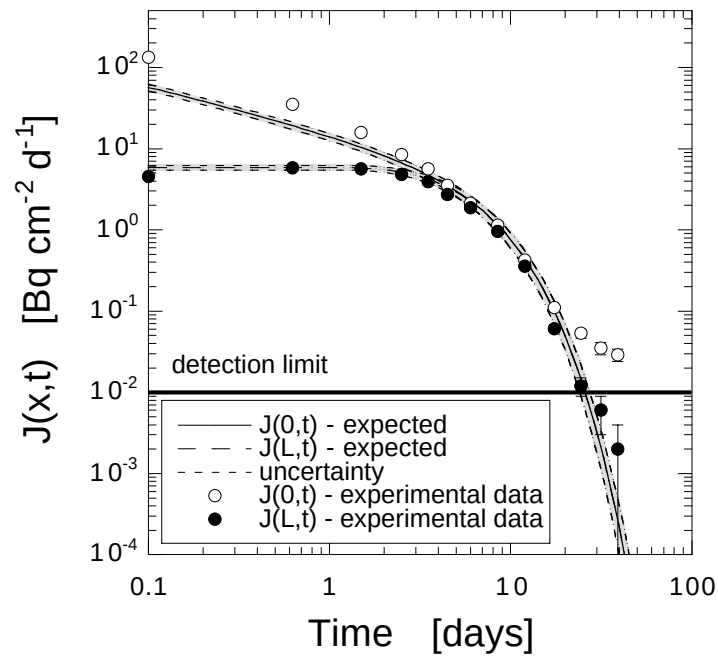


Figure A9-1.3: Out-diffusion of HTO in Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

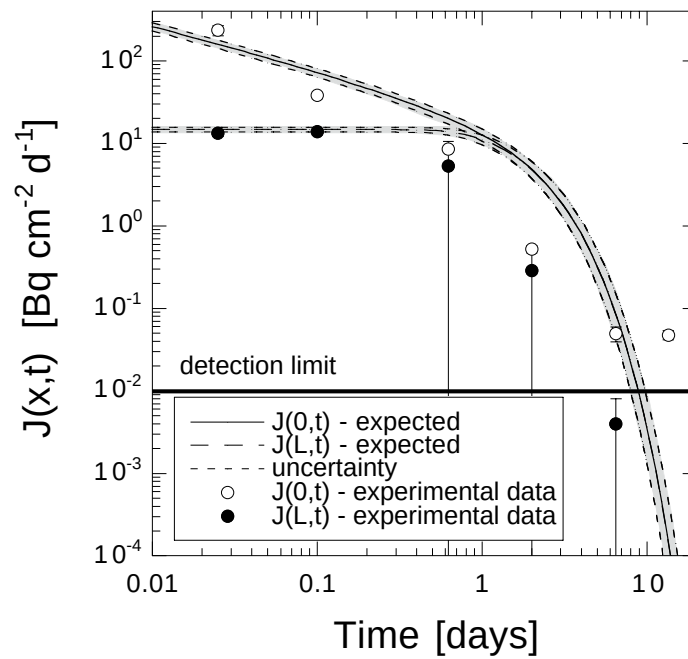


Figure A9-1.4: Out-diffusion of HTO in Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

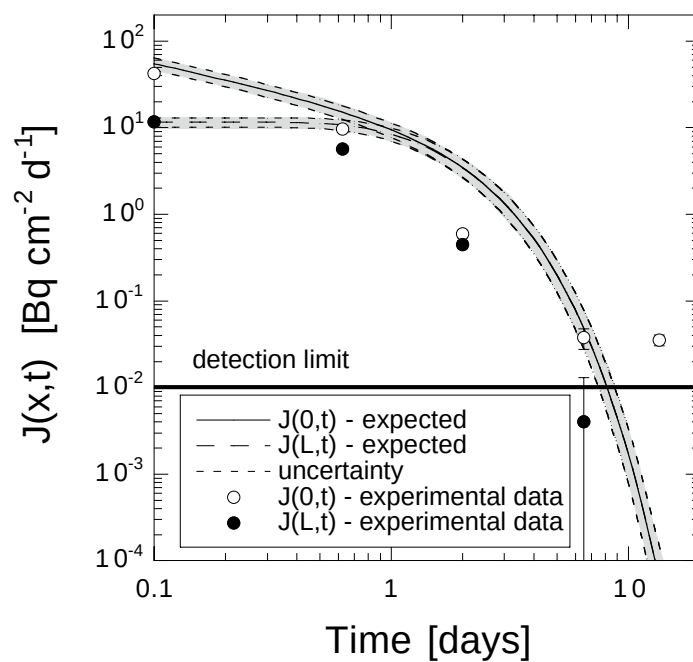


Figure A9-1.5: Out-diffusion of HTO in Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

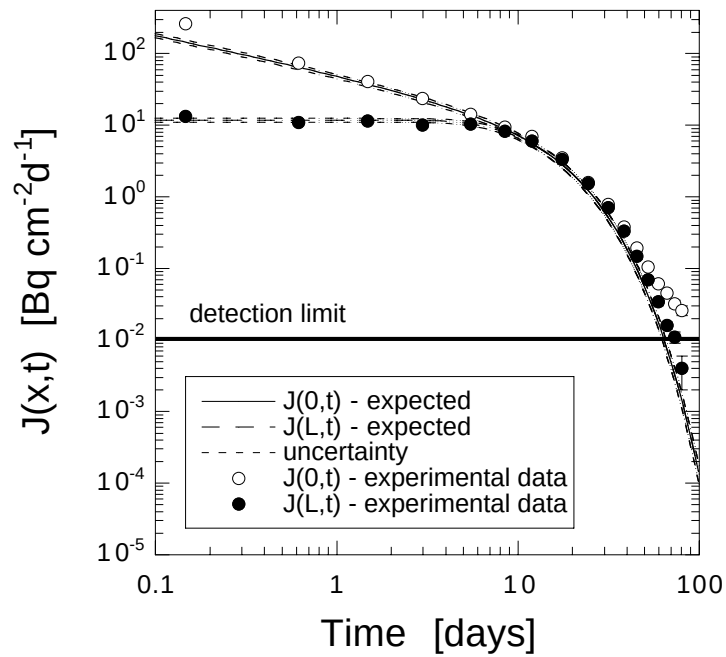
APPENDIX 10: OUT-DIFFUSION OF $^{22}\text{Na}^+$ 

Figure A10-1.1: Out-diffusion of $^{22}\text{Na}^+$ in Opalinus Clay from Benken at a confining pressure of 4 MPa (BE-2a).

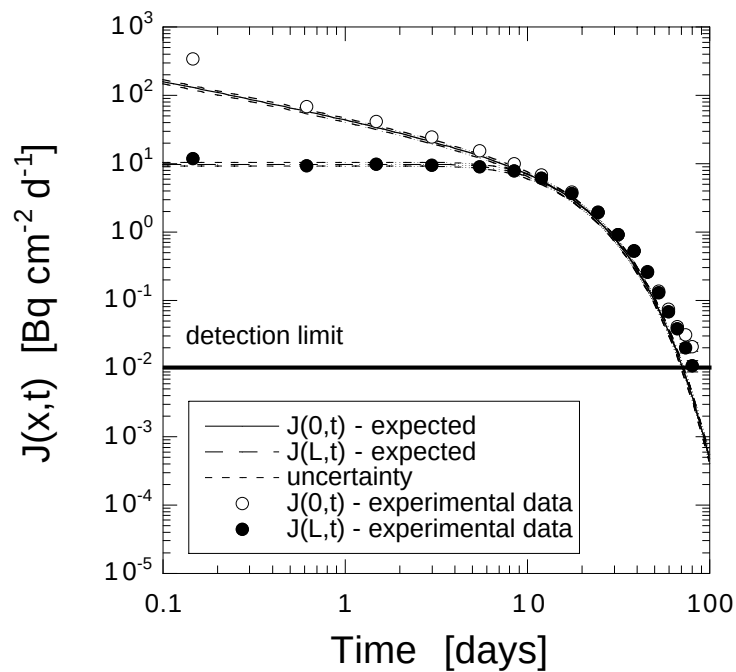


Figure A10-1.2: Out-diffusion of $^{22}\text{Na}^+$ in Opalinus Clay from Benken at a confining pressure of 8 MPa (BE-3a).

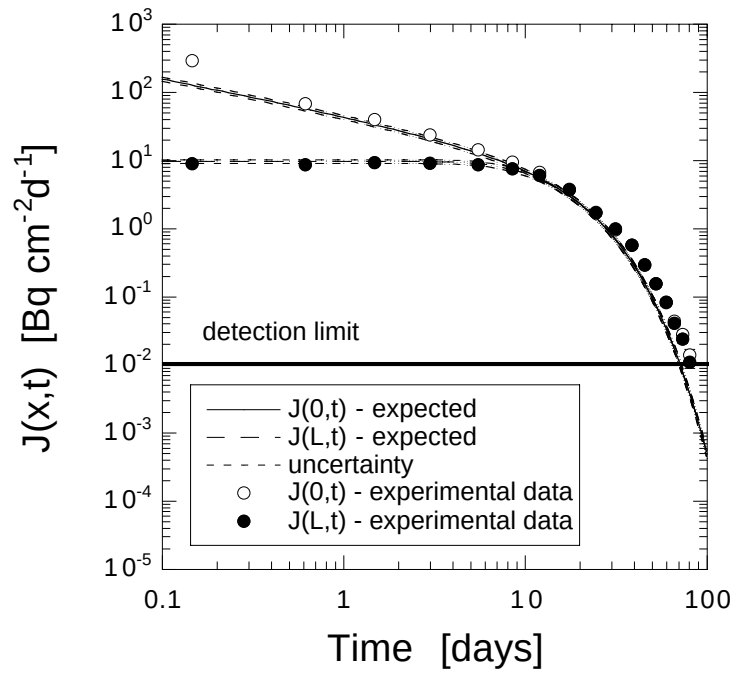


Figure A10-1.3: Out-diffusion of ²²Na⁺ in Opalinus Clay from Benken at a confining pressure of 15 MPa (BE-4a).

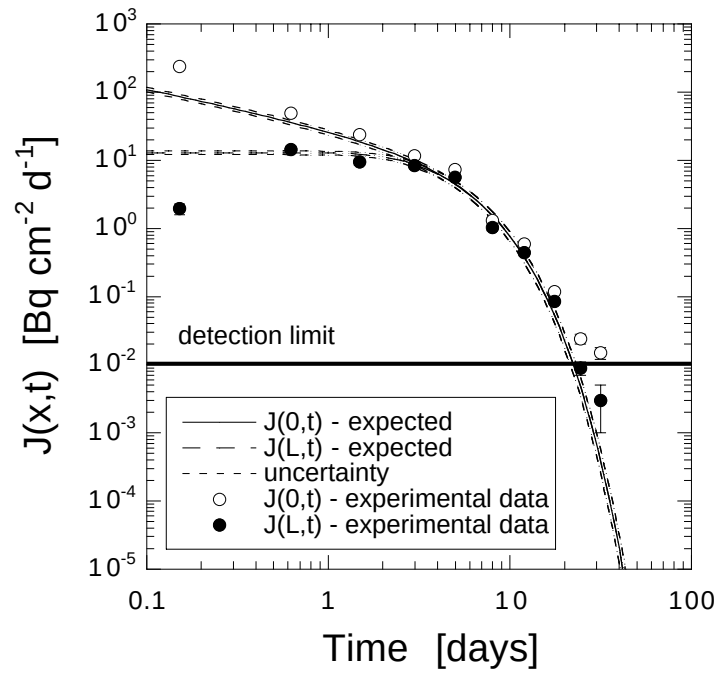


Figure A10-1.4: Out-diffusion of ²²Na⁺ in Opalinus Clay from Mont Terri at a confining pressure of 1 MPa (MT-1a).

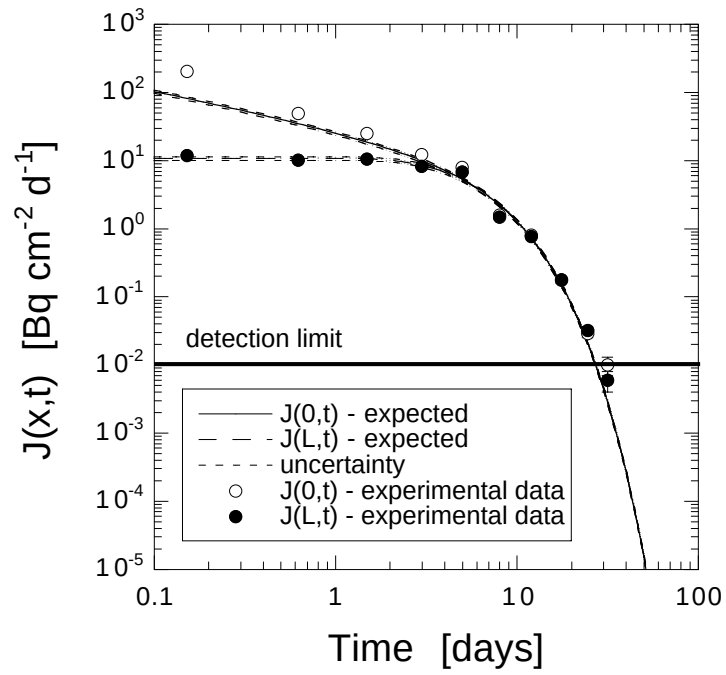


Figure A10-1.5: Out-diffusion of $^{22}\text{Na}^+$ in Opalinus Clay from Mont Terri at a confining pressure of 5 MPa (MT-3a).

APPENDIX 11: THROUGH-DIFFUSION OF HTO (OTHER DEPTHS)

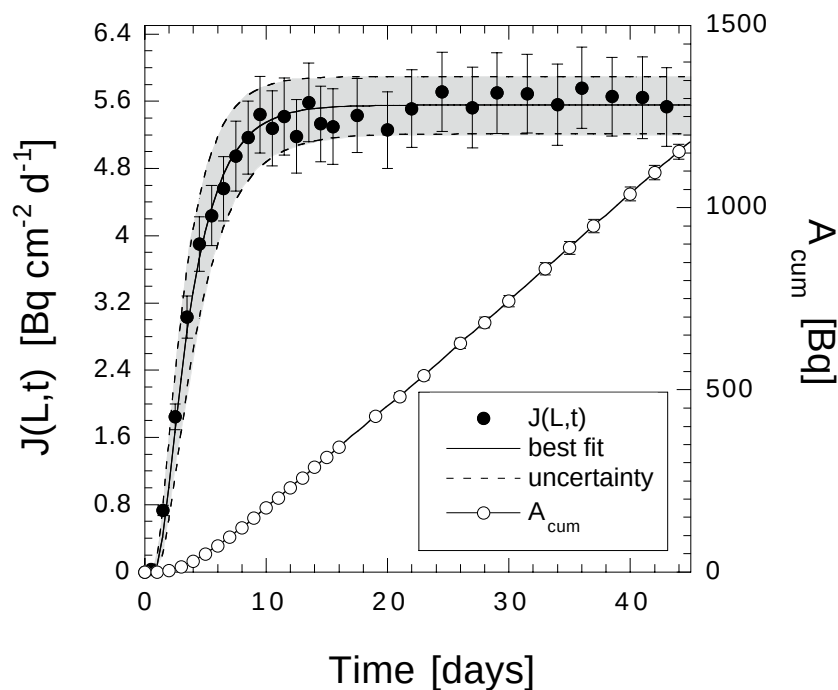


Figure A11-1.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken (BE-S0: -564 m) for diffusion perpendicular to bedding.

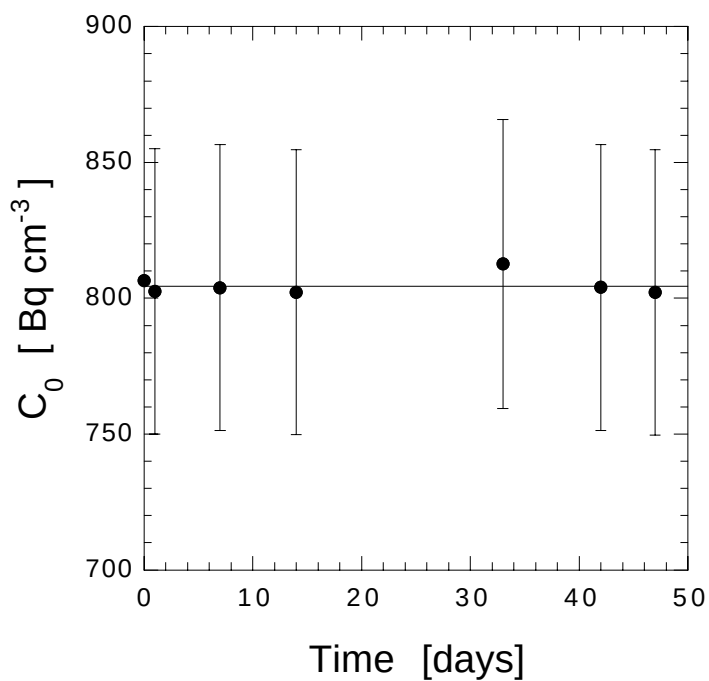


Figure A11-1.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-S0: -564 m).

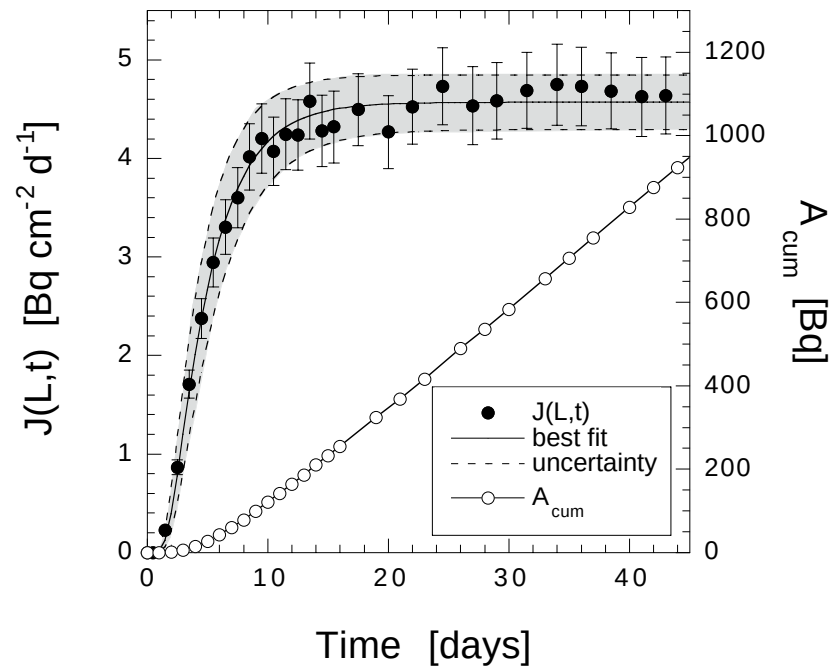


Figure A11-2.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken (BE-S1: -589 m) for diffusion perpendicular to bedding.

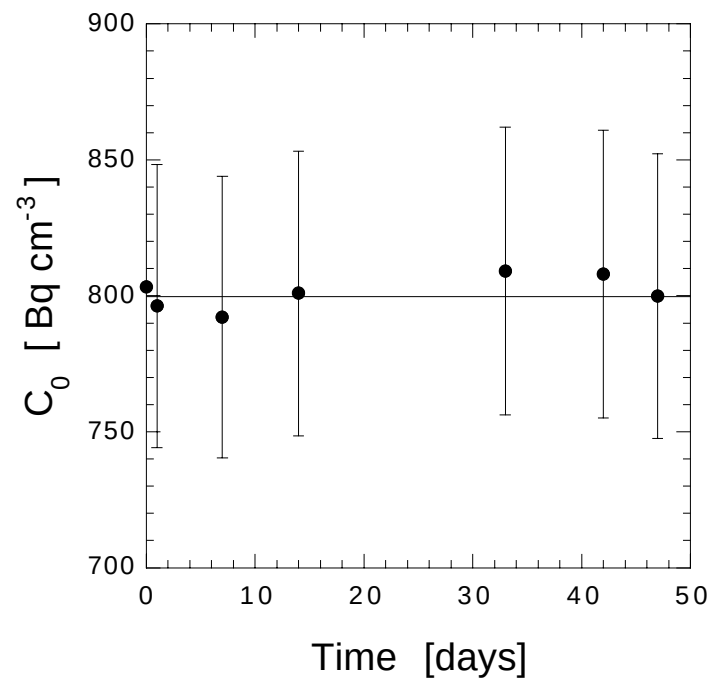


Figure A11-2.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-S1: -589 m).

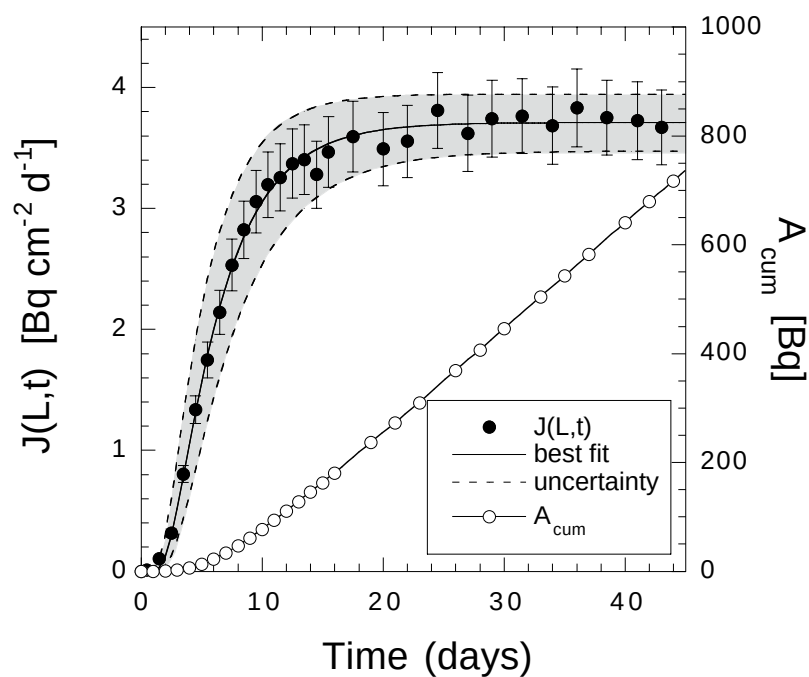


Figure A11-3.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken (BE-S2: -651 m) for diffusion perpendicular to bedding.

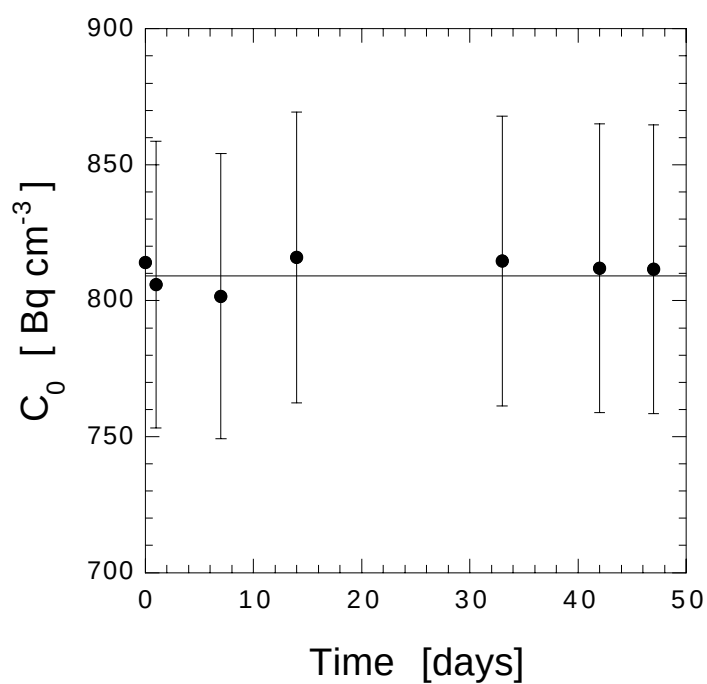


Figure A11-3.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-S1: -651 m).

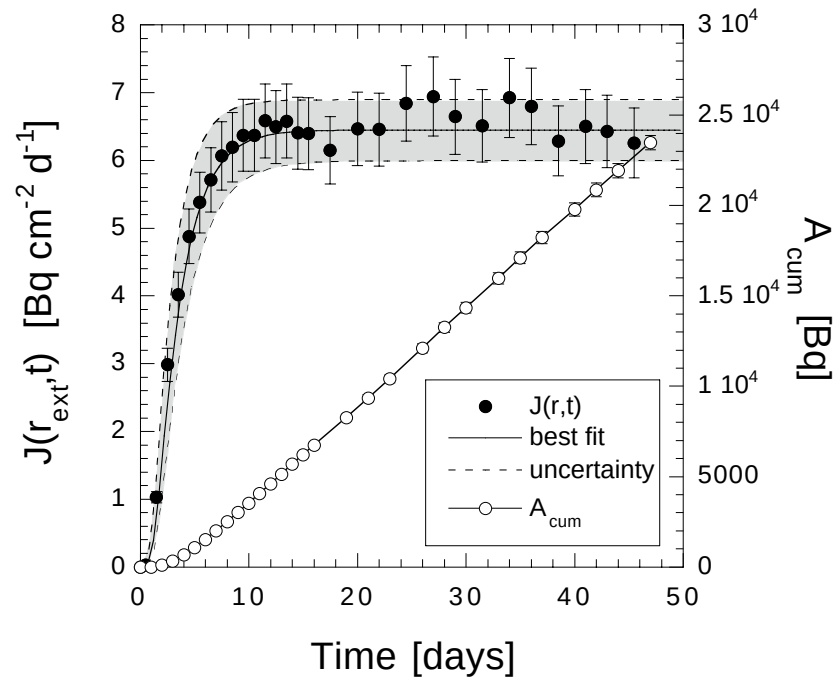


Figure A11-4.1: Flux and total diffused activity of HTO through Opalinus Clay from Benken (BE-R3: -564 m) for diffusion parallel to bedding.

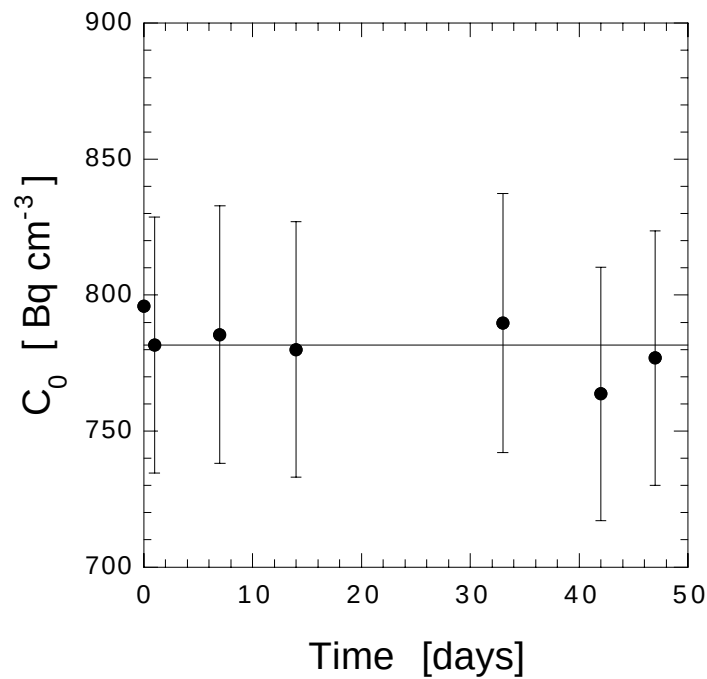


Figure A11-4.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-R3: -564 m).

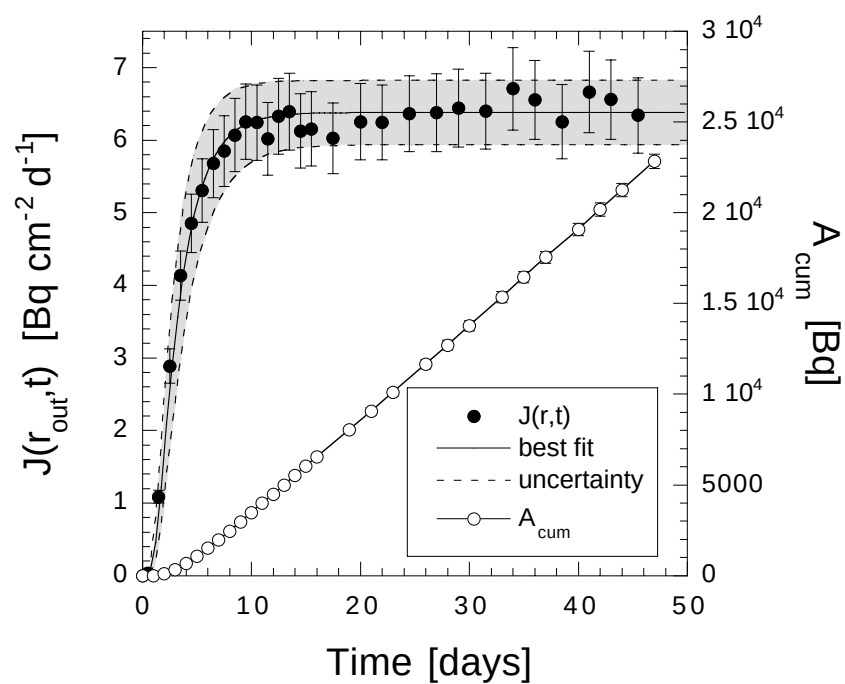


Figure A11-5.1: Flux and total diffused activity of HTO diffused through Opalinus Clay from Benken (BE-R4: -589 m) for diffusion parallel to bedding.

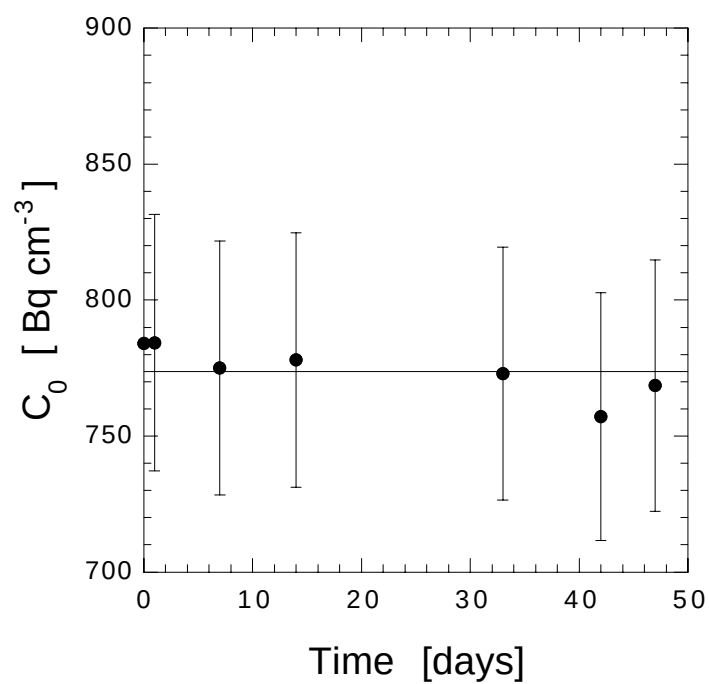


Figure A11-5.2: Concentration C_0 of HTO in the high-concentration container as a function of time (BE-R4: -589 m).

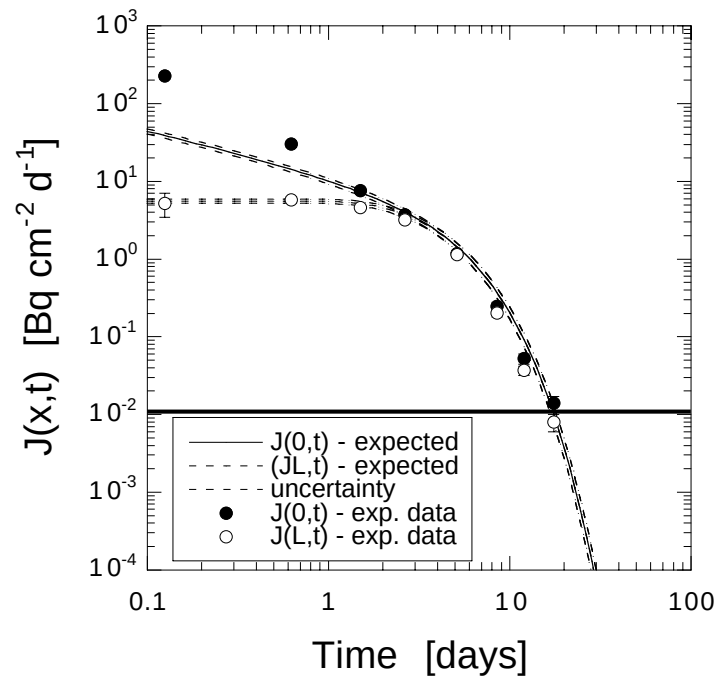
APPENDIX 12: OUT-DIFFUSION OF HTO (OTHER DEPTHS)

Figure A12-1.1: Out-diffusion of HTO in Opalinus Clay from Benken (BE-S0: -564 m) for diffusion perpendicular to bedding.

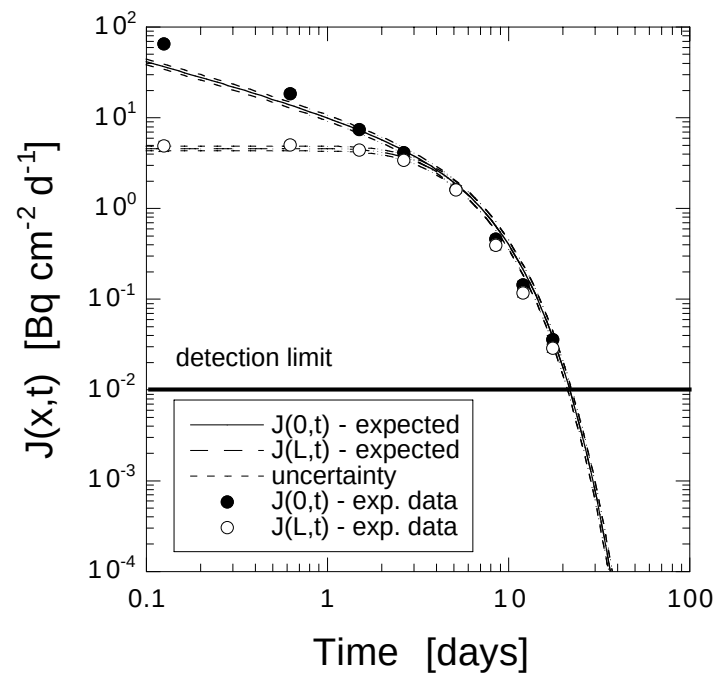


Figure A12-1.2: Out-diffusion of HTO in Opalinus Clay from Benken (BE-S0: -589 m) for diffusion perpendicular to bedding.

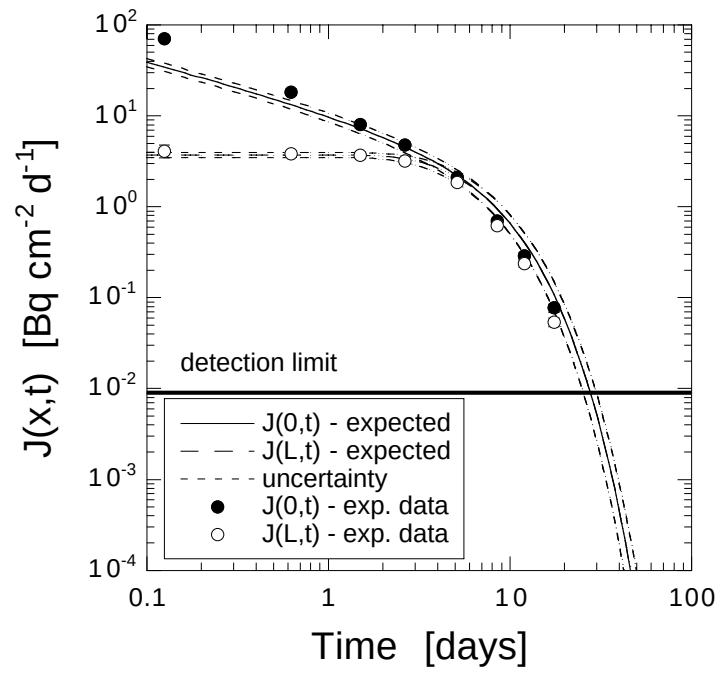


Figure A12-1.3: Out-diffusion of HTO in Opalinus Clay from Benken (BE-S0: -651 m) for diffusion perpendicular to bedding.